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Overview and Applications of the Reproducing Kernel Particle Methods

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Summary

Multiple-scale Kernel Particle methods are proposed as an alternative and/or enhancement to commonly used numerical methods such as finite element methods. The elimination of a mesh, combined with the properties of window functions, makes a particle method suitable for problems with large deformations, high gradients, and high modal density. The Reproducing Kernel Particle Method (RKPM) utilizes the fundamental notions of the convolution theorem, multiresolution analysis and window functions. The construction of a correction function to scaling functions, wavelets and Smooth Particle Hydrodynamics (SPH) is proposed. Completeness conditions, reproducing conditions and interpolant estimates are also derived. The current application areas of RKPM include structural acoustics, elastic-plastic deformation, computational fluid dynamics and hyperelasticity. The effectiveness of RKPM is extended through a new particle integration method. The Kronecker delta properties of finite element shape functions are incorporated into RKPM to develop a C^m kernel particle finite element method. Multiresolution and hp-like adaptivity are illustrated via examples.

1. INTRODUCTION

In solving engineering problems, an analyst is challenged by various complex physical phenomena. Due to the difficulties involved in obtaining analytical solutions for most practical problems, numerical methods have become a necessary alternative. This shift of interest has also been complemented by rapid progress in computer technology. The finite element method (FEM) has been one of the most widely used computational methods. However, FEM exhibits a number of shortcomings in handling problems involving large deformations, high gradients, or systems with a wide range of response spectrum.

One such shortcoming, mesh entanglement in large deformation analyses, can in part be alleviated by methods such as the marker-and-cell (MAC) method [Hirt (1975), Hirt (1983), Nichols and Hirt (1971), Harlow and Welch (1965)], the volume of fluid method [Harlow *et al.* (1976), Amsden and Harlow (1970)], the moving mesh technique [Subbiah et. al (1989)] and the Arbitrary Lagrangian-Eulerian (ALE) formulation [Liu *et al.* (1988), (1991), Huerta and Liu (1988), Belytschko and Kennedy (1978), Belytschko (1983)]. Though rather effective in handling large deformations and keeping track of moving boundaries, these methods require extensive computational effort. Consequently, particle methods have

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emerged as an attractive alternative by eliminating the need for a mesh. The Smooth Particle Hydrodynamics (SPH) method is a grid-free computational method based on a Lagrangian formulation [Lucy (1977), Monaghan and Gingold (1983), Monaghan (1982, 1988, 1992), Gingold and Monaghan (1977, 1982)]. The continuum is represented by a set of particles eliminating the need for a mesh, and consequently, alleviating mesh distortion problems. Moreover, the high accuracy characteristics of the Lagrangian formulation is retained. The SPH methods perform well in the absence of domain boundaries [Libersky and Petschek (1990)]. However, in finite domains, and when the number of unknowns in the problem decreases, [Johnson, Peterson, and Stryrk (1993)] the SPH methods are no longer as accurate as the traditional finite element methods. The element-free Galerkin method (EFG) [Belytschko (1994b, 1994c)] was developed by adding more accurate derivatives in the formulation, and by enforcing the boundary conditions through Lagrange multipliers. EFG proved to be more accurate than SPH, and some other particle methods such as the moving least-square method by Lancaster and Salkauskas (1981) and the diffuse element method (DEM) by Nayroles *et al.* (1992).

The processing of signals and data has been an important task in many fields ranging from mathematics to electrical engineering, from quantum physics to seismology, and many other fields. The focus on frequency analysis, which eventually led to wavelets, originated with J. Fourier in 1807. The Haar function [Haar (1910)] can be considered as the first wavelet with a compact support. Wavelets became an important milestone in signal processing field following the works by Grossman and Morlet (1984), Mallat (1989), Chui (1992) and Daubechies (1992). The multiresolution analysis facilitates the construction of a window function which can translate and dilate a function simultaneously. This zooming feature can be extended to solving problems with different scales. By a translation, a desired portion of the spectrum can be isolated and focused on, whereas the dilation determines the width of the window. Consequently, a space-time window function can span the entire space-time domain.

The versatility of wavelets makes them very attractive in the discretization of differential equations. Glowinski *et al.* (1990) explored the idea of using wavelets instead of piecewise polynomial trial functions in FEM type of methods. However, the straight use of wavelets as trial functions poses a number of difficulties. Low order wavelets cannot be employed due to the lack of regularity. The orthogonality of wavelets, as required in signal processing, is not important. Most significantly, the boundary conditions cannot be easily imposed in Dirichlet problems.

The shortcomings of the standard FEM can be overcome through the application of particle-based methods. Wavelets exhibit attractive features in handling problems with various scales. These two ideas can be combined into the reproducing kernel particle methods (RKPM) where the difficulties are alleviated by introducing a boundary correction function. The resolution of scales through multi-scale decomposition was introduced by Liu, Zhang and Ramirez. (1991a) This concept was later expanded into RKPM in solving structural acoustics problems. The recent developments proposed by Liu (1995) and Liu *et al.* (1995b, 1995c) included a boundary correction term to handle different boundary conditions, completeness conditions and error estimates. The method has been applied in various fields such as structural acoustics, large deformation problems, computational fluid dynamics and recently, micromechanics [Shodja *et al.* (1995)].

In addition to the above referenced methods, exotic topics such as hp clouds [Duarte and Oden (1995)], partition of unity finite element method [Babuška and Melenk (1995)], finite points method [Oñate *et al.* (1995)] and free mesh method [Yagawa *et al.* (1995)] were proposed recently.

The goal of this paper is to introduce the concepts involved in the development and use of RKPM, and show the effectiveness through several applications. Furthermore, several modifications are proposed to enhance the method, and extend its applicability.

The organization of this paper is as follows. The fundamental notions in the development of the reproducing kernel particle methods such as the convolution theorem, multiresolution analysis and window functions are introduced in Section 2. The concept of reproducing conditions and enrichment procedures are also included. Section 3 is devoted to the mathematical formulation of the reproducing kernel particle method. The need for a correction function is discussed, and the guidelines on how to construct one are given along with completeness conditions and error estimates. The application of the reproducing kernel particle method to structural acoustics is discussed in Section 4. Starting from the Fourier analysis, the effectiveness of RKPM is explored. The analysis of the wave equation and a Timoshenko beam are also presented together with several examples. The formulation of RKPM in large deformation applications is introduced in Section 5. The kernel stability is analyzed, and a one-dimensional elastic-plastic example is given. The formulation is also extended to hyperelasticity. To enhance the efficiency of RKPM, a new particle integration method is proposed in Section 6. A procedure to incorporate the Kronecker delta properties of finite element shape functions into RKPM is introduced in Section 7. The formulation of the resulting C^m kernel particle finite element method is also given in the same Section. In Section 8, a domain decomposition procedure is formulated to increase the efficiency of RKPM. Multiresolution analysis is reviewed, and an hp-like adaptivity approach in RKPM is outlined in Section 9. Future research areas are discussed in Section 10, followed by conclusions.

2. OVERVIEW OF RKPM

2.1 Introduction

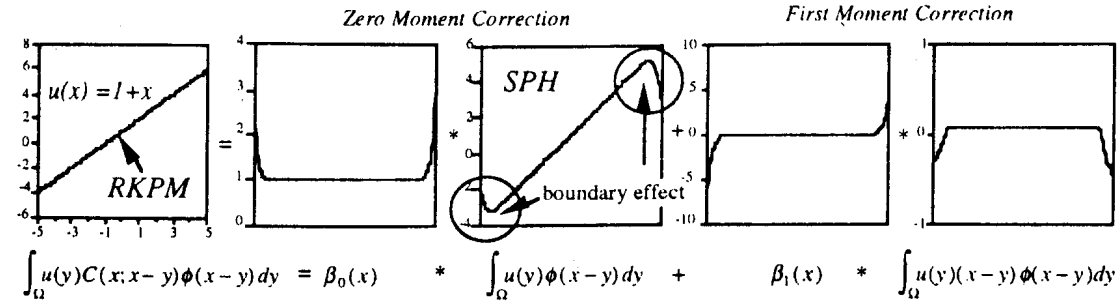
We have been working on multiple scale projection *meshless* methods, which are based on smooth particle hydrodynamics, discrete and continuous moving least square interpolation, continuous and discrete convolution, reproducing kernel and wavelet analysis. From our studies, we found that there is a close relationship among all these methods. This observation has also been pointed out by Belytschko (1994).

To improve the attributes of particle methods, moving least square interpolation, and wavelet analysis, we employ the concepts of reproducing kernels and multiresolution analysis from signal/image processing theories [Poularikas and Seely]. This section aims to provide the researcher with an overview and to share the authors experience with these methods. Due to their construction, artificial boundaries are generally needed in wavelet and SPH analysis. Both wavelet and SPH methods can be shown to belong to a class of reproducing kernel methods where the "reproduced" function $u^R(x)$ is given by:

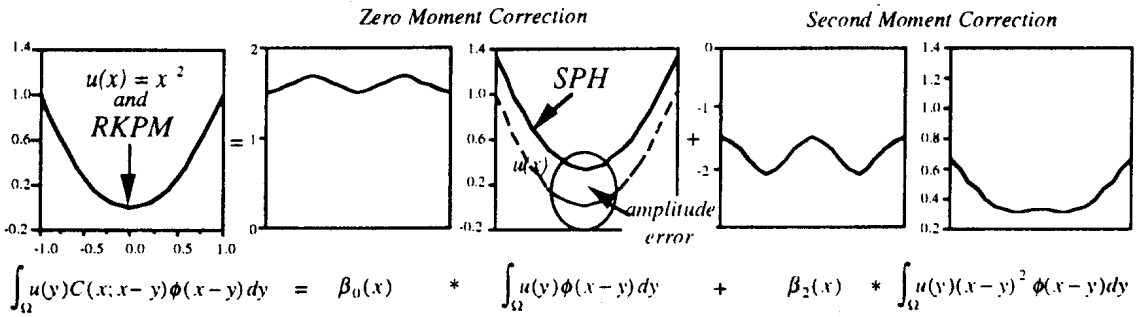
$$u^R(x) = \int_{-\infty}^{+\infty} u(y)\phi(x-y)dy \tag{2.1}$$

In Eq. (2.1), $\phi(x)$ can also be viewed as a projection operator. In references, [Liu *et al.* (1993, 1995a, 1995b), Liu and Chen (1995), Liu and Oberste (1993)] Liu and his coworkers, showed that with the proper construction of a correction function to SPH, scaling functions, and wavelets, not only can artificial boundaries be eliminated, but the accuracy of discrete solutions can also be enhanced throughout the entire domain. Hence, Eq. (2.1) can be rewritten for a bounded domain as:

$$u^R(x) = \int_{\Omega} u(y)C(x;x-y)\phi(x-y)dy \tag{2.2}$$



(a) Boundary correction



(b) Amplitude correction

Figure 2.1. a) Boundary correction. b) Amplitude correction

where $C(x; x-y)$ is the correction function and Ω is the computational domain. The effect of the correction function is demonstrated in Figures 2.1(a) and 2.1(b). Using a flexible time-frequency/space-wave number analysis, as described below, we show how the correction function,

$$C(x; x-y) = \beta_0(x) + \beta_1(x)(x-y) + \beta_2(x)(x-y)^2 + \dots \tag{2.3}$$

not only compensates for the boundary-effect in finite domain function recovery, but also corrects the amplitude error caused by the conventional convolution theorem; in a discrete convolution, phase error can also be corrected. Next, we show how the discrete boundary correction function can eliminate or minimize these errors. As it will be shown later, the $\beta's$ are defined in terms of moments of $\phi(x)$:

$$m_k(x) = \int_\Omega (x-y)^k \phi(x-y)dy \qquad k = 0, 1, 2, \dots \tag{2.4}$$

As can be seen in the Figure (2.1), without the correction function, SPH cannot satisfy consistency. In the same figures, the benefits of the correction function can be illustrated by a reconstruction of $u(x) = 1 + x$ and x^2. The reconstruction shown in these figures reviews the intricacies. Notably, it shows the difference between the RKPM and SPH methods.

Since SPH and wavelet analysis are based on the discretization of a continuous reproducing kernel, we call this class of methods reproducing kernel particle methods (RKPM). Eqs. (2.1) and (2.2) can be viewed as projections. Using the well known convolution theorem and time-frequency/space-wave number analysis [Poularikas and Seely] from signal processing, the intrinsic properties of this class of projection methods can also be revealed. Furthermore, from this basic understanding, better projection operators, $\phi(x)$, can be developed for a larger class of computational mechanics problems.

Multiple scale analysis has its origin in signal analysis. Wavelet analysis is a contemporary science in image processing. A complete description of wavelets can be found in Chui (1992) and Daubechies (1992). A good introduction to wavelets can be found in the articles by Strang (1989) and Williams (1994). However, one major drawback in its application to computational mechanics is its inability to handle large deformation and complex domains. With the help of the correction function for the scaling functions and wavelets and the introduction of a dilation parameter **a**, we are able to perform multiresolution analysis of an arbitrary domain using only a set of nodes (or particles). The incorporation of this dilation parameter into the reproducing kernel Eq.(2.2) gives:

$$u^{R_a}(x) = \int_\Omega u(y) C_a(x; x-y) \phi_a(x-y) dy = \int_\Omega u(y) \bar{\phi}_a(x; x-y) dy \tag{2.5}$$

One can view $\bar{\phi}_a(x)$ as a projection operator, or window function, so that the integral window transform of $\bar{\phi}_a(x)$ and $u(x)$, equivalent to the convolution $\bar{\phi}_a(x) * u(x)$ and denoted by $u^{R_a}(x)$, is a 'reproduction' of the original $u(x)$ with resolution of $\bar{\phi}_a(x)$. In other words, with the proper design of $\bar{\phi}_a(x)$, $u^{R_a}(x)$ will conserve all the resolutions and properties of the original solution $u(x)$ up to scale **a**. For this reason, we prefer to label $\bar{\phi}_a(x)$ and **a** as the scaling function and the scaling (or refinement) parameter, respectively. A smaller **a** implies a finer scale solution of $u(x)$. In particular, when **a** approaches zero, $\bar{\phi}_a(x)$ approaches the Dirac delta function so that $\lim_{a \to 0} u^{R_a}(x) \to u(x)$. Indeed, this is a key concept employed in the SPH methods. However, researchers in the SPH community have not made use of the well-known convolution theorem in the design of $\bar{\phi}_a(x)$, in particular at boundaries.

2.2 Convolution Theorem

Illustrations of the concept of the convolution theorem and the *reproducing kernel* are shown in Figure 2.2. From the convolution theorem, the spatial convolution $\phi(x) * u^R(x)$ denoted by $u^R(x)$, is equivalent to a multiplication in the Fourier transform domain such that

$$\hat{u}^R(\xi) = \hat{u}(\xi)\hat{\phi}(\xi) \tag{2.6}$$

Hence, if $\hat{u}(\xi)$ is band-limited, that is $\hat{u}(\xi) \neq 0$ if $|\xi| \leq \xi_c$ and $\hat{u}(\xi) = 0$ if $|\xi| > \xi_c$, then $\hat{u}^R(\xi) = \hat{u}(\xi)$ if one can design such an optimal $\hat{\phi}(\xi)$. It is noted that ξ_c is the cut-off or band-limited frequency (wave number). In signal processing, $\hat{\phi}(\xi)$ is known as a low pass filter, since $\hat{u}^R(\xi)$ contains only the $\hat{\phi}(\xi)$ part of the frequency (wave number) content of $\hat{u}(\xi)$.

Remark

If $\hat{\phi}(\xi)$ is replaced by $\hat{\phi}_a(\xi)$, then the physical interpretation of the scaling parameter **a** is more transparent. It can be viewed as the frequency, the wave number content, or the resolution scale of $u^{R_a}(x)$.

One obvious choice of $\hat{\phi}(\xi)$ is a perfect rectangular window function, the sinc function. Liu, Zhang and Ramirez (1991a) employed the sinc function as the reproducing kernel in Eq.(2.1) to develop multiple scale finite element methods. In this development, the time step used in the integration of the structural dynamic equations depends only on the band-width of the forcing function. It has been recognized in the signal processing literature that this ideal rectangular window function does not have a compact support in the spatial/temporal domain. Furthermore, the derivative of the sinc function is not square integrable so it is not useful in computational mechanics. This gives rise to the

famous Daubechies scaling functions and wavelets [Chui (1992), Daubechies (1992)], in which the functions are compactly supported temporally (and/or spatially); the higher order Daubechies window functions are very close to perfect rectangular windows in the Fourier transform domain. Hence, Daubechies scaling functions and wavelets are very effective in image processing. A typical Daubechies scaling function is depicted in Figure 2.3. However, because of the enforcement of orthogonality and compact support conditions, lower order Daubechies scaling functions are not smooth and their derivatives can be very irregular. For this reason, we do not plan to use Daubechies scaling functions in Eqs.(2.1-2.3), instead we shall incorporate the well known multiresolution analysis into these equations.

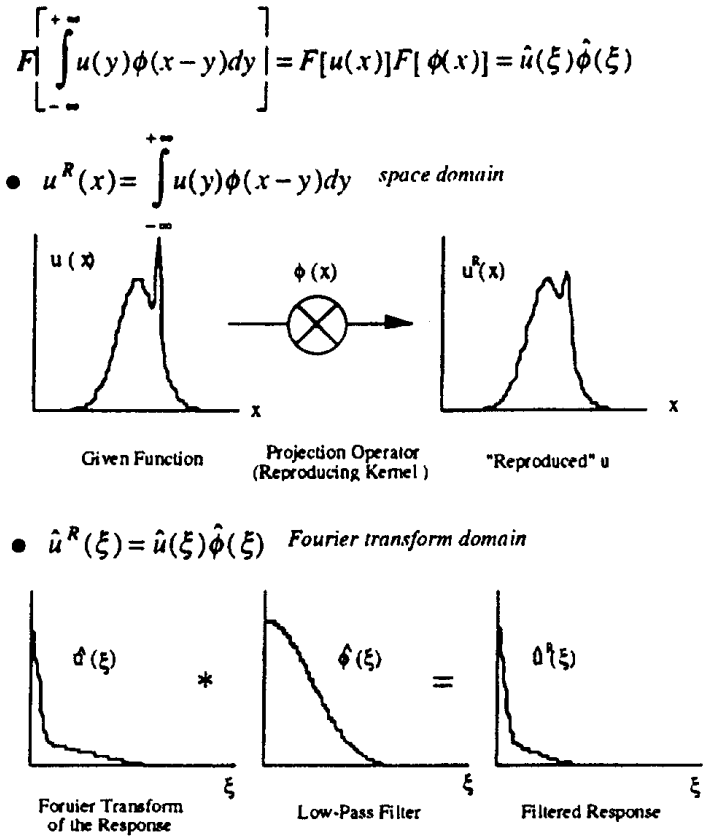


Figure 2.2. Convolution

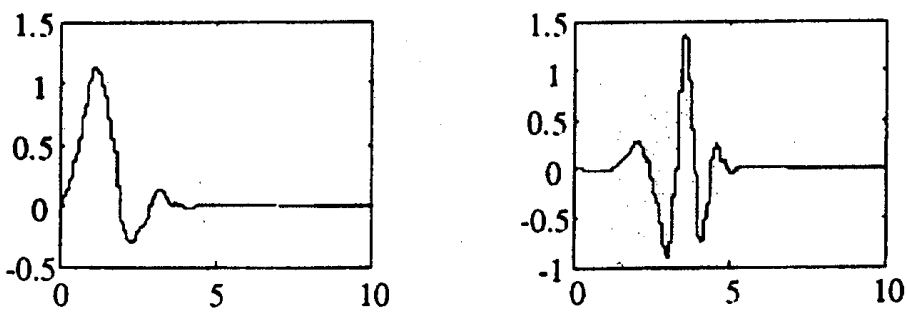


Figure 2.3. Daubechies scaling function and wavelet

2.3 Multiresolution Analysis

Illustrations of multiple scale projection methods, or the multiresolution analysis, are presented in Figure 2.4. We define a projection operator P_a through the reproducing kernel method in the form of:

$$P_a u(x) = \int_\Omega u(y)\bar{\phi}_a(x; x-y)dy \tag{2.7}$$

Based on the resolution of the projection operator $P_{a/2^n}$ a hierarchical representation of a function $u(x)$ is defined as:

$$\begin{aligned} u(x) = \lim_{n\to\infty} P_{a/2^n}u(x) \supset \ldots \supset P_{a/4}u(x) \supset P_{a/2}u(x) \supset P_a u(x) \supset P_{2a}u(x) \supset P_{4a}u(x) \\ \ldots \supset \lim_{n\to-\infty} P_{a/2^n}u(x) = 0 \end{aligned} \tag{2.8}$$

The framework of a multiresolution analysis, as illustrated by the scaling function projection in Figure 2.4, consists of nested closed subspaces defined in Eq.(2.8). If a complementary projection operator Q_{2a} is defined, then the higher scale projected solution $P_a u$ can be represented by the sum of the low and high scale projections:

$$P_a u = P_{2a}u + Q_{2a}u \tag{2.9}$$

The Q_{2a} projection can be viewed as a "peeled off" scale or the "rate of variation" of $P_a u$. It is also known as a wavelet projection given by:

$$Q_{2a}u(x) = \int_\Omega u(y)\bar{\Psi}_{2a}(x; x-y)dy \tag{2.10}$$

where $\bar{\Psi}_{2a}(x; x-y)$ is the 2**a** scale wavelet. It is defined as:

$$\bar{\Psi}_{2a}(x; x-y) = \bar{\phi}_a(x; x-y) - \bar{\phi}_{2a}(x; x-y) \tag{2.11}$$

with the well-known properties:

$$\int_\Omega y^k \bar{\Psi}_a(x; x-y)dy = 0 \qquad \text{for } k = 1, 2, \ldots, n \tag{2.12}$$

where n is the order of the polynomial based wavelet. The wavelet projection, Eq.(2.10), together with the recursive two-level decomposition, Eq.(2.9), constitute the back bone of the wavelet RKPM [Liu and Chen (1995), Liu and Oberste (1993)].

A typical example of the shape of the scaling function and wavelet is shown in Figure 2.5.

As pointed out by Liu *et al.* (1995b), we wish to study the response u in a rectangular time-frequency or space-wave number window given by:

$$[\bar{x} - aw_x, \bar{x} + aw_x] \times \left[\bar{\xi} - \frac{w_\xi}{a}, \bar{\xi} + \frac{w_\xi}{a}\right] \tag{2.13}$$

where w_x and w_ξ are the window widths, and $\bar{x}$ and $\bar{\xi}$ are the locations of the window functions in space/time and wave number/frequency domain, respectively. As illustrated in Figure 2.6, the window narrows to pick up the high frequency or high wave number phenomena of u and widens to study the low frequency or low wave number response. This suggests that we employ a flexible time-frequency or space-wave number reproducing kernel window to define adaptive refinement of the localized response of u around any point x

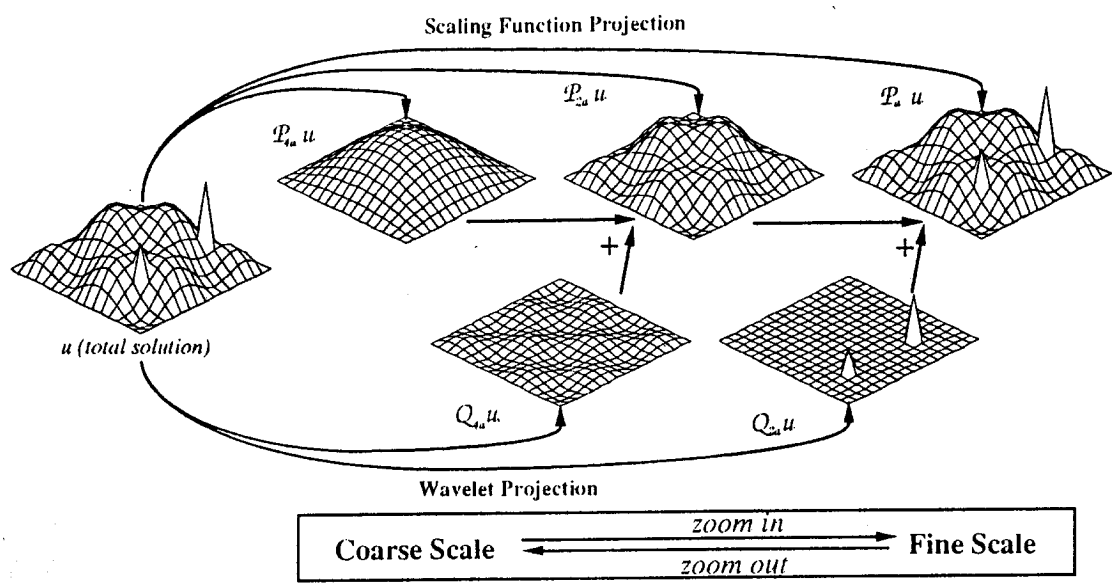


Figure 2.4. Multiresolution analysis

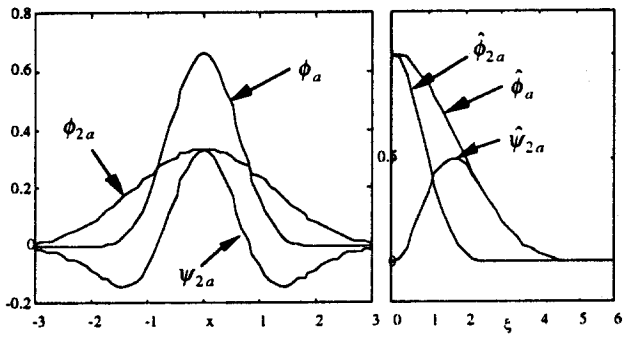


Figure 2.5. Scaling function and wavelet

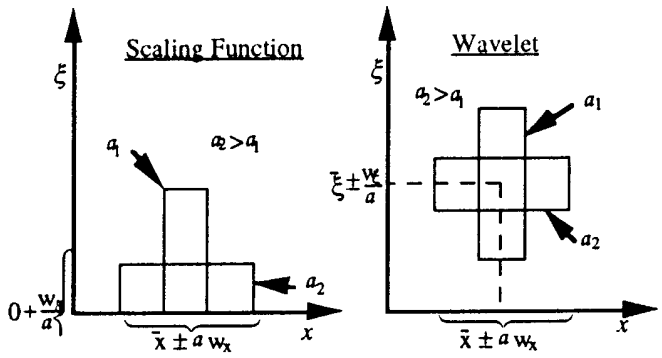


Figure 2.6. Time-frequency/space-wave number localization

[Liu and Chen (1995)]. Depending on the choice of **a** and the wavelet function, the integral wavelet transform, Eq.(2.10), is a measure of the amount of change of u at the location $\bar{x}$ with the zoom-in (smaller **a**) and zoom-out (larger **a**) capability. Hence, the wavelet projection, Eq.(2.10), can also be interpreted as the "peeled off" frequency/wave number band solution of $P_a u$.

With reference to Figure 2.4, the multiresolution analysis of $u(x)$ is given by:

$$P_a u = P_{2a} u + Q_{2a} u = P_{4a} u + Q_{4a} u + Q_{2a} u \tag{2.14}$$

Therefore, this class of reproducing kernel method permits the response of a system to be separated into different scales via projection operators. These scales can be viewed as the wave numbers corresponding to spatial variables, the frequencies corresponding to temporal variables and the fine-scale structure, or the edges of a discontinuity or the singular scales of the response. Through a multiresolution analysis, the image of each scale response can be magnified, examined, recorded and identified. This complete characterization of the unknown response is performed through the integral window transform. A space-scale and time-frequency localization process is achieved by dilating the flexible multiple scale window function. We believe the zoom in and zoom out capability of the window function is especially useful in examining complex flow phenomena, flow induced vibration, structural acoustics, wave propagation problems, dynamic stability of flow-structure interaction, turbulence structures, high frequency structural dynamics response and system identification problems.

2.4 Discrete Convolution Theorem

For computational mechanics, system discretization is inevitable. When a system is discretized, high frequency replicas (also known as aliasing) are introduced into the response. For a complex system, high frequency replicas may interact with the correct high frequency content of the response and it becomes impossible to separate them clearly. Hence, reducing the effect of high frequency replicas is a major step toward improving the solution.

The discretized version of Eqs.(2.1) and (2.6) are denoted by a superscript "h". Their graphical illustrations are shown in Figure 2.7. It is noted that $u(x_j)$ are the discrete nodal values and $SF[u(x)]$ denotes the semi-discrete Fourier transform of $u(x_j)$ given by:

$$SF[u(x)] = \hat{u}^s(\xi) = \sum_{j=-\infty}^{+\infty} u(x_j) e^{-i\xi x} \Delta x_j \tag{2.15a}$$

and

$$\hat{u}^{R_h}(\xi) = \hat{u}^s(\xi)\hat{\phi}(\xi) \tag{2.15b}$$

where $i = \sqrt{-1}$. Similar to the continuous convolution, the low-pass filter retains only the $\hat{\phi}(\xi)$ part of the frequency content of $\hat{u}^s(\xi)$. Consequently, by proper choice of $\hat{\phi}(\xi)$, the high frequency replicas can be eliminated. Moreover, using this local aliasing information, local refinement or hp-like adaptive meshless refinement can be carried out without the help of the exact solution.

Liu and Chen (1995) have developed an error estimation technique based on multiresolution analysis which is especially useful for local refinement and convergence studies. With the dilation parameter, we are able to separate the numerical noise from the high frequency/high wave number component of the physical phenomena. The flexible space-scale window function can be constructed to resemble the well-known unstructured multi-level/multi-grid and hp-adaptive finite element methods. However, the multiple scale adaptive refinements are performed simply by inserting nodes into the highest wavelet scale solution region, and at the same time narrowing the window function. Hence, hp-like adaptive refinements can be performed without a mesh.

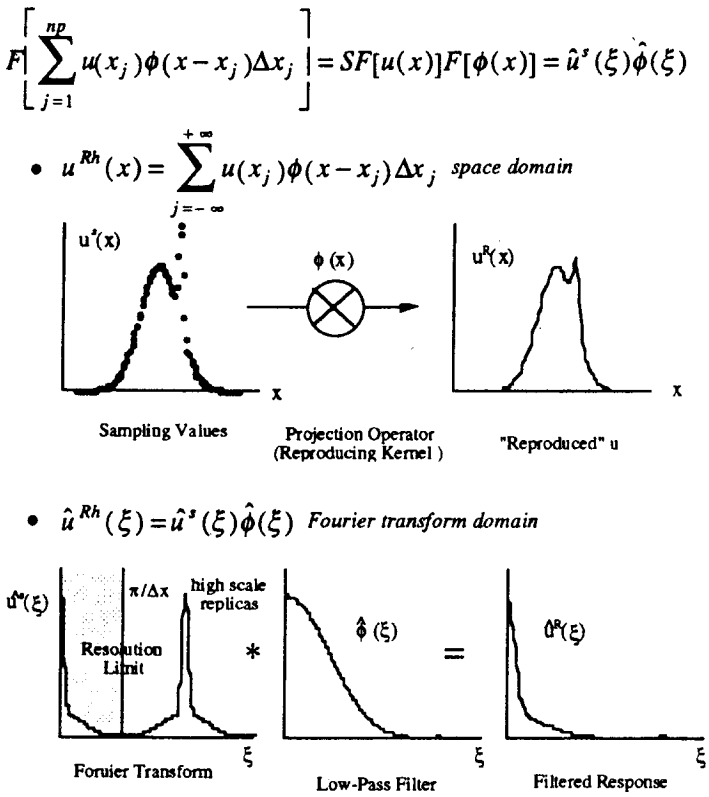


Figure 2.7. Discrete convolution

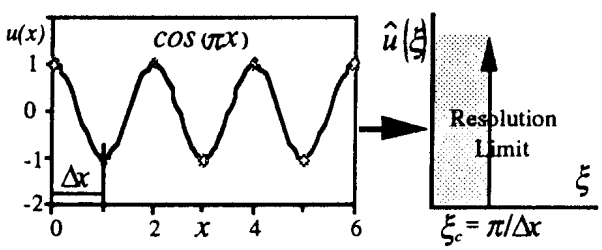


Figure 2.8. Resolution limit

2.5 Discrete System Resolution Limit

From the Shannon theorem [Poularikas and Seely], for a given sampling rate Δx (resolution h), the highest frequency/wave number that can be resolved is $\pi/\Delta x$. It is our intention to construct $\bar{\phi}_a(x)$ with an optimal scaling parameter **a** such that ξ_c is maximized. In other words, what is the minimum number of points (nodes or particles) that is required to accurately represent a sine/cosine wave as illustrated in Figure 2.8? With this construction of RKPM, can Galerkin methods or Galerkin least square procedures or new solution methods be devised so that a $\cos(\pi x)$ wave can be represented by only three nodes ($2\Delta x$)? It is noted from Figure 2.8 that $\xi_c = \pi/\Delta x$ is the resolution limit, and hence, the minimum number of nodes to represent a sine/cosine wave is three.

2.6 Reproducing Conditions

We now employ the well established theories in the construction of wavelets and scaling functions to determine a correction function which is valid at any point within the bounded domain.

As can be imagined from Eq.(2.6), for $\hat{u}^{R_a}(\xi)$ to approach $\hat{u}(\xi)$, $\hat{\phi}_a(\xi)$ must be constructed such that $\hat{\phi}_a(\xi) = 1$ for all ξ; however, this will violate the Riemann-Lebesgue Lemma (i.e. finite energy criterion) that $\hat{\phi}_a(\xi) = 0$ when ξ approaches $\pm\infty$. In general, a compact support scaling function such as the spline family, does not satisfy this condition. Therefore, we propose to add a correction to $\hat{\phi}(\xi)$ via:

$$\hat{\bar{\phi}}(\xi) = \sum_{k=0}^{n} \alpha_k \hat{\phi}^{(k)}(\xi) \tag{2.16}$$

For the spline family, since $\hat{\phi}(0) = 1$, $\hat{\phi}^{(1)}(0) = 0$ and $\hat{\phi}^{(k)}(0) \neq 0$ for $k \geq 2$, $\hat{\phi}(\xi)$ is not close to a rectangular filter. This suggests that we should determine a set of α_i such that $\hat{\phi}(0) = 1$ and $\hat{\phi}^{(k)}(0) = 0$ for $k \geq 1$. It can be shown that

$$m_k(x) = i^k \hat{\phi}^{(k)}(0) \qquad \text{for} \qquad k \in [0, n] \tag{2.17}$$

Hence Eq.(2.16) becomes:

$$\bar{\phi}(x; x-y) = \sum_{k=0}^{n} \beta_k(x)(x-y)^k \phi(x-y) \tag{2.18}$$

Eqs. (2.16) and (2.18) will help us design $\bar{\phi}(x)$ and $\hat{\bar{\phi}}(\xi)$ with compact support simultaneously in both space and wave number (scale) domains. From Table 2.1, by constraining $\hat{\bar{\phi}}(\xi)$ equal to 1 and all derivatives equal to zero at $\xi = 0$, we can set up the moment equation to solve for $\beta's$:

$$\begin{bmatrix} m_0(x) & m_1(x) & \dots & m_n(x) \\ m_1(x) & m_2(x) & \dots & m_{n+1}(x) \\ \vdots & & & \vdots \\ m_n(x) & m_{n+1}(x) & \dots & m_{2n}(x) \end{bmatrix} \begin{bmatrix} \beta_0(x) \\ \beta_1(x) \\ \vdots \\ \beta_n(x) \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \tag{2.19}$$

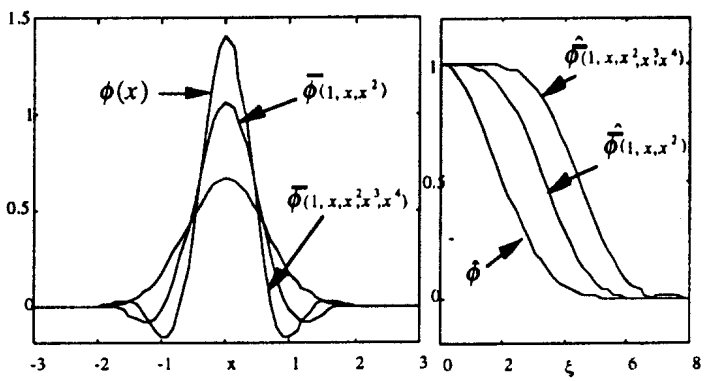


Figure 2.9. Compact supported windows in spatial and Fourier transform domains

Figure 2.9 clearly shows the effect of the correction function as defined in Eq. (2.18) on a cubic spline window function in the Fourier transform domain. As we impose $\hat{\phi}^{(2)}(0) = 0$ and $\hat{\phi}^{(4)}(0) = 0$, the modified $\hat{\bar{\phi}}(\xi) = 0$ becomes flatter at $\xi = 0$. However, all $\hat{\bar{\phi}}^{(k)}(\xi)'$s decay to zero at $\pi/\Delta x$ by the proper choice of the dilation parameter a. It can be imagined from the same figure that when $k \to \infty$, the modified window $\hat{\bar{\phi}}(\xi)$ will approach an ideal rectangular filter. However, this window will have compact support in both space and frequency domains. This window function is suitable for computational mechanics since it is square integrable in its derivatives, and they can be computed by simple differentiation.

2.7 Discrete Reproducing Conditions

The above derivation of the moment matrix equations also holds for all x in the bounded domain. The moments $\tilde{m}_k(x)$ have to be computed by numerical integration, denoted by:

$$\tilde{m}_k(x) = \sum_{j=1}^{np} (x - x_j)^k \phi(x - x_j) \Delta x_j \tag{2.20}$$

where NP is the number of quadrature points used.

Unlike the case of continuous moments, this set of discrete moments, as shown in Table 2.2, gives rise to amplitude and phase errors of the "reproduction" of $u(x) = 1, x, x^2, \ldots$ etc. However, for the spline family, the amplitude and phase errors can be minimized or even eliminated via a technique called energy error ratio developed by Liu and Chen (1995). This is performed by choosing the dilation parameter **a** such that $\hat{\bar{\phi}}(2n\pi/\Delta x) \approx 0$ and $\hat{\bar{\phi}}^{(k)}(2n\pi/\Delta x) \approx 0$. It is noted that the enforcement of $\tilde{m}_0(x) = \hat{\bar{\phi}}(0) = 1$ and $\tilde{m}_k(x) = \hat{\bar{\phi}}^{(k)}(0) = 0$ for $k \geq 1$ is a sufficient condition for the consistency condition, or completeness, similar to that of finite element method.

2.8 Sampling Shape Functions

Through the implementation of the above described window function and the exploitation of the Fourier transform, we develop a new type of sampling shape function given by:

$$u^{R_h}(x) \equiv \sum_{j=1}^{np} u(x_j) C_a(x; x - x_j) \phi_a(x - x_j) \Delta x_j \tag{2.21}$$

for the reproducing kernel particle method (RKPM) that is closely related to finite elements. In particular, if all Δx_j are set equal to one in Eqs.(2.18) through (2.21), the resulting sampling shape functions are reduced to the moving least square interpolants; however, in a more transparent form. They also possess better numerical stability due to the choice of the normalized moments as defined in Eq. (2.20) [Liu and Oberste (1993), Liu *et al.* (1995b)]. These new shape functions can still be used in the usual Galerkin or other formulations. The derivative of the shape functions can be obtained by direct differentiation without a finite element mapping. The two parameters, **a** and x_j, in the scaling function provide the ability to translate and dilate the window function. Translation is required to move the window function around the domain, since the window functions themselves have a compact support. The *ability to translate replaces the need to define elements.* The *dilation parameter is used to provide refinement.* This dilation parameter also controls the convergence rate of the multiple scale RKPM.

In Figure 2.10, we show the effect of the dilation parameter **a** on the number of nodes covered under the window. The dE/E ratio, the sampling shape functions and their derivatives are also presented.

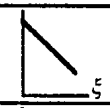
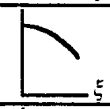
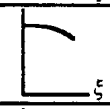
basis	Reproducing Kernel			Reproducing Conditions in terms of	
	Convolution	in terms of moments	in FT Domain @ $\xi = 0$	Moments	in FT domain @ $\xi = 0$
1	$= \int_{-\infty}^{+\infty} 1 \cdot \phi\left(\frac{x-y}{a}\right) dy$	$= m_0(x)$	$= \hat{\phi}(0)$	$m_0(x) = 1$	$\hat{\phi}(0) = 1$ 
x	$= \int_{-\infty}^{+\infty} y \cdot \phi\left(\frac{x-y}{a}\right) dy$	$= x m_0(x) - m_1(x)$	$= x\hat{\phi}(0) - ia\,\hat{\phi}'(0)$	$m_1(x) = 0$	$\hat{\phi}'(0) = 0$ 
x^2	$= \int_{-\infty}^{+\infty} y^2 \cdot \phi\left(\frac{x-y}{a}\right) dy$	$= x^2 m_0(x) - 2x m_1(x) + m_2(x)$	$= x^2\hat{\phi}(0) - i\,2ax\hat{\phi}'(0) - a^2\hat{\phi}''(0)$	$m_2(x) = 0$	$\hat{\phi}''(0) = 0$ 
x^n	$= \int_{-\infty}^{+\infty} y^n \cdot \phi\left(\frac{x-y}{a}\right) dy$	$= \sum_{k=0}^{n} (-1)^k C_k^n x^{n-k} m_k(x)$	$= \sum_{k=0}^{n} (-i)^k C_k^n x^{n-k} a^k \hat{\phi}^{(k)}(0)$	$m_n(x) = 0$	$\hat{\phi}^{(n)}(0) = 0$

Table 2.1 Reproducing conditions for continuous system

basis	Reproducing Kernel			Reproducing Conditions int terms of Moments
	Discrete Convolution	in terms of moments	in FT Domain @ $\xi = 0$ and $2m\pi/\Delta x$	
1	$= \sum_{j=-\infty}^{+\infty} 1 \cdot \phi\left(\frac{x-x_j}{a}\right)\Delta x_j$	$= \tilde{m}_0(x)$	$= \hat{\phi}(0) + (APET)_0$	$\tilde{m}_0(x) = 1$
x	$= \sum_{j=-\infty}^{+\infty} x_j \cdot \phi\left(\frac{x-x_j}{a}\right)\Delta x_j$	$= x\tilde{m}_0(x) - \tilde{m}_1(x)$	$= x\hat{\phi}(0) - ia\hat{\phi}'(0) + (APET)_0 - ia(APET)_1$	$\tilde{m}_1(x) = 0$
x^2	$= \sum_{j=-\infty}^{+\infty} x_j^2 \cdot \phi\left(\frac{x-x_j}{a}\right)\Delta x_j$	$= x^2\tilde{m}_0(x) - 2x\tilde{m}_1(x) + \tilde{m}_2(x)$	$= x^2\hat{\phi}(0) - i\,2ax\hat{\phi}'(0) - a^2\hat{\phi}''(0) + x^2(APET)_0 - i\,2ax(APET)_1 - a^2(APET)_2$	$\tilde{m}_2(x) = 0$
x^n	$= \sum_{j=-\infty}^{+\infty} x_j^n \cdot \phi\left(\frac{x-x_j}{a}\right)\Delta x_j$	$= \sum_{k=0}^{n} (-1)^k C_k^n x^{n-k} \tilde{m}_k(x)$	$= \sum_{n=0}^{k} (-i)^k C_k^n x^{n-k} a^k \hat{\phi}^{(k)}(0) + \sum_{n=0}^{k} (-i)^k C_k^n x^{n-k} a^k (APET)_k$	$\tilde{m}_n(x) = 0$
			Amplitude and Phase Error Term $(APET)_k = \sum_{m \in Z^{\pm}} \hat{\phi}^{(k)}\left(\frac{2m\pi}{\Delta x}\right) e^{i(2m\pi/\Delta x)}$	

Table 2.2. Discrete reproducing conditions for discrete system

As can be seen in Figure 2.10, by simply changing the scale **a** or equivalently the dE/E ratio, we obtain a variable p-order sampling shape function. It is striking to see that the sampling shape functions reduce to the linear finite element shape functions for $dE/E = 14.47$ % with $\mathbf{a} = 0.5$ (for a cubic spline). RKPM has a feature similar to the adaptive p-finite element (p-FE) methods. However, the high convergence rate, or equivalently, the high value of "p", can be achieved simply by increasing the size of the refinement parameter of a *single* window function. This avoids the restrictive requirement of the traditional p-finite element methods, and there is no apprehension of the compatibility (or continuity) along the element boundaries of the hp-finite element (hp-FE) mesh, since RKPM requires only a set of nodes and the global shape functions can be C^∞! As depicted in Figure 2.11, the rate of convergence of the L_2 norm (and H_1 norm) of a smooth Laplacian solution varies from 2 to 16 (and 1 to 15) for a Gaussian window by simply changing the *refinement* or dilation parameter of a single window function. This combination of translation and dilation produces a *meshless p-like adaptive variable-node multiple scale RKPM.*

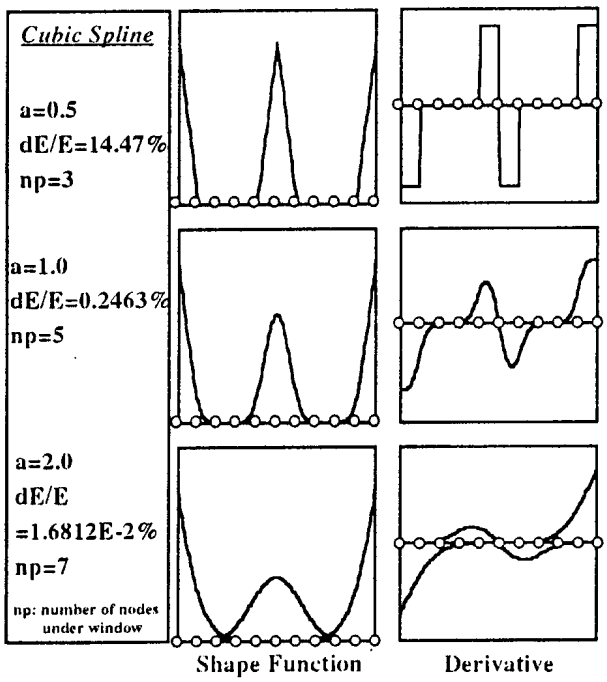


Figure 2.10. The effect of dilation parameter

2.9 Generalized RKPM–Arbitrary Window Function

The reproducing conditions can be extended to a set of generalized linearly independent window functions including finite element shape functions as the window functions. However, in general this arbitrary set of window functions

$$\nu^h = \{\phi_1, \phi_2, \ldots, \phi_{np}\} \tag{2.22}$$

does not satisfy the consistency conditions as defined in the discrete reproducing conditions, Table 2.2. Therefore, we apply the discretized version of the correction function $C(x; x - x_j)$

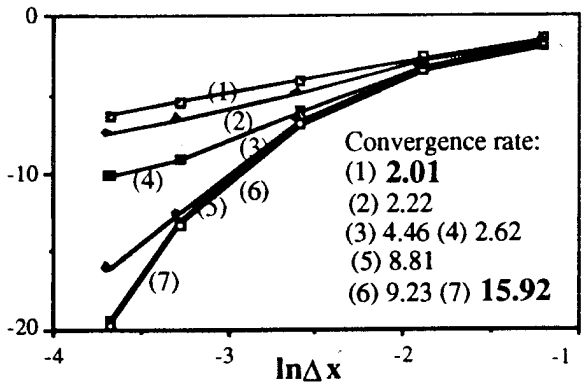


Figure 2.11. p-type behavior

of Eq.(2.18) [Liu *et al.* (1995d)]:

$$\bar{\phi}_j(x) = \left[\sum_{k=0}^{n} \beta_k(x)(x-x_j)^k\right] \phi_j(x) = C(x; x-x_j)\phi_j(x) \tag{2.23}$$

With this modification of window functions, the generalized reproducing conditions are given as:

$$\tilde{m}_0(x) = \sum_{j=1}^{np} 1 \cdot \bar{\phi}_j(x)\Delta x_j = 1 \tag{2.24}$$

$$\tilde{m}_k(x) = \sum_{j=1}^{np} (x-x_j)^k \bar{\phi}_j(x)\Delta x_j = 0 \tag{2.25}$$

and the $\beta's$ can be determined by Eq. (2.19).

Remark:

In a large deformation analysis, the finite element nodes and finite elements can be replaced by RKPM shape functions to avoid mesh distortion.

2.10 Computation of $u(x_j)$ and $\hat{u}^s(\xi)$

We have explored the reconstruction of a function $u(x)$ using signal processing theories by assuming that $u(x_j)$ are given. In computational methods, $u(x_j)$ needs to be determined through Galerkin or other solution methods, such as least-square projection or Galerkin Least-Square (GLS). We wish to illustrate that the frequency or wave number content of $\hat{u}^s(\xi)$ can be affected by these solution methods. It can be seen from the discrete convolution theorem, Eq. (2.15b), that if we can obtain the correct frequency content or the correct scale of $u(x_j)$, the sampling shape function interpretation Eq. (2.21) will converge to the reproducing kernel solution $u^R(x)$. The effect of the Galerkin and GLS solution procedure on the frequency spectrum of $\hat{u}^s(\xi)$ is illustrated in Figure 2.12. As can be seen, GLS eliminates the incorrect high frequency content and hence, reduces the phase error when compared to that obtained using a Galerkin procedure. It should be noted that both Galerkin and GLS provide amplitude correction of $u(x_j)$, and hence unlike in signal processing no special treatment of the data is required.

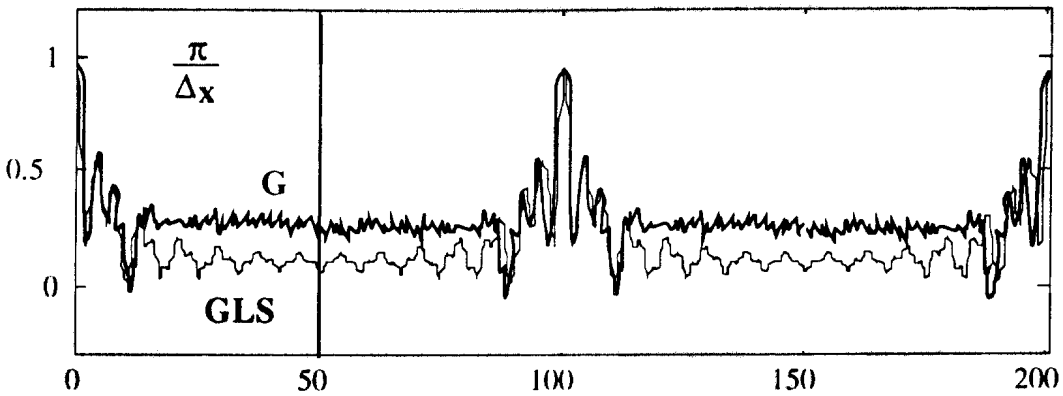


Figure 2.12. Effect of Galerkin and GLS

3. MATHEMATICAL FORMULATIONS OF RKPM

In this section, some fundamental concepts introduced in Section 2 are revisited. The equations which play an important role in the derivations are rewritten in multiple space dimensions. The fundamental concepts are elaborated further. The authors believe that this repetition will provide fluidity in the developments presented in this section.

3.1 Fundamental Concepts

As indicated in Section 2.1, the solution to a differential equation, or a set of differential equations can be expressed as

$$u^R(\mathbf{x}) = \int_{-\infty}^{+\infty} u(\mathbf{y})\phi(\mathbf{x};\mathbf{x}-\mathbf{y})d\mathbf{y} \tag{3.1}$$

where ϕ is a kernel function which acts like a projection operator and u^R is the "reproduced" solution of $u(\mathbf{x})$ This form is one of the fundamental developments in many interpolation methods, and will be referred to as the reproducing kernel method. Widely used methods such as SPH, and wavelet methods also belong to this class of methods. One major drawback is the need for a special boundary treatment in finite domains. Through proper construction of a boundary correction term, the artificial boundaries required by the SPH and wavelet methods can be eliminated, and the accuracy of the discrete solution is improved [Liu (1995)].

In problems involving finite domains, the SPH solutions reveal amplitude and phase errors mainly due to the boundary effects. This phenomenon can be attributed to the failure to meet the completeness requirements.

In a finite domain, Eq.(3.1) can be written as:

$$u^R(\mathbf{x}) = \int_{\Omega} u(\mathbf{y})\bar{\phi}(\mathbf{x};\mathbf{x}-\mathbf{y})d\mathbf{y} \tag{3.2}$$

where

$$\bar{\phi}(\mathbf{x};\mathbf{x}-\mathbf{y}) = C(\mathbf{x};\mathbf{x}-\mathbf{y})\phi(\mathbf{x}-\mathbf{y}) \tag{3.3}$$

The correction function $C(\mathbf{x};\mathbf{x}-\mathbf{y})$ can be constructed in such a way to avoid the difficulties mentioned above. Since the integral defined in Eq. (3.2) is too complicated to be carried out analytically, it is generally discretized either by a grid (as in the finite element

method), or by particles (SPH methods). The methods involving mesh-free Lagrangian particles concerned with the solution of Eq. (3.2) are referred to as reproducing kernel particle methods (RKPM). Before proceeding with the construction of a consistent correction function, a final word is given on the discrete convolution concept. By spatial/temporal discretization, high-frequency replicas (commonly referred to as aliasing) are introduced into the system response. In complex mechanical systems, these non-physical frequencies may interact with the physical frequencies. The elimination of aliasing is very important in representing the true frequency content of the system. A comprehensive discussion and an error estimation procedure to separate physical and non-physical frequencies are given in Liu and Chen (1995).

3.2 The Necessity of a Correction Function

The construction of an appropriate correction function to compensate for the boundary effects and to minimize the amplitude and phase errors is as follows:

$$\bar{\phi}(\mathbf{x};\mathbf{x}-\mathbf{y}) = C(\mathbf{x};\mathbf{x}-\mathbf{y})\phi(\mathbf{x}-\mathbf{y}) = \sum_{k=0}^{n} \beta_k(\mathbf{x})(\mathbf{x}-\mathbf{y})^k \phi(\mathbf{x}-\mathbf{y}) \tag{3.4}$$

The moments of $\phi(\mathbf{x})$ are defined as:

$$m_k(\mathbf{x}) = \int_\Omega (\mathbf{x}-\mathbf{y})^k \phi(\mathbf{x}-\mathbf{y}) d\mathbf{y} \qquad k = 0, 1, 2, \ldots \tag{3.5}$$

The unknown constants $\beta' s$ are determined by imposing the reproducing conditions which will be introduced in Section 3.5. Although wavelet methods enjoy a wide range of applications in signal and image processing, they are not suitable for handling complex problems in computational mechanics. However, a dilation parameter, $\mathbf{a}$, a concept encountered in wavelet theory, can be embedded in the correction function. Through the introduction of this parameter, a multiresolution analysis can be performed on a complex domain. Consequently, Eq. (3.2) can be rewritten in terms of $\mathbf{a}$:

$$u^{R_a}(\mathbf{x}) = \int_\Omega u(\mathbf{y}) C_a(\mathbf{x};\mathbf{x}-\mathbf{y}) \phi_a(\mathbf{x}-\mathbf{y}) d\mathbf{y} \equiv \int_\Omega u(\mathbf{y}) \bar{\phi}_a(\mathbf{x};\mathbf{x}-\mathbf{y}) d\mathbf{y} \tag{3.6}$$

where $\bar{\phi}_a(\mathbf{x};\mathbf{x}-\mathbf{y})$ can be interpreted not only as a projection operator, but as a window function as well. This projection operator/window function is in fact a scaling function designed to reproduce all the inherent properties, up to the scale $\mathbf{a}$. Since a smaller value of $\mathbf{a}$ implies a finer resolution, as $\mathbf{a}$ approaches zero, the reproduced solution $u^R(\mathbf{x})$ would approach $u(\mathbf{x})$.

3.3 The Construction of the Reproducing Kernel

For an arbitrary kernel, $\bar{\phi}_a(\mathbf{x};\mathbf{x}-\mathbf{y})$ the reproducing equation for $u^{R_a}(\mathbf{x})$ is given by:

$$u^{R_a}(\mathbf{x}) = \int_\Omega u(\mathbf{y}) \bar{\phi}_a(\mathbf{x};\mathbf{x}-\mathbf{y}) d\mathbf{y} \tag{3.7}$$

Consider the Taylor series expansion of $u(\mathbf{y})$ around $\mathbf{x}$ up to the nth order,

$$u(\mathbf{y}) = \sum_{|\alpha|=0}^{n} \frac{(-1)^\alpha}{\alpha!} D^\alpha u(\mathbf{x})(\mathbf{x}-\mathbf{y})^\alpha + \sum_{|\alpha|=n+1} \frac{(-1)^\alpha}{\alpha!} D^\alpha u(\bar{\mathbf{x}})(\mathbf{x}-\mathbf{y})^\alpha \tag{3.8}$$

where D denotes a differential operator $\bar{\mathbf{x}} = \mathbf{x} + \theta(\mathbf{y}-\mathbf{x})$ and $0 < \theta \leq 1$.

It is noted that in order to get the above simplified expression, a new kind of notation system, *multi-index notation*, is applied. Below are some definitions of this notation for ND dimensions.

$$(i) \quad |\alpha| \equiv \sum_{i=1}^{ND} \alpha_i \qquad\qquad (ii) \quad \mathbf{x}^\alpha \equiv x_1^{\alpha_1} x_2^{\alpha_2} \dots x_{ND}^{\alpha_{ND}}$$

$$(iii) \quad D^\alpha u(\mathbf{x}) \equiv \partial_{x_1}^{\alpha_1} \partial_{x_2}^{\alpha_2} \dots \partial_{x_{ND}}^{\alpha_{ND}} u(x) \qquad (iv) \quad \alpha! \equiv \alpha_1! \alpha_2! \dots \alpha_{ND}!$$

The expansion in Eq.(3.8) can be substituted into Eq.(3.7) to obtain

$$u^{R_a}(\mathbf{x}) = \sum_{|\alpha|=0}^{n} \partial^\alpha u(\mathbf{x}) \bar{m}_\alpha(\mathbf{a}, \mathbf{x}) + \sum_{|\alpha|=n+1} \partial^\alpha u(\bar{\mathbf{x}}) \bar{m}_\alpha(\mathbf{a}, \mathbf{x}) \tag{3.9}$$

where

$$\partial^\alpha u(\mathbf{x}) = \frac{(-1)^\alpha}{\alpha!} D^\alpha u(\mathbf{x}) \qquad 0 \le |\alpha| \le n+1 \tag{3.10}$$

$$\bar{m}_\alpha(\mathbf{a}, \mathbf{x}) = \int_\Omega (\mathbf{x} - \mathbf{y})^\alpha \bar{\phi}_\alpha(\mathbf{x}; \mathbf{x} - \mathbf{y}) d\mathbf{y} \tag{3.11}$$

The γth derivative of the reproducing equation for $u^{R_a}(\mathbf{x})$ is given by:

$$D^\gamma u^{R_a}(\mathbf{x}) = \int_\Omega u(\mathbf{y}) D^\gamma \bar{\phi}_a(\mathbf{x}; \mathbf{x} - \mathbf{y}) d\mathbf{y} \tag{3.12}$$

Applying a Taylor series expansion of $u(\mathbf{y})$ again, Eq.(3.12) can be rewritten as:

$$D^\gamma u^{R_a}(\mathbf{x}) = \sum_{|\alpha|=0}^{n} \partial^\alpha u(\mathbf{x}) \bar{m}_{\alpha,\gamma}(\mathbf{a}, \mathbf{x}) + \sum_{|\alpha|=n+1} \partial^\alpha u(\bar{\mathbf{x}}) \bar{m}_{\alpha,\gamma}(\mathbf{a}, \mathbf{x}) \tag{3.13}$$

where

$$\bar{m}_{\alpha,\gamma}(\mathbf{a}, \mathbf{x}) = \int_\Omega (\mathbf{x} - \mathbf{y})^\alpha D^\gamma \bar{\phi}_\alpha(\mathbf{x}; \mathbf{x} - \mathbf{y}) d\mathbf{y} \tag{3.14}$$

It can be seen that during the procedure of constructing the reproducing kernel, the Taylor series expansion for multiple dimensions plays an important role. Moreover, the following derivations for reproducing conditions and interpolant error estimates are derived with the help of this useful tool.

3.4 Conditions for Completeness and Consistency

Lemma 1a) Completeness/consistency conditions

$u^{R_a}(\mathbf{x})$ reproduces $u(\mathbf{x})$ up to the nth order if

$$\boxed{\bar{m}_\alpha(\mathbf{a}, \mathbf{x}) = \delta_{\alpha 0}} \qquad 0 \le |\alpha| \le n \tag{3.15}$$

and the remainder is given by:

$$\boxed{u^{R_a}(\mathbf{x}) - u(\mathbf{x}) = \sum_{|\alpha|=n+1} \partial^\alpha u(\bar{\mathbf{x}}) \bar{m}_\alpha(\mathbf{a}, \mathbf{x})} \tag{3.16}$$

Proof: By construction. Set $\bar{m}_0(\mathbf{a}, \mathbf{x}) = 1$ and $\bar{m}_k(\mathbf{a}, \mathbf{x}) = 0$ $(k = 1, n)$ in Eq.(3.9), that is $\bar{m}_\alpha(\mathbf{a}, \mathbf{x}) = \delta_{\alpha 0}$. Then $u(\mathbf{x})$ can be reproduced up to the nth order.

Remarks

a) Lemma 1a implies the determination of $\beta(\mathbf{x})$ such that $\bar{m}_\alpha(\mathbf{a},\mathbf{x}) = \delta_{\alpha 0}$.
b) The reproducing kernel $\bar{\phi}_a(\mathbf{x};\mathbf{x}-\mathbf{y}) = C(\mathbf{a},\mathbf{x})\phi_a(\mathbf{x}-\mathbf{y})$ can reproduce a complete set of polynomials up to the nth order.

Lemma 1b) Super completeness and consistency conditions

$D^\gamma u^{R_a}(\mathbf{x})$ reproduces $D^\gamma u(\mathbf{x})$ up to the nth order if

$$\boxed{\bar{m}_{\alpha,\gamma}(\mathbf{a},\mathbf{x}) = (-1)^\alpha \alpha! \delta_{\alpha,\gamma}} \qquad 1 \le |\alpha|, \gamma \le n \tag{3.17}$$

and the remainder is given by:

$$\boxed{D^\gamma u^{R_a}(\mathbf{x}) - D^\gamma u(\mathbf{x}) = \sum_{|\alpha|=n+1} \partial^\alpha u(\bar{\mathbf{x}}) \bar{m}_{\alpha,\gamma}(\mathbf{a},\mathbf{x})} \tag{3.18}$$

Proof: By construction. Applying Eq.(3.17) to Eq.(3.13) $D^\gamma u^{R_a}(\mathbf{x}) = D^\alpha u(\mathbf{x}) \times \delta_{\alpha\gamma} = D^\gamma u(\mathbf{x})$. This means that the derivatives $D^\gamma u(\mathbf{x})$ can be reproduced up to the nth order.

Remark

Lemma 1b implies the determination of $D^\gamma \beta(\mathbf{x})$ such that $\bar{m}_{\alpha,\gamma}(\mathbf{a},\mathbf{x}) = (-1)^\alpha \alpha! \delta_{\alpha,\gamma}$.

3.5 The Construction of the Correction Function

Corollary 1a. Determination of $\beta(\mathbf{x})'s$

$\bar{m}_\alpha(\mathbf{a},\mathbf{x}) = \delta_{\alpha 0}$ implies:

$$\boxed{\mathbf{M}\beta = \mathbf{P}(\mathbf{0})} \quad \text{or} \quad \boxed{\beta = \mathbf{M}^{-1}\mathbf{P}(\mathbf{0})} \tag{3.19}$$

where the components of the moment matrix $\mathbf{M}$ are:

$$M_i = \int_\Omega (\mathbf{x}-\mathbf{y})^\alpha \phi_a(\mathbf{x};\mathbf{x}-\mathbf{y})(\mathbf{x}-\mathbf{y})^\beta d\mathbf{y} \quad 0 \le \alpha + \beta = i \le n \tag{3.20}$$

and $\mathbf{P}$ is the basis function defined as $\mathbf{P}(\mathbf{x}-\mathbf{y}) = [\mathbf{1}, (\mathbf{x}-\mathbf{y}), \ldots, (\mathbf{x}-\mathbf{y})^n]^T$ and $\mathbf{P}(\mathbf{0}) = [\mathbf{1}, \mathbf{0}, \ldots, \mathbf{0}]^T$.

Proof: Substituting the definition of the correction function into the moment Eq. (3.11), Eq.(3.19) can be obtained.

Remark:

Eq.(3.19) can be written in matrix form as:

$$\begin{bmatrix} M_0(x) & M_1(x) & \ldots & M_n(x) \\ M_1(x) & M_2(x) & \ldots & M_{n+1}(x) \\ \vdots & & & \vdots \\ M_n(x) & M_{n+1}(x) & \ldots & M_{2n}(x) \end{bmatrix} \begin{bmatrix} \beta_0(x) \\ \beta_1(x) \\ \vdots \\ \beta_n(x) \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \tag{3.21}$$

The construction of the correction function, as given in Eq.(3.4), is now complete.

Corollary 1b. Determination of the derivative of $\beta(\mathbf{x})'s$ *(i.e.* $\mathbf{D}^\gamma\beta(\mathbf{x})$*)*

$\bar{m}_{\alpha,\gamma}(\mathbf{a},\mathbf{x}) = (-1)^\alpha \alpha! \delta_{\alpha\gamma}$ implies

$$\boxed{D^\gamma[\mathbf{M}(\mathbf{a},\mathbf{x})\beta(\mathbf{a},\mathbf{x})] = D^\gamma[\mathbf{P}(\mathbf{0})]} \tag{3.22}$$

or in matrix form:

$$\begin{bmatrix} \mathbf{M} & \mathbf{0} & \mathbf{0} & \dots & \mathbf{0} \\ D\mathbf{M} & \mathbf{M} & & & \mathbf{0} \\ D^2\mathbf{M} & 2D\mathbf{M} & \mathbf{M} & & \mathbf{0} \\ \vdots & & & \ddots & \vdots \\ C_\gamma^0 D^\gamma\mathbf{M} & C_\gamma^1 D^{\gamma-1}\mathbf{M} & \dots & \dots & C_\gamma^\gamma\mathbf{M} \end{bmatrix} \begin{bmatrix} \beta \\ D\beta \\ D^2\beta \\ \vdots \\ D^\gamma\beta \end{bmatrix} = \begin{bmatrix} \mathbf{P}(\mathbf{0}) \\ \mathbf{0} \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix} \tag{3.23}$$

where $C_\gamma^m = \frac{\gamma!}{(\gamma-m)!m!} \quad (m = 0, 1, 2, \dots, \gamma)$

Proof: Eqs.(3.22) and (3.23) can be proved by induction. However, the detailed algebra is not presented here.

Remark:

The significance of ***Corollary 1b.*** emerges in problems where the derivatives of the reproduced functions are needed.

3.6 Interpolant Estimates

The second terms in the right side of Eq.(3.9) and Eq.(3.13) are the truncation errors for the reproduced function and its derivatives, that is:

$$u^{R_a}(\mathbf{x}) - u(\mathbf{x}) = \sum_{|\alpha|=n+1} \partial^\alpha u(\bar{\mathbf{x}})\bar{m}_\alpha(\mathbf{a},\mathbf{x}) \tag{3.24}$$

and

$$D^\gamma u^{R_a}(\mathbf{x}) - D^\gamma u(\mathbf{x}) = \sum_{|\alpha|=n+1} \partial^\alpha u(\bar{\mathbf{x}})\bar{m}_{\alpha,\gamma}(\mathbf{a},\mathbf{x}) \tag{3.25}$$

The interpolation estimates in Hilbert Space can be defined via the following theorem.

Theorem 3.1

Assume $u(\mathbf{x})$ *and* $\phi(\mathbf{x}) \in C^{n+1}(\bar{\Omega}) \cap W_2^{n+1}(\Omega)$ *are in* $\mathbf{R}^{ND}$*. The interpolation kernel function* $\mathcal{K}_a(\mathbf{x};\mathbf{x}-\mathbf{y})$ *is generated by an* m*-order,* ℓ*-component polynomial base vector* $\{(\mathbf{x}-\mathbf{y})^\alpha, 0 \le |\alpha| \le l\}$*. The following interpolation estimates hold*:

$$\boxed{\|u^{R_a}(\mathbf{x}) - u(\mathbf{x})\|_{L^2} \le C_1 a^{n+1} \|\phi'(\mathbf{x})\|_{L^2} |u(\mathbf{x})|_{W_\infty^{n+1}} \|C(a)\|_{L^2}} \tag{3.26}$$

$$\boxed{\|u^{R_a}(\mathbf{x}) - u(\mathbf{x})\|_{H^k} \le C_2 a^{n+1-k} \|\phi'(\mathbf{x})\|_{H^k} |u(\mathbf{x})|_{W_\infty^{n+1}} \|C(a)\|_{H^k}} \tag{3.27}$$

where C_1, C_2 are constants, which are independent of the dilation parameter $\mathbf{a}$. ϕ' is defined as $\phi' = E(a)\phi$, in which $E(a)$ is a normalization factor and $\int_\Omega \phi' d\Omega = 1$. For a complete set of polynomial base vectors, the following relationship holds:

$$2-D: \ \ell = \frac{1}{2}m(m+1) \qquad 3-D: \ \ell = \sum_{k=1}^{m} \frac{1}{2}k(k+1)$$

In numerical computation, moments and their derivatives have to be computed by numerical integration, which are denoted by the following expressions, respectively.

$$\bar{m}_\alpha(\mathbf{a},\mathbf{x}) \leftarrow \tilde{m}_\alpha(\mathbf{a},\mathbf{x}) \equiv \sum_{J=1}^{NP} (\mathbf{x}-\mathbf{x}_J)^\alpha \kappa_a(\mathbf{x};\mathbf{x}-\mathbf{y})\Delta\mathbf{V}_J \tag{3.28}$$

$$\bar{m}_{\alpha,\gamma}(\mathbf{a},\mathbf{x}) \leftarrow \tilde{m}_{\alpha,\gamma}(\mathbf{a},\mathbf{x}) \equiv \sum_{J=1}^{NP} (\mathbf{x}-\mathbf{x}_J)^\alpha D^\gamma \kappa_a(\mathbf{x};\mathbf{x}-\mathbf{y})\Delta\mathbf{V}_J \tag{3.29}$$

The dilation parameter $\mathbf{a}$ can be defined as the product of the nodal spacing h and the refinement parameter r, dictating the number of nodes covered under the window function.

Correspondingly, the interpolation estimates in terms of h for RKPM are given as follows:

$$\boxed{\|u^{R_a^h}(\mathbf{x}) - u(\mathbf{x})\|_{L^2} \leq C_1(rh)^{n+1}\|\phi'(\mathbf{x})\|_{L^2}|u(\mathbf{x})\|_{W_\infty^{n+1}}\|C(rh)\|_{L^2}} \tag{3.30}$$

$$\boxed{\|u^{R_a}(\mathbf{x}) - u(\mathbf{x})\|_{H^k} \leq C_2(rh)^{n+1-k}\|\phi'(\mathbf{x})\|_{H^k}|u(\mathbf{x})\|_{W_\infty^{n+1}}\|C(rh)\|_{H^k}} \tag{3.31}$$

Proof: It is a straight-forward exercise to bound Eqs. (3.24) and (3.25) using the moment definition to give Eqs. (3.26) and (3.27).

Remarks:

a) It can be shown that $\|C(a)\| \sim 0(1)$ for ***exact*** integration of moments.
b) $\|C(a)\|_{L^2} = \|C(r,h)\|_{L^2}$.

Figure 3.1 indicates: (i) $\|C(r,h)\|_{L^2}$ is unbounded for $\mathbf{r} < \mathbf{0.5}$, which is also known as the *kernel instability*; [Liu *et al.* (1995b)] (ii) $\|C(r,h)\|_{L^2} > 0(1)$ for $\mathbf{0.5} \leq \mathbf{r} \leq \mathbf{1} - \varepsilon$; (iii) $\|C(r,h)\|_{L^2} \sim 0(1)$ for $\mathbf{r} > \mathbf{1}$, which is independent of r and h. In addition, Liu and Chen (1995) pointed out that optimal $\mathbf{r} \approx \mathbf{1.17}$ which was determined via an energy error ratio.

3.7 Interpolation Function

In a numerical computation, the reproducing kernel Eq.(3.6) is discretized by trapezoidal rules (particle integration) with NP particles (nodes) to give:

$$u^{R_h}(x) = \sum_{j=1}^{NP} u(x_j)C_a(x;x-x_j)\phi_a(x-x_j)\Delta x_j \tag{3.32a}$$

where

$$C_a(x;x-x_j) = \sum_{k=0}^{n} \beta_k(x)(x-x_j)^k \tag{3.32b}$$

$$\bar{\phi}_j(x) = C_a(x;x-x_j)\phi_j(x) \tag{3.32c}$$

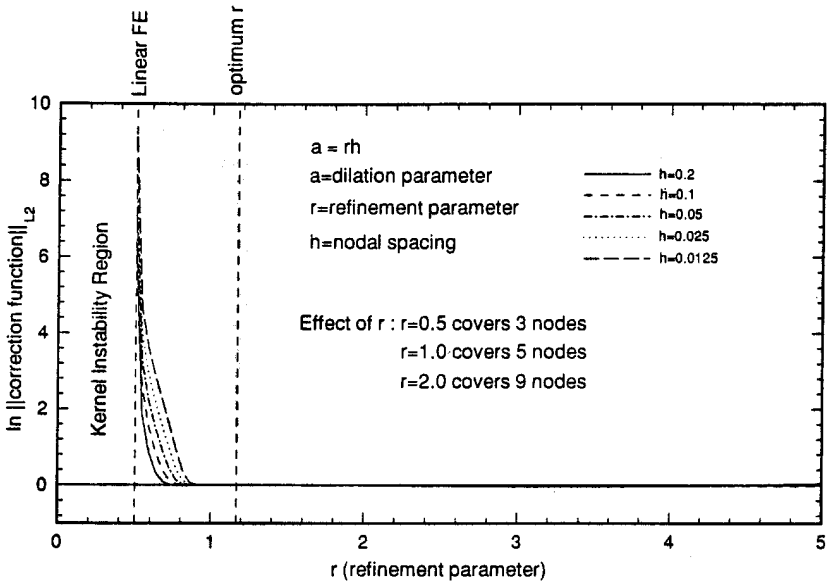


Figure 3.1. Exponential behavior of $ln\|C\|_{L^2}$ with respect to refinement parameter

As expected, the generalized reproducing conditions are given as:

$$\tilde{m}_0(x) = \sum_{j=1}^{NP} 1 \cdot \bar{\phi}_j(x)\Delta x_j = 1 \tag{3.33a}$$

$$\tilde{m}_k(x) = \sum_{j=1}^{NP} (x - x_j)^k \bar{\phi}_j(x)\Delta x_j = 0 \tag{3.33b}$$

3.8 Relationship between RKPM and Moving Least Square Kernel Method

It can be shown that, even though the derivation of the RKPM is completely different from the Moving Least Square Kernel Method given by Liu *et al.* (1995c), the end results seem to give the same interpolation error estimates as in Eq.(3.30) and Eq.(3.31). At the present time, we are only able to prove that this formulation is equivalent to the standard Moving Least Square approximation as well as the Element Free Galerkin method given by Belytschko *et al.* (1994a, 1994b) if:

1) all $\Delta V_j = 1$; and 2) linear basis only.

Readers are urged to consult Liu *et al.* (1995b, 1995c).

4. REPRODUCING KERNEL PARTICLE METHODS FOR ACOUSTIC PROBLEMS

Several computational formulations have been proposed to solve vibration and wave propagation problems. Due to the approximations introduced by these methods, the computed waves exhibit distortions of the physical wave motion. In this section, the use of RKPM in structural acoustics is introduced. Wave propagation in bars and the Timoshenko beam is formulated and some examples are presented. Multiresolution anlysis and hp-like adaptivity are discussed in Section 9.

4.1 Wave Equation

In one dimensional space, the wave equation is given as:

$$\frac{1}{c^2}\ddot{u} = u_{,xx} \quad x \in [0, L] \tag{4.1}$$

where c is the wave speed, a superposed dot indicates the temporal derivative, and a subscript comma denotes the spatial derivative. In computational analysis, appropriate approximation functions are employed to represent the solution:

$$u^h(x,t) = \sum_{I=1}^{NP} N_I(x) u_I(t) \tag{4.2}$$

where a superscript h indicates the approximate solution, NP refers to the total number of grid points (as in the FEM) or particles (as in the RKPM), N_I are the approximation functions and u_I denotes the approximate solution at each grid point or particle.

The wave characteristics of Eq.(4.1) are studied by considering a general harmonic wave solution, given as

$$u = u_0 e^{i(\omega t - kx)} \qquad i = \sqrt{-1} \tag{4.3}$$

where u_0 is the amplitude, ω is the circular frequency, and k is the wave number. By using Eq.(4.3), the frequency equation for the continuum wave equation is obtained

$$\left(\frac{\omega^2}{c^2}\right) - k^2 = 0 \tag{4.4}$$

This equation indicates that there is no dispersion. For $\omega = 0$, which is representative of rigid body motion, the corresponding wave number is zero. The discrete form of Eq.(4.3) for the I-th grid point or particle becomes:

$$u_I(t) = u_0 e^{i(\omega t - k x_I)} \tag{4.5}$$

The characteristic equation is obtained by focusing on the Ith grid point/particle

$$\begin{aligned} &-\left(\frac{\omega}{c}\right)^2 \left\{ (1-\alpha) M_{II}^L + \alpha \left[\sum_{m=1}^{n_L} M_{I,I-m} e^{ik\Sigma_{j=1}^{m} \Delta x_{I-j}} + M_{II} + \sum_{m=1}^{n_R} M_{I,I+m} e^{-ik\Sigma_{j=0}^{m-1} \Delta x_{I+j}} \right] \right\} + \\ &+ \sum_{m=1}^{n_L} K_{I,I-m} e^{ik\Sigma_{j=1}^{m} \Delta x_{I-j}} + K_{II} + \sum_{m=1}^{n_R} K_{I,I+m} e^{-ik\Sigma_{j=0}^{m-1} \Delta x_{I+j}} = 0 \end{aligned} \tag{4.6}$$

where Δx_I is the spacing between the Ith and $(I+1)$th mesh point/particle locations; n_R and n_L are the number of points to the right and to the left of the Ith location which fall under the window function and $\alpha \in [0, 1]$ which determines the amount of lumped mass M_{II}^L versus the consistent mass M_{II}.

In a standard finite element analysis with linear Lagrange interpolants, the Ith stiffness term is coupled to the $(I-1)$th and $(I+1)$th terms only such that $n_L = n_R = 1$. By employing linear approximation functions, a lumped mass, and constant spacing h the wave characteristic equation, Eq.(4.6) can be rewritten as:

$$-\frac{\omega^2}{c^2} + \frac{4}{h^2} \sin^2\left(\frac{1}{2}kh\right) = 0 \tag{4.7}$$

In the reproducing kernel particle method, the shape functions are defined in terms of window functions:

$$N_I(x) = \left[c_1(x) + c_2(x)\frac{x - x_I}{a}\right]\frac{1}{a}\phi\left(\frac{x - x_I}{a}\right)\Delta x_I \tag{4.8}$$

where $\phi(x)$ is a window function, **a** is a dilation parameter which determines the width of the window and the function c_1 and c_2 are the boundary correction functions.

For uniform spacing between the particles, the general matrix form of the discretized wave equation, Eq. (4.5), becomes

$$\begin{aligned} &-\left(\frac{\omega}{c}\right)^2\left\{(1-\alpha)M_{II}^L + \alpha\left[\sum_{m=1}^{n} M_{I,I-m}e^{ikmh} + M_{II} + \sum_{m=1}^{n} M_{I,I+m}e^{ikmh}\right]\right\} + \\ &+\sum_{m=1}^{n_L} K_{I,I-m}e^{ikmh} + K_{II} + \sum_{m=1}^{n_R} K_{I,I+m}e^{-ikmh} = 0 \end{aligned} \tag{4.9}$$

The spurious mechanisms associated with the existing discretization methods can be studied by comparing their characteristic equation Eq. (4.7) with Eq. (4.4).

The comparison of the RKPM with varying **a** and the analytic dispersion (frequency vs wave number) relation is depicted in Figure 4.1. An increase of the dilation parameter reduces the error between the analytical and computational (RKPM) results.

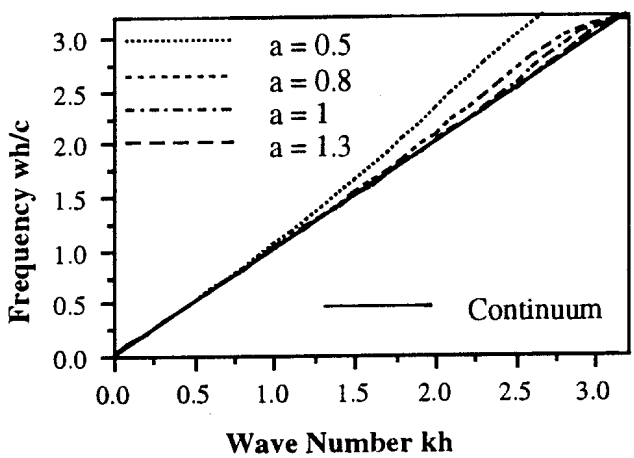


Figure 4.1. Dispersion relationship for wave equation with varying **a** (consistent mass)

4.2 Multiple Scale RKPM Timoshenko Beam

The coupled equations for the Timoshenko beam can be expressed in a matrix operator form as:

$$\begin{bmatrix} \rho A\frac{\partial^2}{\partial t^2} - \kappa GA\frac{\partial^2}{\partial x^2} & \kappa GA\frac{\partial}{\partial x} \\ \kappa GA\frac{\partial}{\partial x} & -\rho I\frac{\partial^2}{\partial t^2} + EI\frac{\partial^2}{\partial x^2} - \kappa GA \end{bmatrix}\begin{bmatrix} w \\ \theta \end{bmatrix} = 0 \tag{4.10}$$

where E and G are the Young's modulus and the shear modulus, A and ρ are the cross-sectional area and density, I is the moment of inertia and κ is the shear correction factor. The characteristic equation which describes the relationship between frequency and wave

number is obtained by substituting the harmonic wave equation solution, Eq.(4.3) into Eq.(4.10):

$$a\eta^4 + b\eta^2 + c = 0 \tag{4.11}$$

where

$$a = \frac{EI}{\rho A} \tag{4.12a}$$

$$b = -\frac{I}{A}\left(1 + \frac{E}{\kappa G}\right)\omega^2 \tag{4.12b}$$

$$c = \frac{\rho I}{GA\kappa}\omega^4 - \omega^2 \tag{4.12c}$$

The eigenvalues of the continuous Timoshenko beam is the solution of the 4th order equation, Eq. (4.11):

$$\eta = \pm\left\{\left[\frac{\rho}{G\kappa} + \frac{\rho}{E}\right]\omega^2 \pm \sqrt{\frac{\rho A}{EI}\omega^2 + \frac{1}{4}\left[\frac{\rho}{G\kappa} - \frac{\rho}{E}\right]^2\omega^4}\right\}^{\frac{1}{2}} \tag{4.13}$$

Following a similar procedure as indicated in Section 4.1, the discrete wave equation, Eq. (4.5) is substituted into eq. (4.10) to obtain the semi-discrete RKPM equations (note that a uniform particle spacing is assumed, and the equations are written for an interior node j):

$$\begin{aligned}\rho A m_j^w \ddot{w}_j = &-\kappa GA(k^w_{j,j-n_w} w_{j-n_w} + \ldots + k^w_{j,j-1} w_{j-1} + k^w_{j,j} w_j + k^w_{j,j+1} w_{j+1} + \ldots + k^w_{j,j+n_w} w_{j+n_w}) \\ &+ \kappa GA(k^{w\theta}_{j,j-n_{w\theta}} \theta_{j-n_{w\theta}} + \ldots + k^{w\theta}_{j,j-1} \theta_{j-1} + k^{w\theta}_{j,j+1} \theta_{j+1} + \ldots + k^{w\theta}_{j,j+n_{w\theta}} \theta_{j+n_{w\theta}})\end{aligned} \tag{4.14}$$

$$\begin{aligned}\rho A m_j^\theta \ddot{\theta}_j = &-EI(k^\theta_{j,j-n_\theta} \theta_{j-n_\theta} + \ldots + k^\theta_{j,j-1} \theta_{j-1} + k^\theta_{j,j} \theta_j + k^\theta_{j,j+1} \theta_{j+1} + \ldots + k^\theta_{j,j+n_\theta} \theta_{j+n_\theta}) \\ &+ \kappa GA(k^{\theta w}_{j,j-n_{w\theta}} w_{j-n_{w\theta}} + \ldots + k^{\theta w}_{j,j-1} w_{j-1} + k^{\theta w}_{j,j+1} w_{j+1} + \ldots + k^{\theta w}_{j,j+n_{w\theta}} w_{j+n_{w\theta}}) \\ &- \kappa GA(k^{\theta\theta}_{j,j-n_\theta} \theta_{j-n_\theta} + \ldots + k^{\theta\theta}_{j,j-1} \theta_{j-1} + k^{\theta\theta}_{j,j} \theta_j + k^{\theta\theta}_{j,j+1} \theta_{j+1} + \ldots + k^{\theta\theta}_{j,j+n_\theta} \theta_{j+n_\theta})\end{aligned} \tag{4.15}$$

where the superscripts w and θ denote the terms related to the transverse displacement and rotation, respectively, m_j^w and m_j^θ are evaluated as lumped masses and $k^w_{i,j}$, $k^\theta_{i,j}$, $k^{w\theta}_{i,j}$, $k^{\theta w}_{i,j}$ and $k^{\theta\theta}_{i,j}$ are the non-zero coefficients of the stiffness matrices $\mathbf{K}^w_{IJ}$, $\mathbf{K}^\theta_{IJ}$, $\mathbf{K}^{w\theta}_{IJ}$, $\mathbf{K}^{\theta w}_{IJ}$ and $\mathbf{K}^{\theta\theta}_{IJ}$, respectively. It is noted that n_w, n_θ, $n_{w\theta}$ and $n_{\theta w}$, denote the number of nodes covered under the support of the window functions. The dilation parameters associated with rotations and transverse displacements are indicated by a_θ and a_w, respectively.

The discrete Fourier matrix operators corresponding to the RKPM formulation can be shown to be:

$$\begin{bmatrix} \rho A m_j^w \omega^2 - \kappa GA\alpha^w & i\ \kappa GA\alpha^{w\theta} \\ i\ \kappa GA\alpha^{\theta w} & \rho I m_j^\theta \omega^2 - EI\alpha^\theta - \kappa GA\alpha^{\theta\theta} \end{bmatrix} \begin{bmatrix} w_j \\ \theta_j \end{bmatrix} = 0 \tag{4.16}$$

where

$$m_j^w = \int_0^L N_I^w dx \tag{4.17a}$$

$$m_j^\theta = \int_0^L N_I^\theta dx \tag{4.17b}$$

$$\alpha^\theta = \sum_{m=-n_\theta}^{n_\theta} k_{j,j+m}^\theta \cos m\eta h \tag{4.17c}$$

$$\alpha^w = \sum_{m=-n_w}^{n_w} k_{j,j+m}^w \cos m\eta h \tag{4.17d}$$

$$\alpha^{w\theta} = \sum_{m=-n_{w\theta}}^{n_{w\theta}} k_{j,j+m}^{w\theta} \sin m\eta h \tag{4.17e}$$

$$\alpha^{\theta w} = \sum_{m=-n_{\theta w}}^{n_{\theta w}} k_{j,j+m}^{\theta w} \sin m\eta h \tag{4.17f}$$

$$\alpha^{\theta\theta} = \sum_{m=-n_{\theta\theta}}^{n_{\theta\theta}} k_{j,j+m}^{\theta\theta} \cos m\eta h \tag{4.17g}$$

The frequency equation is obtained by setting the determinant of Eq. (4.16) equal to zero:

$$a_r\omega^4 + b_r\omega^2 + c_r = 0 \tag{4.18}$$

where

$$a_r = \rho^2 AIm_j^w m_j^\theta \tag{4.19a}$$

$$b_r = -[\rho Am_j^w(EI\alpha^\theta + \kappa GA\alpha^{\theta\theta}) + \kappa GA\alpha^w \rho Im_j^\theta] \tag{4.19b}$$

$$c_r = [\kappa GA\alpha^w(EI\alpha^\theta + \kappa GA\alpha^{\theta\theta})(\kappa GA)^2\alpha^{w\theta}\alpha^{\theta w}] \tag{4.19c}$$

In order to facilitate the use of consistent, lumped or mixed mass matrix formulation m_j^w and m_j^θ are replaced by $\tilde{m}^w$ and $\tilde{m}^\theta$.

$$\tilde{m}^w = [(1-r)m_j^w + r\tilde{\alpha}^{ww}] \tag{4.20a}$$

$$\tilde{m}^\theta = [(1-r)m_j^\theta + r\tilde{\alpha}^{\theta\theta}] \tag{4.20b}$$

where

$$\tilde{\alpha}^{ww} = \sum_{m=-n_{ww}}^{n_{ww}} m_{j,j+m}^{ww} \cos m\eta h \tag{4.21a}$$

$$\tilde{\alpha}^{\theta\theta} = \sum_{m=-n_{\theta\theta}}^{n_{\theta\theta}} m_{j,j+m}^{\theta\theta} \cos m\eta h \tag{4.21b}$$

The degeneration of $\tilde{m}^w$ and $\tilde{m}^\theta$ to a lumped mass formulation occurs when $r = 0$, and to a consistent mass formulation when $r = 1$.

Note that Eq. (4.18) is not only a function of the wave number, η, and frequency ω it is also a function of the *two dilation parameters for the transverse displacement and rotation*, a_w and a_θ, respectively. The optimal values for the dilation parameters can be selected such that the discrete frequency relation Eq. (4.18) is the same as that of the continuum frequency equation. In other words, by performing a dispersion analysis of the RKPM for the Timoshenko beam, a set of dilation parameters for the transverse displacements and

rotations can be determined. This optimal set dictates that the interpolation functions for the transverse displacements and rotations are constructed with a correct match between the bending and transverse shear frequencies (wave numbers).

By substituting the exact wave number into both the discrete frequency relation, Eq.(4.18), and the continuum frequency equation, the optimal values of the dilation parameters, a_w and a_θ, can be found so that the frequency error is minimized. The comparison of the RKPM and the analytical dispersion relation for the Timoshenko beam is depicted in Figure 4.2.

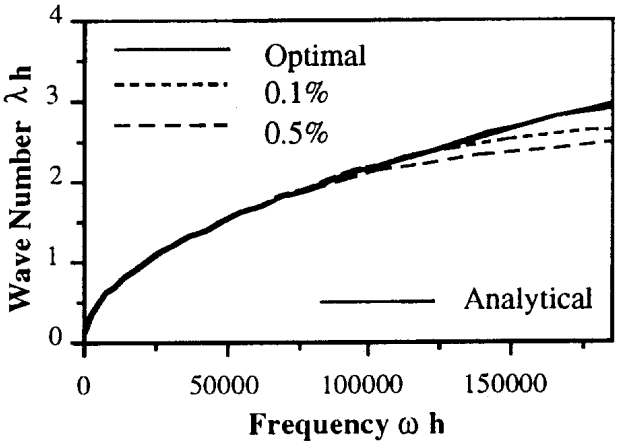


Figure 4.2. Dispersion relation with varying a_θ

4.3 Numerical Results

4.3.1 Radiation for an arch

Consider the Helmholtz equation in a polar coordinate system with the following problem statement:

$$\nabla^2\phi + \kappa^2\phi = 0 \qquad \text{in} \quad \Omega \tag{4.22}$$

where

$$\nabla^2 = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) \tag{4.23}$$

The boundary conditions are:

$$\phi(1) = \begin{cases} 1 & -\pi/6 \le \theta \le \pi/6 \\ 0 & \text{otherwise} \end{cases} \tag{4.24}$$

and the Sommerfeld radiation condition at infinity is:

$$\lim_{r\to\infty} \sqrt{r}(\phi_{,r} - i\kappa\phi) = 0 \tag{4.25}$$

In this example, the Sommerfeld condition is replaced by a 1st order DtN mapping on $r = 2$, which is given as:

$$B_1\phi|_{r=2} = 0 \tag{4.26}$$

where

$$B_1 = \frac{\partial}{\partial r} - ir + \frac{1}{2r} \tag{4.27}$$

The meshless hp-adaptivity is given in Figure 4.3. As it can be seen in the figure, only a set of nodes is required for adaptivity, and there is no apprehension of a compatibility problem.

4.3.2 Wave propagation in bars

The governing equation for the one dimensional wave propagation, in a bar of length L, in the frequency domain is given as:

$$\frac{d^2\phi}{dx^2} + k^2\phi = F(x) \tag{4.28}$$

where k is the wave number, and Dirichlet boundary conditions are assumed at both ends:

$$\phi(0) = \phi_0 \tag{4.29}$$

and

$$\phi(L) = \phi_L \tag{4.30}$$

Two numerical examples are performed to compare the results of the RKPM with the available analytical solutions:

Case (i): A homogeneous differential equation with nonhomogeneous boundary conditions, i.e.

$$F(x) = 0, \quad \phi(0) = 1 \quad \text{and} \quad \phi(L) = 0 \tag{4.31}$$

It can be shown that the analytical solution for Case (i) is:

$$\phi(x) = \frac{\sin(k(L-x))}{\sin(kL)} \tag{4.32}$$

Case (ii): A nonhomogeneous differential equation with homogenous boundary conditions, i.e.

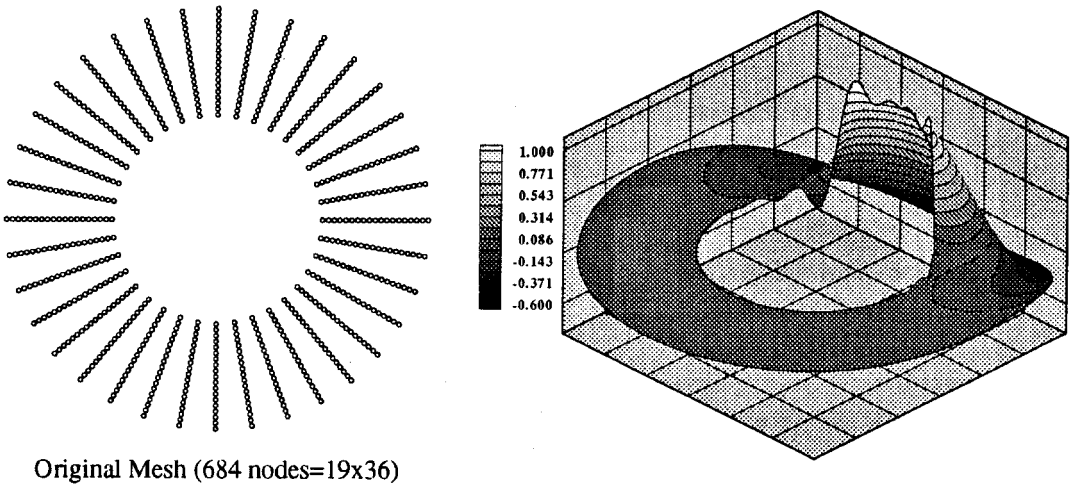
$$F(x) = 1, \quad \phi(0) = 0 \quad \text{and} \quad \phi(L) = 0 \tag{4.33}$$

The analytical solution for this case is given as:

$$\phi(x) = \frac{1}{k^2}\left[1 - \cos(kx) + \frac{\cos(kL) - 1}{\sin(kL)}\sin(kx)\right] \tag{4.34}$$

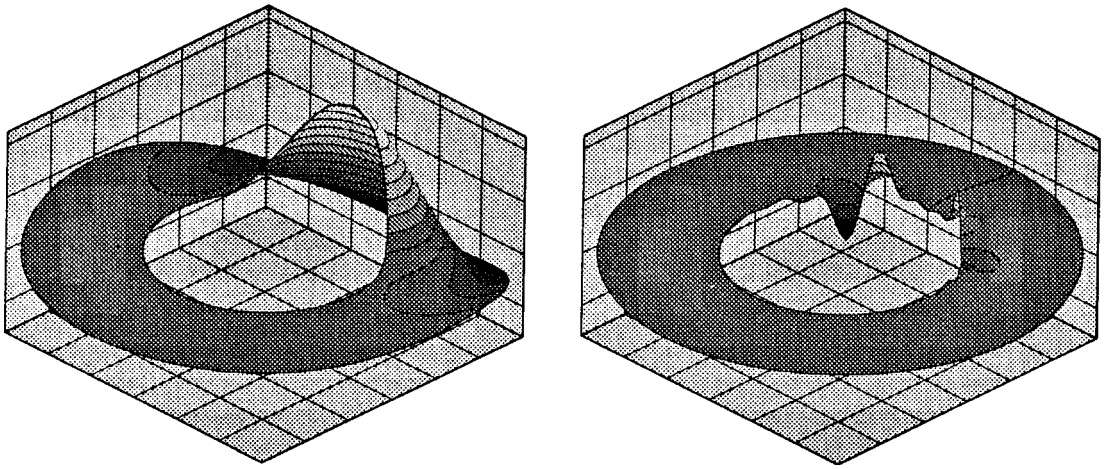
In numerical examples, the bar length is taken as 10 units. The RKPM is used with 11, 21 and 41 particles to obtain a measure of the convergence rate. In order to show the significance of the dilation parameter **a**, several cases are considered by keeping the number of particles constant and varying **a**. The comparison of the mid-point displacement between the analytical and the RKPM solutions is depicted in Figure 4.4. For both numerical examples presented here, the RKPM solution yields excellent agreement for a wide range of wave numbers showing some deviation near the resolution limit $k\ h = \pi$. The mid-point displacements presented in Figure 4.4 are normalized according to

$$U = \frac{u^{RKPM}}{u^{Analytical}} \tag{4.35}$$

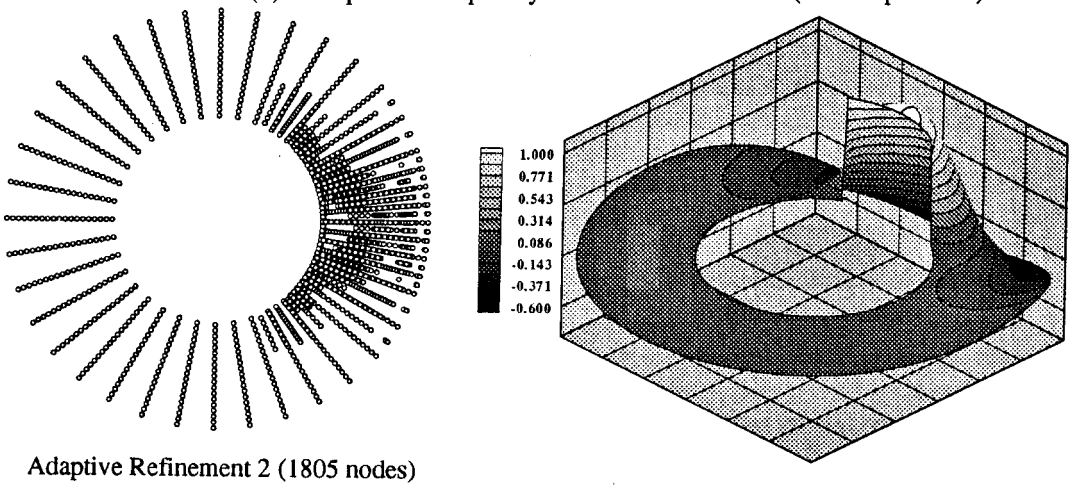


Original Mesh (684 nodes=19x36)

(a) An hp-like Adaptivity in Acoustic Problem (Mesh and Solution)



(b) An hp-like Adaptivity in Acoustic Problem (Decomposition)



Adaptive Refinement 2 (1805 nodes)

(c) An hp-like Adaptivity in Acoustic Problem (Adaptivity)

Figure 4.3.

As the dilation parameter is increased the RKPM solution accurately represents a larger portion of the wave number spectrum. However, the difference between the analytical and the RKPM solutions starts deteriorating beyond the value of 1.17 for the dilation parameter. This implies the existence of an optimum value of the dilation parameter.

The convergence of the RKPM solution to the analytical solution is studied by increasing the number of particles. Figure 4.5 depicts the relationship between the error in L^2 norm and the particle distance (equivalent to mesh parameter in finite element method). As seen in Table 4.1, the convergence rate is higher for increasing wave numbers which also implies high frequencies. Even with a relatively small number of particles, a convergence rate of 4 is obtained, indicating the effectiveness of the RKPM.

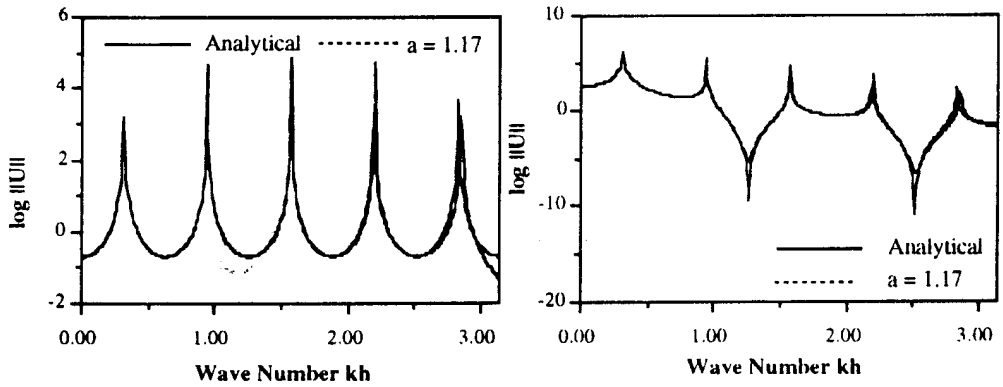


Figure 4.4. Wave propagation in bar, case (i) and case (ii)

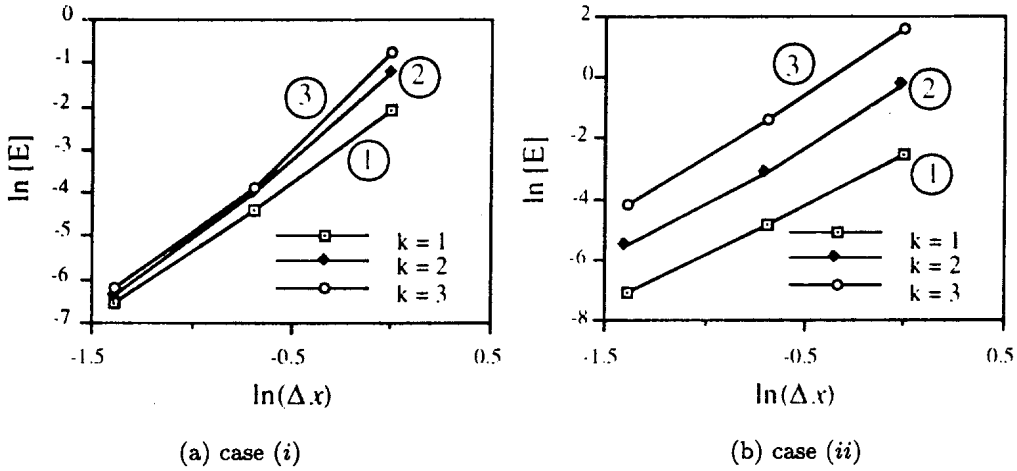


Figure 4.5. Convergence Rate for Different Wave Numbers k (E is the L^2 norm error, Δx is the particle distance)

Curve Number	Wave Number	Case (i)	Case (ii)
1	1	3.08	3.31
2	2	3.42	3.82
3	3	3.37	4.18

Table 4.1. Convergence rates for the example cases (i) and (ii) given in Figures 4.5

4.3.3 Time harmonic analysis of a Timoshenko beam

The governing equation of a Timoshenko beam has been given in Eq.(4.10). Due to its relative simplicity, the analytical solution for the dynamic response is determined from the following equation [Dym and Shames (1973)]:

$$EI\frac{\partial^4 w}{\partial x^4} + \rho A\frac{\partial^2 w}{\partial t^2} - \rho I\left(1+\frac{E}{\kappa G}\right)\frac{\partial^4 w}{\partial x^2 \partial t^2}\frac{\rho^2 I \partial^4 w}{\kappa G \partial t^4} = q + \frac{\rho I \partial^2 q}{\kappa A G \partial t^2} - \frac{EI\partial^2 q}{\kappa A G \partial x^2} \tag{4.36}$$

where q is the distributed force acting on the beam and ω is the oscillation frequency.

The analytical and computational results are compared for the following cases:

(i) Steady state solution for a beam simply supported on both ends
(ii) Steady state solution for a beam clamped on both ends
(iii) Frequency solution for a beam simply supported on both ends
(iv) Frequency solution for a beam clamped on both ends

The same geometrical and material characteristics (see Table 4.2) are employed in all these cases.

E	200 MPa
G	80 MPa
ρ	8000 kg/m^3
L	1 m
thickness	0.001 m
thickness	0.0001 m

Table 4.2. Geometrical and material characteristics of the Timoshenko beam

The analytical solution for each case is given as follows:

Case (i):

$$w(x) = \frac{1}{2\kappa GA}\left[\left(x-\frac{L}{2}\right)^2 - \frac{L^2}{4}\right] - \frac{1}{EI}\left[\frac{1}{24}\left(x-\frac{L}{2}\right)^4 - \frac{L^2}{16}\left(x-\frac{L}{2}\right)^2 + \frac{5L^4}{384}\right] \tag{4.37}$$

Case (ii):

$$w(x) = \frac{1}{2\kappa GA}x(x-L) - \frac{1}{24EI}x^2(x-L)^2 \tag{4.38}$$

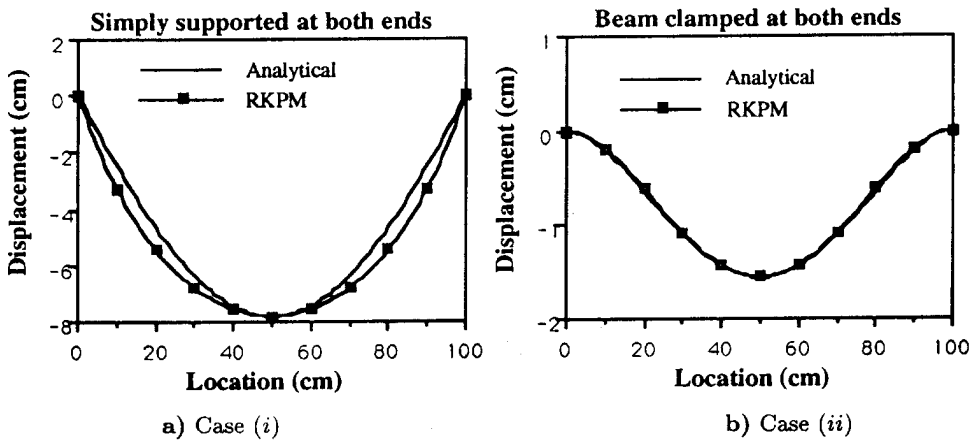
Case (iii) and Case(iv): (see Table 4.3).

For all the numerical results, a uniform load is applied to the entire beam. For both Case (i) and Case (ii), the thickness is selected as 0.01 cm and 11 particles are used in the RKPM. The comparison of results are presented in Figure 4.6.

In both Case (iii) and Case (iv), a thickness of 0.1cm and a frequency of 10000 rad/sec is used in the RKPM with 51 particles such that the analytical wave number corresponds to five particles per wavelength. The results on the comparison between the RKPM and the analytical solution for a beam with both simply supported and clamped end, are shown in Figure 4.7a and 4.7b, respectively. In both cases, the phase is well represented over the entire beam and the amplitude errors at the wave tips are less than 5%.

	Simply Supported (Case (*ii*))	*Clamped Ends* (Case (*iv*))
$w(x)$	$A_p + A\cos\lambda x + B\sin\lambda x + C\cosh\lambda x + D\sinh\lambda x$	
A_p	$\dfrac{q_0 \kappa G}{-\omega^2 \rho A \kappa G + \omega^4 \rho I}$	
λ	$\left[\dfrac{-b+\sqrt{b^2-4ac}}{2a}\right]^{1/2}$	
a	EI	
b	$\omega^2 \rho I(1+\dfrac{E}{\kappa G})$	
c	$-\omega^2 \rho A(1-\dfrac{\omega^2 \rho I}{\kappa G A})$	
$BC's$	$w(0)=w(L)=w''(0)=w''(L)=0$	$w(0)=w(L)=w'(0)=w'(L)=0$
A	$-\dfrac{A_p}{2}$	$\dfrac{(\cosh\lambda L-1)(\cos\lambda L+1)-\sin\lambda L\sinh\lambda L}{2(1-\cos\lambda L\cosh\lambda L)}A_p$
B	$\dfrac{\cos\lambda L-1}{2\sin\lambda L}A_p$	$\dfrac{\sinh\lambda L(\cos\lambda L-1)+\sin\lambda L(\cosh\lambda L-1)}{2(1-\cos\lambda L\cosh\lambda L)}A_p$
C	$-\dfrac{A_p}{2}$	$-A_p - A$
D	$\dfrac{\cosh\lambda L-1}{2\sinh\lambda L}A_p$	$-B$

Table 4.3. The Analytical Solution for Cases (*iii*) and (*iv*)



a) Case (*i*) b) Case (*ii*)

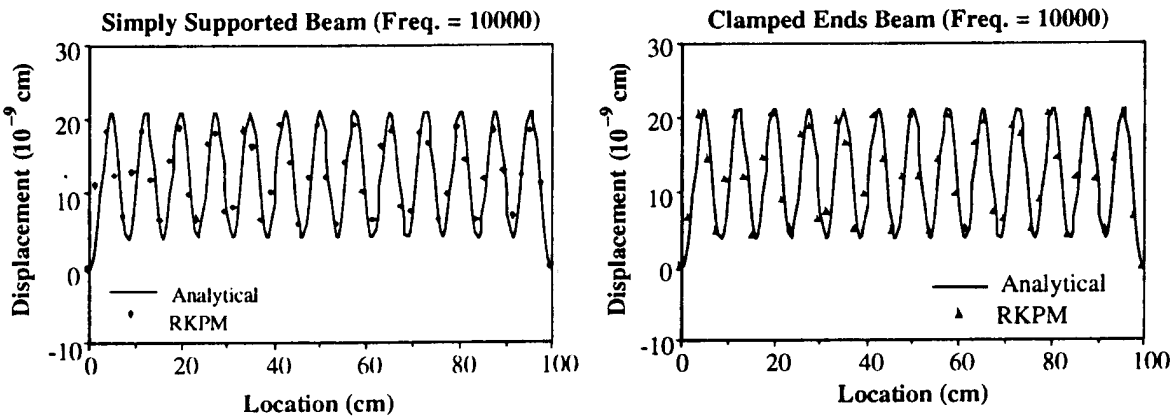
Figure 4.6. Uniformly loaded Timoshenko beam – Steady state solutions

5. RKPM FOR LARGE DEFORMATION ANALYSIS

5.1 Large Deformation and Stability Analysis

An updated Lagrangian formulation for the RKPM has been proposed by Liu *et al.* (1995a). Therefore, we briefly review the highlight of this approach. Readers are urged to consult Liu *et al.* (1995a).

The reproducing kernel particle method is implemented into a Galerkin formulation as a weighted residual method [Liu *et al.* (1995a)]. The major difference between the RKPM and the finite element method construction is the loop which occurs over nodes instead of



a) Case (*iii*) b) Case (*iv*)

Figure 4.7.

elements. The following variables are used: $\mathbf{t}$ is the traction, $\mathbf{b}$ is the body force, $\boldsymbol{\sigma}$ is the stress tensor, and $\ddot{u}$ denotes acceleration.

$$\int_{\Omega} \rho \delta u_i \ddot{u}_i dV + \int_{\Omega} \delta u_{i,j} \sigma_{ij} dV = \int_{\Omega} \delta u_i b_i dV + \int_{\Gamma} \delta u_i t_i dS \tag{5.1}$$

In a large deformation formulation, the internal force matrix for Jth node is given as below:

$$\boldsymbol{f}_J^{\text{int}} = \sum_{k=1}^{NG} \sigma(x_k) N_{J,x}(x_k) \Delta x_k \tag{5.2}$$

where NG is the total number of Gaussian quadrature points. The integration weight Δx_k is replaced by $\Delta x_k = \Delta M_k / \rho_k$ in which ΔM_k and ρ_k are respectively the lumped mass and density associated with an integration point x_k. Prior to updating Δx_k and x_k, the nodal coordinate is updated via

$$x_J = X_J + u(x_J) \qquad J = 1, \ldots, NP \tag{5.3}$$

and the shape functions are used to update the Gaussian quadrature points and the mass densities ρ_k as:

$$x_k = \sum_{J=1}^{NP} N_J(x_k) x_J \qquad k = 1, \ldots, NG \tag{5.4}$$

$$\rho_k = \sum_{J=1}^{NP} N_J(x_k) \rho_J \qquad k = 1, \ldots, NG \tag{5.5}$$

The nodal density ρ_J is computed using the continuity equation.

5.1.1 Reproducing kernel stability

The stability of the kernel function is mainly a function of the number of nodes encompassed by the window function. The number of significant nodes in a shape function is controlled by the refinement parameter $\mathbf{a}$. From the aforementioned definitions for the shape of the window function, the following equation can be derived to estimate the number of nodes

under any given window function. If the radius of a given window function Δx_c is defined as the distance from the center to the edge of the window's significant support, a ratio can be defined to relate the height of the window function at the peak to the small value where it is safely approximated to be zero. For a Gaussian window function,

$$R = \frac{\exp(-\Delta x_c^2/a^2)}{\exp(-\Delta x_0^2/a^2)} \tag{5.6}$$

$$\Delta x_c = \Delta x \left(\frac{n-1}{2}\right) \tag{5.7}$$

Using these definitions it is possible to derive the kernel stability stated in Eq. (5.8) where the variables are as follows:

$n =$ the number of nodes covered by the Gaussian function
$j =$ the parameter controlling the dilation
$R =$ the ratio between the value at a node in the tails of the function to the significance of the peak of the Gaussian function

$$n = 1 + 2^{j+1}\sqrt{\frac{-2lnR}{\pi}} \tag{5.8}$$

Theoretically the number of nodes should be at least two in order to maintain the variable connectivity window arrays. This is verified analytically through solutions that were run with values as low as 0.5 for **a** when the support of the Gaussian function is limited. Solutions for the number of nodes in the support of a window function for various height ratios are given by Liu *et al.*(1995a)

From the derivation of correction function, it is easily shown that the stability condition is governed by the determinant which appears in the correction function:

$$Det = \bar{m}_0\bar{m}_{11} - \bar{m}_1^2 > 0 \tag{5.9}$$

Therefore, when the Trapezoidal rule is used to integrate the moments, instability occurs when

$$Det = \bar{m}_0\bar{m}_{11} - \bar{m}_1^2 \leq 0 \tag{5.10}$$

These provide a very nonlinear stability relationship between **a** and Δx. Our numerical experiments have confirmed these stability conditions.

5.1.2 Critical time step

In order to perform analyses on a structural dynamic class of problems, it is very important to understand the relationship between this new shape function and the critical time step. This is determined here by performing a standard eigenvalue analysis which involves solving the determinant of $[\mathbf{K} - \lambda\mathbf{M}]$. This calculation only needs to be performed for the element case since it is known that the maximum element frequency determines the upper bound for the critical time step. The critical time step was found to depend on several parameters including the dilation parameter and the boundary correction function.

Using symbolic manipulation, the determinant of $[\mathbf{K}-\lambda\mathbf{M}]$ is solved for the RKPM shape functions containing several different numbers of nodes. By evaluating the determinant it is found that there is only one non-zero eigenvalue for each Gauss quadrature point, regardless

of the number of nodes in the support of the shape function. It is possible to simplify the result for equally spaced nodes to the following terms.

$$D(x, x_J) = C_1(x) + C_2(x)\left(\frac{x - x_J}{a\Delta x}\right) \tag{5.11}$$

$$D'(x, x_J) = C_1'(x) + C_2'(x)\left(\frac{x - x_J}{a\Delta x}\right) + \frac{C_2(x)}{a\Delta x} \tag{5.12}$$

$$\lambda_{\max} = \left(\frac{c^2}{l^2}\right)\frac{1}{a}\sum_{J=1}^{NP}\frac{[D_J'\mathbf{\Phi}_J + D_J\mathbf{\Phi}_J']^2}{D_J\mathbf{\Phi}_J}\Delta x_J \tag{5.13}$$

where Δx is the length between nodes and Δx_J is the integration weight using Trapezoidal rule C_1 and C_2 are defined in terms of the moments of $\mathbf{\Phi}(x)$ [Liu *et al.* (1995a)]. The critical time step is given for a central difference time integration scheme in Eq.(5.14). By substituting the maximum eigenvalue corresponding to the maximum frequency, the critical time step can be calculated.

$$\Delta t \leq \frac{2}{\omega_{\max}} \tag{5.14}$$

$$\Delta t \leq \frac{2l}{c}a^{1/2}\left[\sum_{J=1}^{NP}\frac{[D_J'\mathbf{\Phi}_J + D_J\mathbf{\Phi}_J']^2}{D_J\mathbf{\Phi}_J}\Delta x_J\right]^{-1/2} \tag{5.15}$$

Results for the critical time step were calculated using Eq.(5.15) for cubic spline window functions and are given by Liu et al (1995a) for 21 nodes. The discretized form of the critical time step was perceived to have a simple translation to the continuous form. The discretized form contains the derivative of the reproducing kernel squared. Then the largest eigenvalue and thus the critical time step is as follows:

$$\lambda = \left(\frac{c^2}{l^2}\right)\frac{1}{a}\int_{B(x)}\frac{k_{,x}^2(x, \tilde{x})}{k(x, \tilde{x})}d\tilde{x} \tag{5.16}$$

where $k(x, \tilde{x})$ is the reproducing kernel function defined by:

$$k(x, \tilde{x}) = D(x, \tilde{x})\mathbf{\Phi}\left(\frac{x - \tilde{x}}{a\Delta x}\right) \tag{5.17}$$

5.1.3 Numerical examples

1. Large deformation problem for 1D elastic-plastic bar

A wave propagation problem of a one-dimensional bar is analyzed using the RKPM. The solution for the elastic/plastic deforming rod was obtained with the spline window function and an explicit Newmark-Beta predictor/corrector time integration algorithm.

Figure 5.1 depicts the stress at the midpoint of the bar which is modeled with an elastic-plastic material under compression: yield stress $\sigma_y = 3 \times 10^4$ psi; $E_p = E/4$; applied stress, $\sigma_0 = 1 \times 10^6$ psi. The total number of particles is 41 and a dilation parameter of $\mathbf{a} = 1.4$ is used. For reference, a finite element solution using 320 elements is compared.

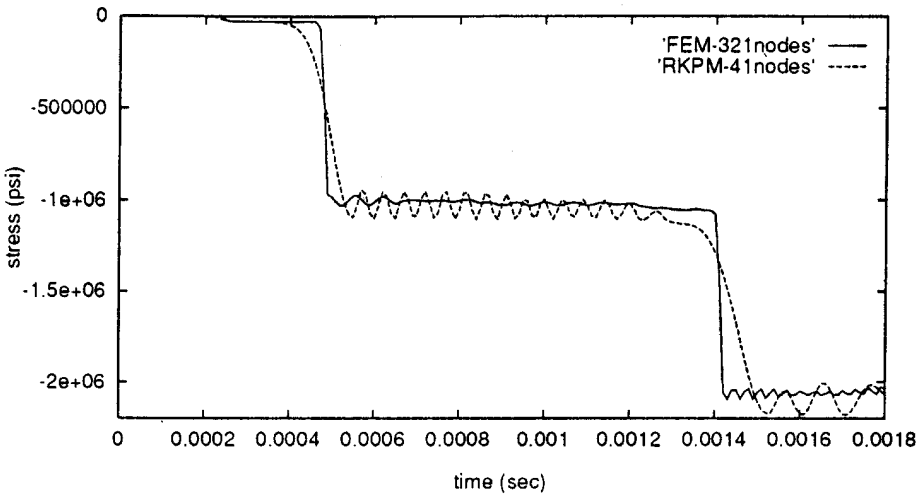


Figure 5.1. Stress at the midpoint of bar for 1D elastic-plastic large deformation problem

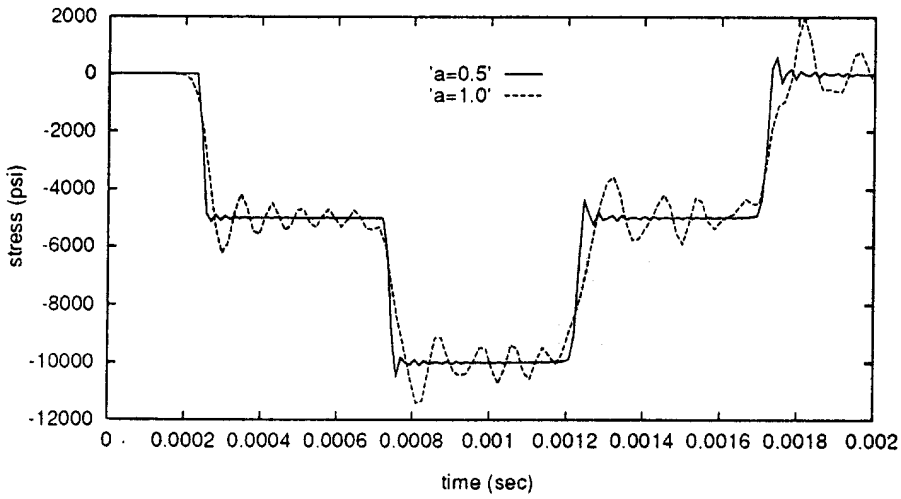


Figure 5.2. RKPM results of 41 nodes : $\mathbf{a} = 0.5$ and $\Delta t = 1.22 \times 10^{-5}$ sec for solid line; $\mathbf{a} = 1.0$ and $\Delta t = 1.65 \times 10^{-5}$ sec for dotted line (Δt is 1.35 times larger than FEM's critical time step)

We have studied before about the effect of dilation parameter on the critical time step and predicted that, for RKPM, the critical time step becomes larger as the dilation parameter becomes larger. The two results of RKPM with different dilation parameters and different time steps are shown in Figure 5.2. Both results employed 41 nodes. Although the high frequency components of the solution in this figure are filtered away, it is interesting to note that the problem was run at 1.35 times the FEM critical time step.

2. Nonlinear elastic ring and underwater explosion simulation

Reproducing kernel particle methods require no explicit mesh, hence there is no mesh entanglement problem due to the lightly restrained movement of nodes (almost no shear) in large deformation. The application of reproducing kernel particle methods for large deformation is illustrated in Figures 5.3 and 5.4. In Figure 5.3, the discretization of a nonlinear elastic ring consists of only a set of nodes or particles, and the equation of motion

is governed by the interaction forces among the particles. Figure 5.4 presents a numerical simulation of an underwater explosion using Lagrangian RKPM Formulation, consisting of a set of particles only.

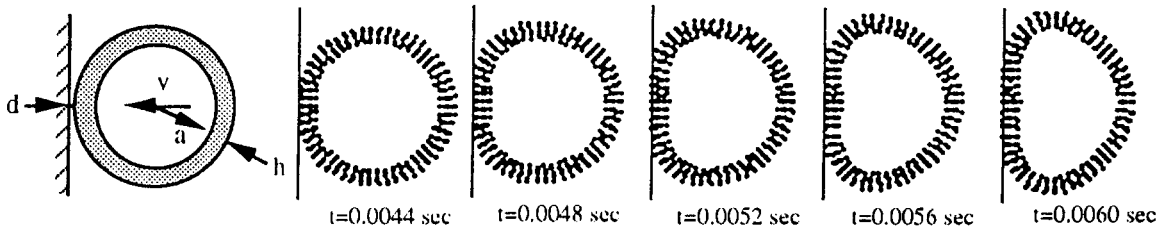


Figure 5.3. Dynamic analysis of a nonlinear elastic ring using RKPM

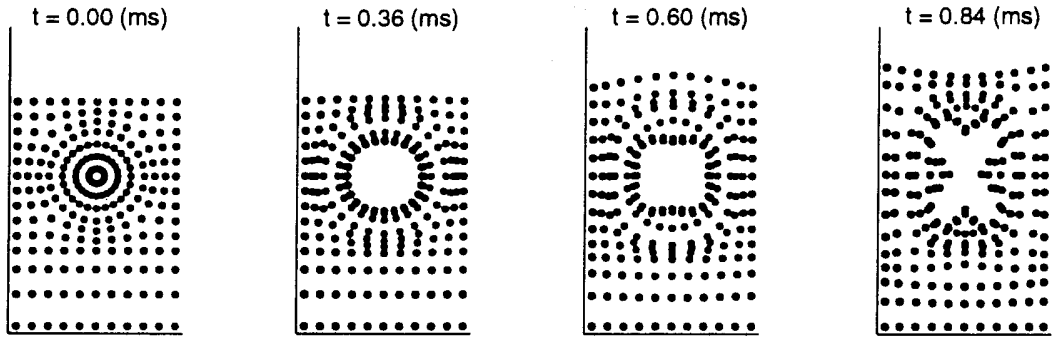


Figure 5.4. Underwater explosion simulation using lagrangian RKPM

5.2 RKPM for Rubber Hyperelasticity

A unique characteristic of rubber is its soft, highly extensible, and highly elastic behavior. In most engineering applications, rubber components experience very large deformations under normal service conditions. The major computational difficulties in the nonlinear analysis of rubber are due to (1) rubber being nearly incompressible and (2) rubber components deform to an extremely large scale, leading to severe mesh distortion. Although many effective finite element formulations [Hughes (1980), Sussman and Bathe (1987), Liu *et al.* (1988), Chen *et al.* (1994, 1995, 1995a, 1995b, 1995c, 1995d)] have been proposed to handle rubber incompressibility, the mesh distortion problem remains unresolved in rubber applications. Therefore, an effective computational technique that can model material incompressibility and large deformations is highly desirable for the analysis of rubber.

The Reproducing Kernel Particle Methods offer several attractive features for the large deformation analysis of elastomers. The major advantage of RKPM is the mesh-free computational environment. Chen *et al.* (1995a) employed a material based Lagrangian kernel function in conjunction with a Total Lagrangian Galerkin approximation to develop a Hyperelastic Reproducing Kernel Particle Method that effectively handles very large distortions in rubber applications. By the use of a material based Lagrangian kernel function, the dilation parameters selected initially can be used throughout the computation without experiencing a tension instability. Chen *et al.* (1995a) also introduced two methods to handle essential boundary conditions for large deformation analysis. One method is based on the extension of the modified variational principle proposed by Lu *et al.* (1994), and the other one is based on a coordinate transformation.

5.2.1 Variational form of hyperelasticity

The variational equation for hyperelasticity is given by

$$\delta\Pi = \int_{\Omega_x} \delta E_{ij} S_{ij} d\Omega - \int_{\Omega_X} \delta u_i b_i d\Omega - \int_{\Gamma_X^h} \delta u_i h_i^0 d\Gamma \tag{5.18}$$

where Ω_X is the domain of the undeformed configuration, $\mathbf{X}$ is the material coordinate, b_i is the body force, h_i^0 is the surface traction on the undeformed natural boundary, Γ_X^h, E_{ij} is the Green-Lagrangian strain, S_{ij} is the 2nd Piola Kirchhoff stress. They are defined by:

$$E_{ij} = \frac{1}{2}(F_{ki}F_{kj} - \delta_{ij}) \tag{5.19}$$

$$F_{ij} = \frac{\delta x_i}{\delta X_j} = \frac{\delta u_i}{\delta X_j} + \delta_{ij} \tag{5.20}$$

$$S_{ij} = \frac{\partial W}{\partial E_{ij}} \tag{5.21}$$

where W is the strain energy density function, u_i is the displacement that satisfies $u_i = g_i$ on the essential boundary Γ_X^g, F_{ij} is the deformation gradient, and δ_{ij} is the Kronecker delta.

For nonlinear problems, linearization is required. The linearization of Eq. (5.18) leads to:

$$\begin{aligned}\Delta\delta\Pi = &\int_{\Omega_X} \frac{\partial \delta u_i}{\partial X_j} H_{ijkl} \frac{\partial \Delta u_k}{\partial X_l} d\Omega \\ &+ \int_{\Omega_X} \left[\frac{1}{2}\left(F_{pi}\frac{\partial \delta u_p}{\partial X_j} + F_{pj}\frac{\partial \delta u_p}{\partial X_i}\right)\right] C_{ijkl} \left[\frac{1}{2}\left(F_{qk}\frac{\partial \Delta u_q}{\partial X_l} + F_{ql}\frac{\partial \Delta u_q}{\partial X_k}\right)\right] d\Omega \\ &- \int_{\Omega_X} \delta u_i \Delta b_i d\Omega - \int_{\Gamma_X^h} \delta u_i \Delta h_i^0 d\Gamma\end{aligned} \tag{5.22}$$

where

$$C_{ijkl} = \frac{\partial^2 W}{\partial E_{ij} \partial E_{kl}} \tag{5.23}$$

$$H_{ijkl} = \delta_{ik} S_{jl} \tag{5.24}$$

The explicit expression for C_{ijkl} for a general Rivlin strain energy density function can be found in Chen *et al.* (1994)

5.2.2 Treatment of essential boundary conditions by a modified variational principle

Since the Reproducing Kernel interpolants do not satisfy the Kronecker delta properties, the enforcement of essential boundary conditions require additional efforts. Lu *et al.* (1995) proposed a modified variational principle which replaces the Lagrangian multiplier of the essential boundary conditions by a physically meaningful surface traction on the essential

boundaries. The extension of this modified variational principle to nonlinear elasticity leads to:

$$\delta\Pi = \int_{\Omega_X} \delta E_{ij} S_{ij} d\Omega - \int_{\Omega_X} \delta u_i b_i d\Omega - \int_{\Gamma_X^h} \delta u_i h_i^0 d\Gamma - \int_{\Gamma_X^g} \delta u_i \bar{h}_i^0 d\Gamma - \int_{\Gamma_X^g} \delta \bar{h}_i^0 (u_i - g_i) d\Gamma \tag{5.25}$$

where Γ_X^g is the essential boundary, $\mathbf{h}^0$ is the surface traction on Γ_X^h and $\bar{\mathbf{h}}^0$ is the surface traction on Γ_X^g that is defined by:

$$\bar{h}_i^0 = \sigma_{ij} n_j^0 \tag{5.26}$$

σ_{ij} is the 1st Piola Kirchhoff stress and $\mathbf{n}^0$ is the surface normal in the undeformed configuration.

Linearization of Eq. (5.24) leads to:

$$\begin{aligned}
\Delta\delta\Pi = &\int_{\Omega_X} \frac{\partial \delta u_i}{\partial X_j} H_{ijkl} \frac{\partial \Delta u_k}{\partial X_l} d\Omega \\
&+ \int_{\Omega_X} \left[\frac{1}{2}\left(F_{pi}\frac{\partial \delta u_p}{\partial X_j} + F_{pj}\frac{\partial \delta u_p}{\partial X_i}\right)\right] C_{ijkl} \left[\frac{1}{2}\left(F_{qk}\frac{\partial \delta u_q}{\partial X_l} + F_{ql}\frac{\partial \delta u_q}{\partial X_k}\right)\right] d\Omega \\
&- \int_{\Gamma_X^g} \delta u_i n_j^0 \left(\bar{C}_{ijkl}\Delta E_{kl} + \bar{H}_{ijkl}\frac{\partial \Delta u_k}{\partial X_l}\right) d\Gamma \\
&- \int_{\Gamma_X^g} \left(\delta E_{ij}\bar{C}_{ijkl} + \frac{\partial \delta u_i}{\partial X_j}\bar{H}_{ijkl}\right) n_l^0 \Delta u_k d\Gamma - \int_{\Omega_X} \delta u_i \Delta b_i d\Omega \\
&- \int_{\Gamma_X^h} \delta u_i h_i^0 d\Gamma - \int_{\Gamma_X^g} \Delta(\delta \bar{h}_i^0) u_i d\Gamma
\end{aligned} \tag{5.27}$$

The last three terms on the right side of Eq. (5.27) are due to the treatment of essential boundaries where

$$\bar{C}_{ijkl} = C_{ipkl} F_{jp} \tag{5.28}$$

$$\bar{H}_{ijkl} = S_{il}\delta_{kj} \tag{5.29}$$

Substituting the RKPM displacement interpolation function Eq.(3.32a) into Eqs.(5.25) and (5.27), the following incremental matrix equation is obtained:

$$\mathbf{K}\Delta\mathbf{u} = \Delta f \tag{5.30}$$

where

$$\mathbf{K} = \mathbf{K}_G + \mathbf{K}_M + \mathbf{K}_U \tag{5.31}$$

$$\mathbf{K}_G = \int_{\Omega_X} \mathbf{B}_G^T \mathbf{H} \mathbf{B}_G d\Omega \tag{5.32}$$

$$\mathbf{K}_M = \int_{\Omega_X} \mathbf{B}_M^T \mathbf{C} \mathbf{B}_M d\Omega \tag{5.33}$$

$$\mathbf{K}_U = \mathbf{K}_u + \mathbf{K}_u^T = \text{stiffness associated with prescribed displacements} \tag{5.34}$$

$$\mathbf{K}_u = -\int_{\Gamma_X^g} \mathbf{g}^T (\overline{\mathbf{C}\mathbf{B}}_M + \bar{\mathbf{H}}\mathbf{B}_G) d\Omega \tag{5.35}$$

$$\mathbf{g} = \begin{bmatrix} n_1^0 \mathbf{N} & \mathbf{0} \\ \mathbf{0} & n_2^0 \mathbf{N} \\ n_2^0 \mathbf{N} & \mathbf{0} \\ \mathbf{0} & n_1^0 \mathbf{N} \end{bmatrix} \tag{5.36}$$

$$\mathbf{B}_M = \begin{bmatrix} \mathbf{F}_{XX}\mathbf{N}_{\prime X} & \mathbf{F}_{YX}\mathbf{N}_{\prime X} \\ \mathbf{F}_{XY}\mathbf{N}_{\prime X} & \mathbf{F}_{YY}\mathbf{N}_{\prime Y} \\ \mathbf{F}_{XX}\mathbf{N}_{\prime Y} + \mathbf{F}_{XY}\mathbf{N}_{\prime X} & \mathbf{F}_{YX}\mathbf{N}_{\prime Y} + \mathbf{F}_{YY}\mathbf{N}_{\prime X} \end{bmatrix} \tag{5.37}$$

$$\bar{\mathbf{B}}_M = \begin{bmatrix} \mathbf{F}_{XX}\mathbf{N}_{\prime X} & \mathbf{F}_{YX}\mathbf{N}_{\prime X} \\ \mathbf{F}_{XY}\mathbf{N}_{\prime X} & \mathbf{F}_{YY}\mathbf{N}_{\prime Y} \\ \frac{1}{2}(\mathbf{F}_{XX}\mathbf{N}_{\prime Y} + \mathbf{F}_{XY}\mathbf{N}_{\prime X}) & \frac{1}{2}(\mathbf{F}_{YX}\mathbf{N}_{\prime Y} + \mathbf{F}_{YY}\mathbf{N}_{\prime X}) \\ \frac{1}{2}(\mathbf{F}_{XX}\mathbf{N}_{\prime Y} + \mathbf{F}_{XY}\mathbf{N}_{\prime X}) & \frac{1}{2}(\mathbf{F}_{YX}\mathbf{N}_{\prime Y} + \mathbf{F}_{YY}\mathbf{N}_{\prime X}) \end{bmatrix} \tag{5.38}$$

$$\mathbf{B}_G = \begin{bmatrix} \mathbf{N}_{\prime X} & \mathbf{0} \\ \mathbf{0} & \mathbf{N}_{\prime Y} \\ \mathbf{N}_{\prime Y} & \mathbf{0} \\ \mathbf{0} & \mathbf{N}_{\prime X} \end{bmatrix} \tag{5.39}$$

$$\Delta \mathbf{f} = \mathbf{f}^{ext} - \mathbf{f}^{int} - \mathbf{f}_U \tag{5.40}$$

$$\mathbf{f}_U = \mathbf{f}_u^1 + \mathbf{f}_u^2 = \text{ force associated with prescribed displacements} \tag{5.41}$$

$$\mathbf{f}_u^1 = \int_{\Gamma_X^g} \begin{bmatrix} \mathbf{N}^T \bar{h}_1^0 \\ \mathbf{N}^T \bar{h}_2^0 \end{bmatrix} d\Gamma \tag{5.42}$$

$$\mathbf{f}_u^2 = \int_{\Gamma_X^g} (\bar{\mathbf{B}}_M^T \bar{\mathbf{C}}^T + \mathbf{B}_G^T \bar{\mathbf{H}}^T) \mathbf{Q} d\Gamma \tag{5.43}$$

$$\mathbf{Q} = \begin{bmatrix} n_1^0(u_1 - g_1) \\ n_2^0(u_2 - g_2) \\ n_1^0(u_2 - g_2) \\ n_2^0(u_1 - g_1) \end{bmatrix} \tag{5.44}$$

Note that the last term in Eq. (5.27) leads to an asymmetric tangent matrix and therefore is neglected.

In the present formulation, since the original configuration is selected as the referential configuration, the kernel function and displacement interpolation function are expressed in the material coordinates as:

$$\mathbf{N} = [N_1, N_2, \ldots, N_{NP}] \tag{5.45}$$

$$N_I(\mathbf{X}) = \mathbf{\Phi}_a(\mathbf{X} - \mathbf{X}_I) C^N(\mathbf{X}, \mathbf{X} - \mathbf{X}_I) \Delta V_I \tag{5.46}$$

$$\mathbf{\Phi}_a(\mathbf{X} - \mathbf{X}_I) = \frac{1}{a}\phi\left(\frac{\|\mathbf{X} - \mathbf{X}_I\|}{a}\right) \tag{5.47}$$

where ϕ_a is the window function, $\|\cdot\|$ is the Euclidean norm of a vector, $C^N(\mathbf{X}; \mathbf{X} - \mathbf{X}_I)$ is the correction function which satisfies the N-th order reproducing condition and NP is the number of particles.

5.2.3 Treatment of essential boundary conditions by method of transformation

As can be seen in Eqs. (5.30)-(5.44), a tremendous amount of computation is devoted to the treatment of prescribed boundary displacements. A transformation method is proposed by Liu and Chen (1995) and Chen *et al.* (1995a) to transform the incremental equation from generalized coordinates to nodal coordinates. Recall the discretized displacement reproducing equation,

$$u(X_I) = \sum_{J=1}^{NP} N_J(X_I) u_J \equiv \mathbf{N}(X_I)\mathbf{u} \tag{5.48}$$

where X_I is the nodal coordinate and $u(X_I) \equiv \hat{u}_I$ is the nodal displacement. One can reconstruct the nodal displacements by using the following equation:

$$\hat{\mathbf{u}} = \mathbf{T}^T \mathbf{u} \tag{5.49}$$

where

$$T_{IJ} = N_I(X_J) \tag{5.50}$$

Consider the following linearized energy equation

$$\delta\mathbf{u}^T(\mathbf{f}_n^{int} + \mathbf{K}\Delta\mathbf{u}) = \delta\mathbf{u}^T\mathbf{f}_{n+1}^{ext} \tag{5.51}$$

where n is the load step counter. Inserting Eq. (5.49) into Eq. (5.51) yields:

$$\delta\hat{\mathbf{u}}^T(\mathbf{T}^{-1}\mathbf{f}_n^{int} + \mathbf{T}^{-1}\mathbf{K}\mathbf{T}^{-T}\Delta\hat{\mathbf{u}}) = \delta\hat{\mathbf{u}}^T\mathbf{T}^{-1}\mathbf{f}_{n+1}^{ext} \tag{5.52}$$

A transformed incremental equilibrium equation is obtained:

$$\hat{\mathbf{K}}\Delta\hat{\mathbf{u}} = \Delta\hat{\mathbf{f}} \tag{5.53}$$

$$\hat{\mathbf{K}} = \mathbf{T}^{-1}\mathbf{K}\mathbf{T}^{-T} \tag{5.54}$$

$$\Delta\hat{\mathbf{f}} = \hat{\mathbf{f}}^{ext} - \mathbf{f}_n^{int} \tag{5.55}$$

where $\hat{\mathbf{f}}^{ext}$ and $\hat{\mathbf{f}}_n^{int}$ are nodal external and internal forces, respectively, defined by

$$\hat{\mathbf{f}}^{ext}\mathbf{T}^{-1}\mathbf{f}^{ext} \tag{5.56}$$

$$\hat{\mathbf{f}}_n^{int}\mathbf{T}^{-1}\mathbf{f}_n^{int} \tag{5.57}$$

By using Eq. (5.53), prescribed displacements can be directly imposed.

5.2.4 Rubber strain energy density functions

Mooney (1940) and Rivlin (1956), expressed the strain energy density function of an elastic medium by the three invariants of the Green deformation tensor $I_1,\ I_2,\ I_3$ such that

$$W(I_1,\ I_2,\ I_3) = \sum_{m+n+k=1}^{\infty} A_{mnl}(I_1-3)^m(I_2-3)^n(I_3-1)^k \tag{5.58}$$

and

$$I_1 = tr(\mathbf{G}), \quad I_2 = \frac{1}{2}\left[(tr\mathbf{G})^2 - tr(\mathbf{G}^2)\right], \quad I_3 = \det(\mathbf{G}) \tag{5.59}$$

where tr designates trace. The second Piola-Kirchhoff stress can be obtained from W by

$$S_{ij} = \frac{\partial W}{\partial I_n}\frac{\partial I_n}{\partial E_{ij}} = 2[K_1\delta_{ij} + K_2(I_1\delta_{ij} - G_{ij}) + K_3 I_3 G_{ij}^{-1}] \quad n = 1,2 \tag{5.60}$$

where

$$K_n = \frac{\partial W}{\partial I_n} \tag{5.61}$$

For incompressible materials Eq. (5.58) reduces to

$$\bar{W}(I_1, I_2) = \sum_{m+n=1}^{\infty} A_{mn}(I_1-3)^m(I_2-3)^n \tag{5.62}$$

This strain energy density function, expressed by an infinite power series, is normally truncated to neo-Hookean or Mooney-Rivlin forms. However, the material constants in these two functions obtained from tensile test data are not applicable to other deformation modes, and do not model large strain accurately. Yeoh (1990) suggested a third order approximation and expressed the strain energy density as a cubic function of I_1

$$\bar{W} = A_{10}(I_1-3) + A_{20}(I_1-3)^2 + A_{30}(I_1-3)^3 \tag{5.63}$$

It was later observed that this cubic function fails to capture shear stress-strain behavior in the small strain range and was modified by Yeoh (1993) by adding an exponential term to improve the small strain performance:

$$\bar{W} = \frac{\alpha}{\beta}[1 - e^{-\beta(I_1-3)}] + A_{10}(I_1-3) + A_{20}(I_1-3)^2 + A_{30}(I_1-3)^3 \tag{5.64}$$

A modified strain energy density function for nearly incompressible materials can be expressed as:

$$W(I_1, I_2, J) = \bar{W}(I_1, I_2) + \tilde{W}(J) \tag{5.65}$$

For nearly incompressible materials, the invariants I_1 and I_2 are not measures of pure distortion (isochoric). Penn (1970) proposed the following reduced invariants to separate the distortional and dilatational deformation:

$$\bar{I}_1 = I_1 I_3^{-1/3}, \quad \bar{I}_2 = I_2 I_3^{-2/3} \tag{5.66}$$

The strain energy density function is expressed as:

$$W(\bar{I}_1, \bar{I}_2, J) = \bar{W}(\bar{I}_1, \bar{I}_2) + \tilde{W}(J) \tag{5.67}$$

where $\tilde{W}(J)$ is the volumetric strain energy density, $J = \sqrt{I_3}$. A widely used form of $\tilde{W}(J)$ is:

$$\tilde{W}(J) = \frac{k}{2}(J-1)^2 \tag{5.68}$$

where k, the bulk modulus, is measured experimentally. It has been discussed by Chen *et al.* (1995d) that the use of $\tilde{W}(J)$ in Eq. (5.68) does not guarantee existence of a solution for compressible and incompressible hyperelasticity. An example of $\tilde{W}(J)$ that satisfies the existence conditions of hyperelasticity given in Chen *et al.* (1995) is:

$$\tilde{W}(J) = \frac{k}{4}(J^2 - 1 - 2\log J) \tag{5.69}$$

The second Piola-Kirchhoff stress associated with Eq. (5.67) is:

$$S_{ij} = 2\left[\bar{K}_1 I_3^{-1/3}\left(\delta_{ij} - \frac{1}{3}I_1 G_{ij}^{-1}\right) + \bar{K}_2 I_3^{-2/3}\left(I_1\delta_{ij} - G_{ij} - \frac{2}{3}I_2 G_{ij}^{-1}\right)\right] + PJG_{ij}^{-1} \tag{5.70}$$

where

$$\bar{K}_n = \frac{\partial \bar{W}}{\partial \bar{I}_n} \quad n = 1, 2 \tag{5.71}$$

$$P = \frac{\partial W}{\partial J} = \frac{\partial \bar{W}}{\partial J} \tag{5.72}$$

The detailed discussion of the physical significance of the reduced invariants and the decomposition of the strain energy density function, numerical implementation procedures, and the study of their performance in tension, compression and shear problems can be found in Chen *et al.* (1994, 1995c) and Chen and Pan (1995). The existence of the solution for nearly incompressible and incompressible hyperelasticity was discussed by Chen *et al.* (1995).

5.2.5 Numerical examples

The cubic strain energy density function with the material properties taken from Chen *et al.* (1995c), is used in the following examples. For carbon black filled rubber, the ratio of bulk modulus to shear modulus is $10^4 \sim 10^5$. A bulk modulus of $k = 10^5$ Mpa is used in the computation. The following examples were discussed in detail [Chen *et al.* (1995a)].

1. Uniaxial tension-compression of rubber

Although the load-displacement behavior of this problem is nonlinear, the displacement field in the domain at any instant of time is linear. Therefore, RKPM with a modified kernel

function that satisfies reproducing conditions for a linear field is expected to provide a good solution for this problem. A 1.0 cm × 0.25 cm plane strain rubber unit is subjected to tension and compression along the axial direction as shown in Figures 5.5-5.6. To generate uniaxial deformation, no end constraints are imposed, and the rubber unit is free to deform in the transverse direction.

Two particle discretizations, one consisting of 4×9 equally spaced particles and another consisting of 36 randomly spaced particles, are used to discretize the domain of the problem. The specimen is first stretched to 1000% in tension. Both regular nodal spacing and irregular spacing particle models, with various dilatation parameters, are used in the analysis. The predicted axial force versus stretch ratio is compared against the incompressible solution in Figure 5.5. The specimen is also compressed until the deformed axial dimension is only 1/10 of the undeformed axial dimension. Significant nonlinearities are observed when the compressive strain is beyond - 80% in compression. The results obtained from both particle models are compared against the analytical solution in Figure 5.6. The final deformations analyzed by the irregularly spaced particles are plotted in Figure 5.7. The numerical errors in this problem are due partly to the finite bulk modulus being used in the analysis.

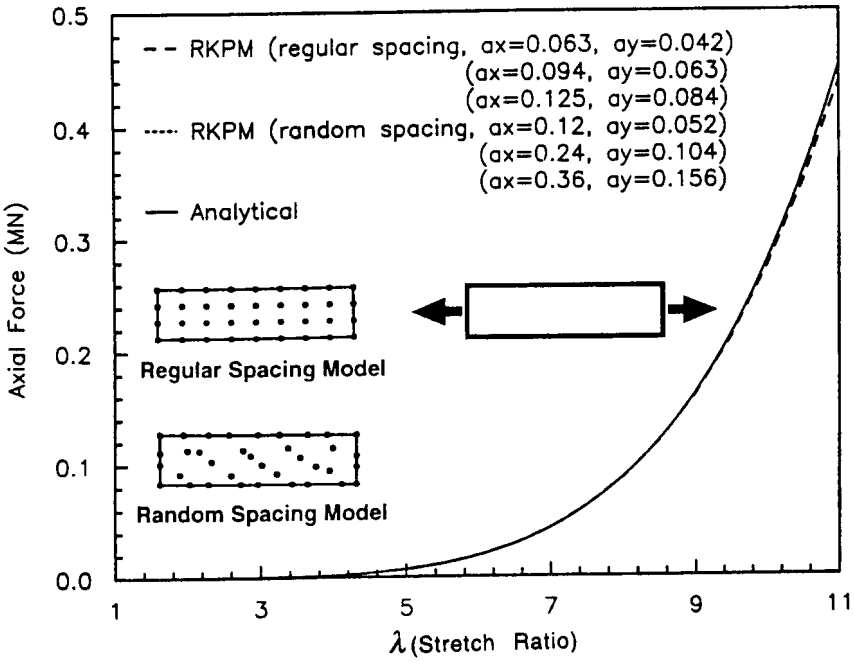


Figure 5.5. Rubber under 1000% uniaxial tension

2. Simple shear test

Simple shear is a fundamental laboratory test to characterize the material properties of rubber. A test specimen of 0.25 cm x 1.0 cm is analyzed by the same particle models as used in Example 1.

For the comparison with available experimental data, the test specimen is sheared to an engineering shear strain of 200%. The minimum dilation parameters are calculated based on the maximum distance between particles. In each analysis, the same dilation parameter is assigned to all particles. Although the particle distances are changing during structural deformation, the dilation parameters do not require readjustment since material

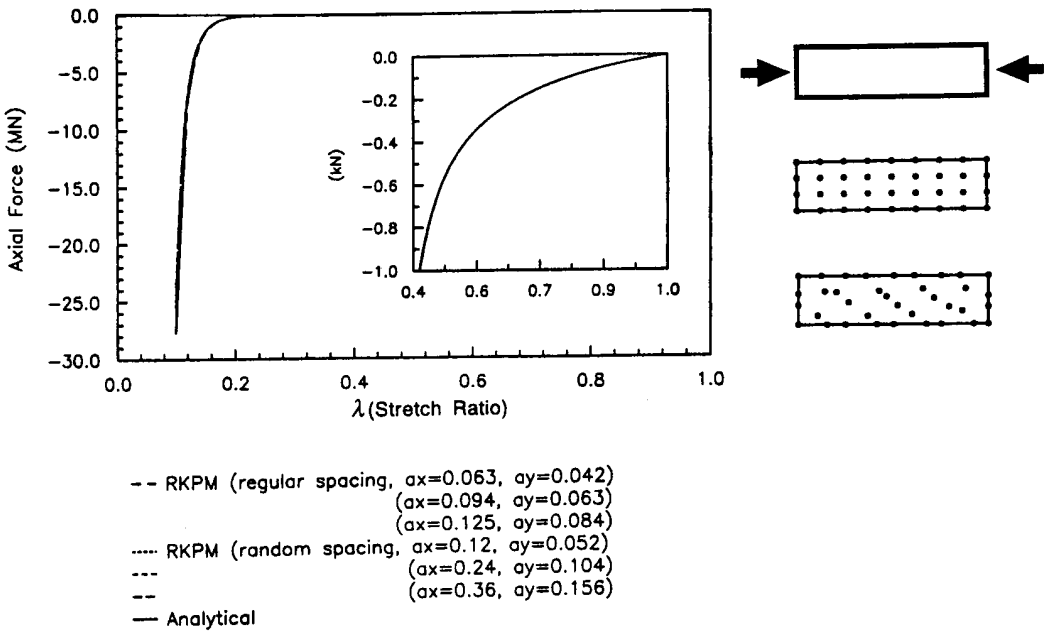


Figure 5.6. Rubber under 90% uniaxial compression

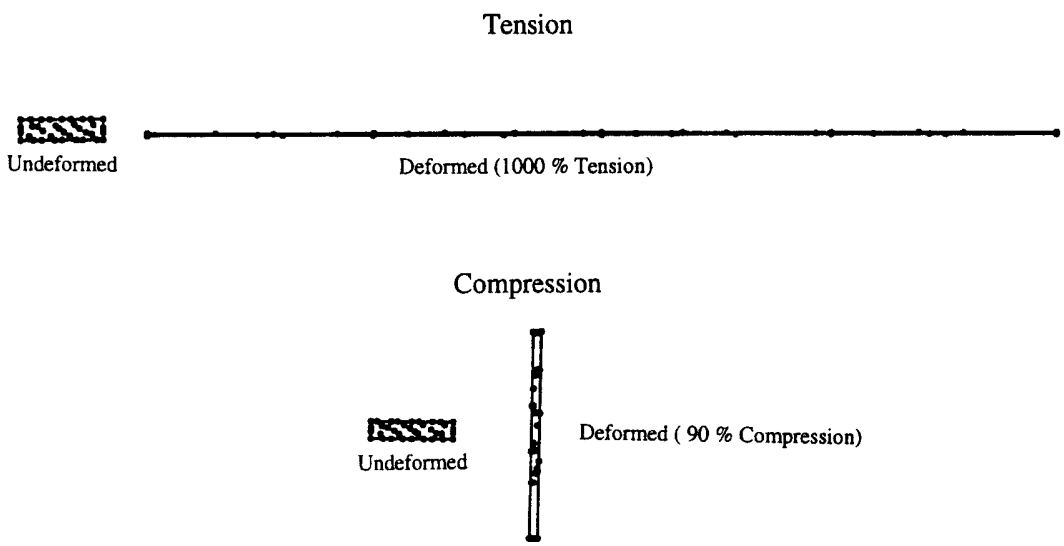


Figure 5.7. Deformation of rubber under 1000% uniaxial tension and 90 % uniaxial compression

based Lagrangian kernel functions are employed in the formulation. The analysis results, obtained from two discrete particle models with various dilation parameters are compared with the simple shear solution and experimental data in Figure 5.8. Very good agreement is observed between the numerical solution, the simple shear solution and the experimental data. The deformed configurations are also plotted in Figures 5.8.

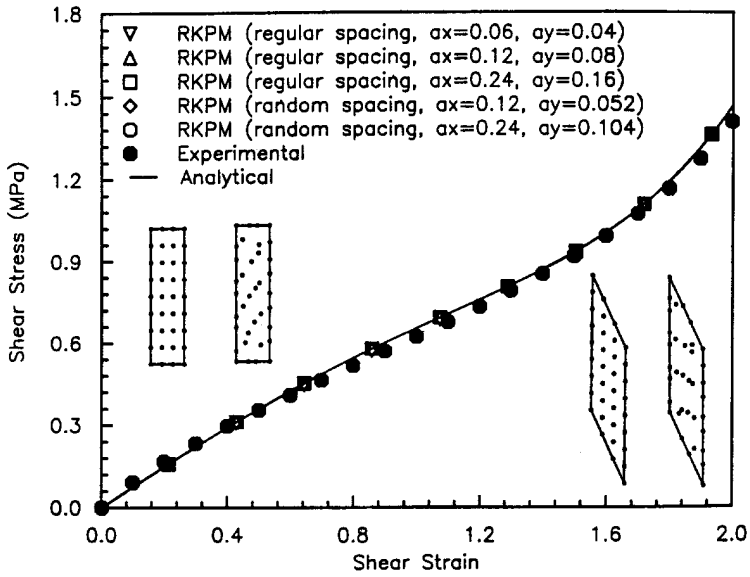


Figure 5.8. Simple shear test

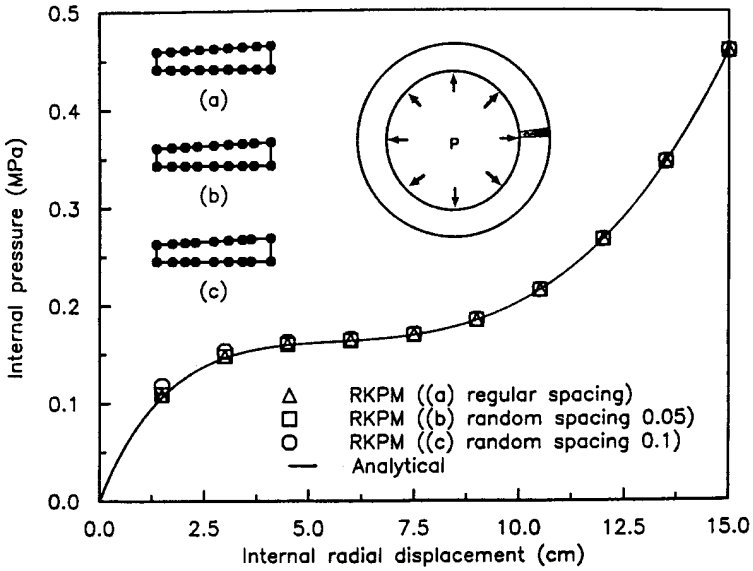


Figure 5.9. Inflation of an infinitely long rubber tube

3. Inflation of an infinitely long rubber tube

A plane strain rubber tube subjected to internal pressure is analyzed by RKPM. The inner and outer radii of the tube are 6cm and 8cm, respectively. Due to axisymmetry, only one wedge of the tube is modeled with proper boundary conditions imposed to enforce axisymmetry. A total of 9 particles are used to discretize the thickness of the tube. Three sets of regularly and irregularly spaced particles as shown in Figures 5.9 are used to model one wedge of the tube.

The pressure is applied to the inner surface of the tube. The tube is inflated until the inflated inner radius reaches 3.5 times its original dimension. The analytical solution of this problem using the cubic strain energy density function was discussed by Chen at el. (1995c). The pressure-displacement curves calculated by the three particle models are compared against analytical solution in Figures 5.9 and the results are satisfactory. The 360 degree undeformed and deformed structures generated by rotating the one-wedge particles are plotted in Figures 5.10 for visualization.

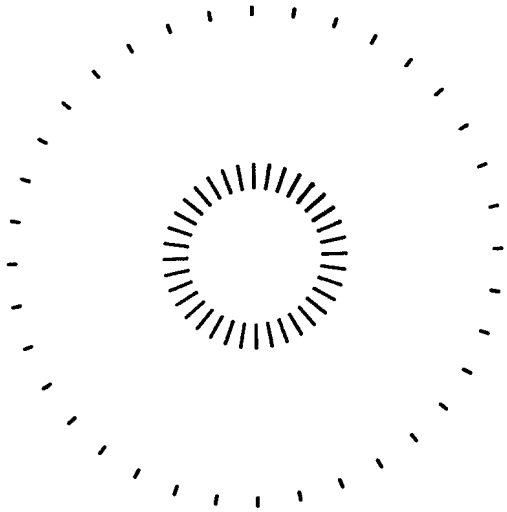


Figure 5.10. Deformation of rubber tube under inflation

4. Bonded rubber deformation

The bonded rubber behavior was studied extensively by Gent *et al.* (1959) and was analyzed by finite element method in detail [Chen *et al.* (1994, 1995c)]. In bearing applications, rubber pads are usually sandwiched between metal layers. The rubber-metal bonding provides high interlaminar stiffness and low shear stiffness. The detailed physical significance and numerical studies of this problem are discussed in Chen *et al.* (1994).

A rubber pad with width to thickness ratio of 10 is bonded to two metal plates at top and bottom surfaces. Only rubber is modeled by regularly spaced and irregularly spaced particles as shown in Figure 5.11. Perfect bonding between rubber-metal interfaces is assumed and metal plates are treated as rigid bodies. The rubber pad is first sheared while holding constant distance between two metal plates. The predicted shear load-deflection curves and the rubber deformations are plotted in Figure 5.11.

Next, the shape factor effect on the Apparent Young's modulus of bonded rubber is analyzed. The analysis results are compared against Gent's approximation [Gent *et al.* (1959)] in Figure 5.12 and the agreement is satisfactory. The deformation of bonded rubber under large tension and compression are shown in Figure 5.13. As have discussed in Chen *et al.* (1995c), finite element analysis of this problem terminated at 500% tension and 30% compression due to mesh distortion occurs near the edges on the rubber-metal interfaces. Using RKPM, this analysis can advance further. The deformed configurations at 1000% tension and 45% compression, analyzed using two particle models are shown in Figure 5.13. Some disagreement in the bulge-out deformation predicted by the two analysis models are observed. The appropriate selection of the dilation parameters for irregular spacing model need to be further studied to correct the disagreement.

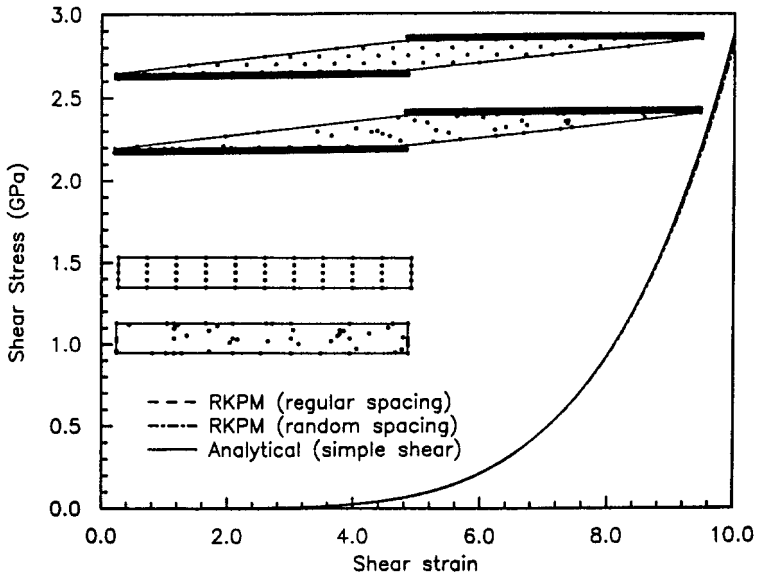


Figure 5.11. Bonded thin rubber pad under shear deformation

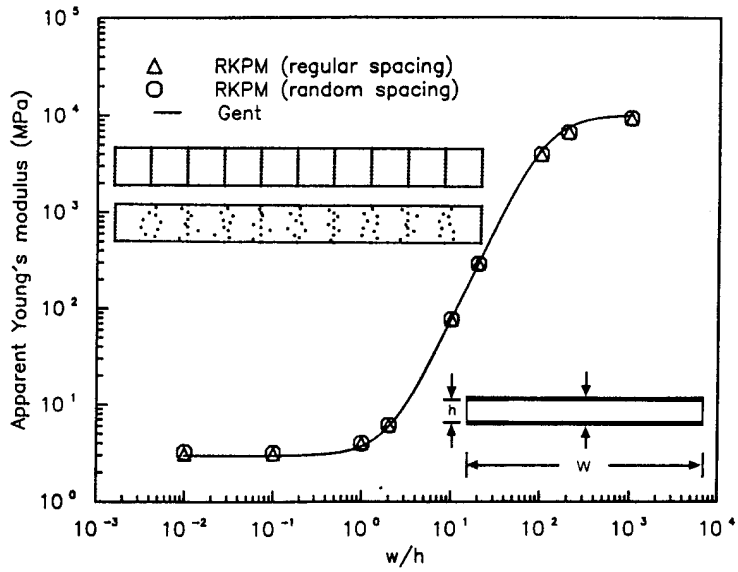


Figure 5.12. Analysis of shape factor effect in bonded rubber components under compression

5. Analysis of engine mount

Engine mounts are designed to support engine weight, isolate engine vibration from being transmitted to the passenger compartment, and to protect the engine from external excitation. In this problem, the deformation of engine mount due to vertical excitation is analyzed using finite element method and RKPM.

In the finite element analysis, only the rubber components are modeled by a finite element mesh shown in Figure 5.14a. The center metal piece, treated as a rigid body, is connected to engine and is forced to move downward in vertical direction. The outer surface of the

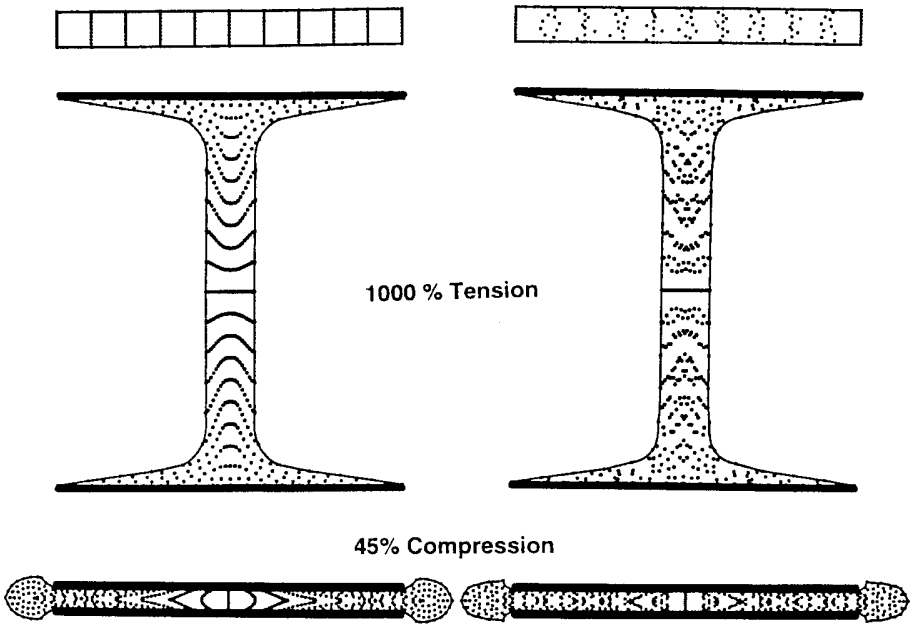


Figure 5.13. Large deformation of bonded thin rubber pad in bearings under tension and compression

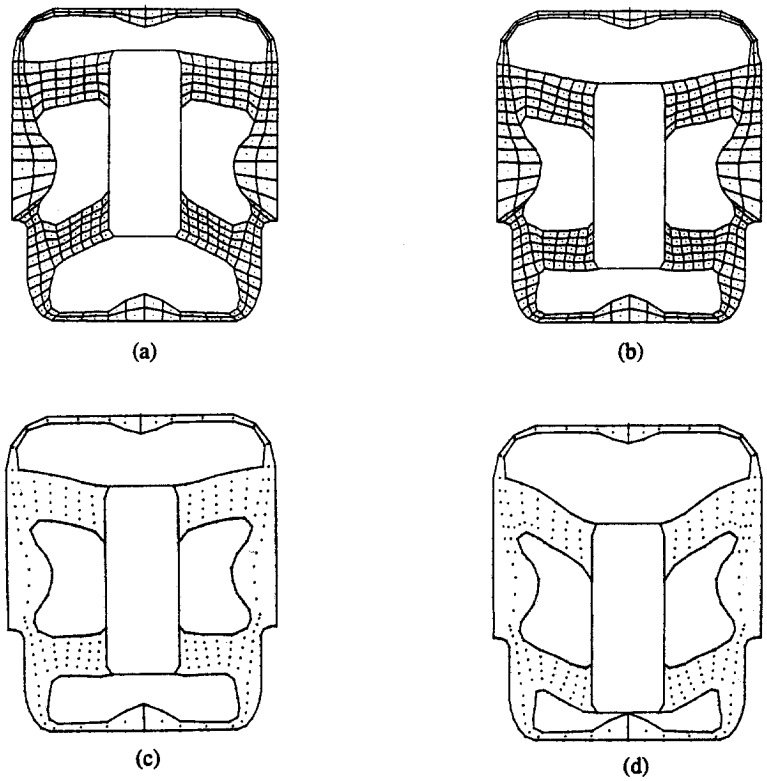


Figure 5.14. Analysis of engine mount using FEM and RKPM

engine mount is bonded on metal casing that is attached to car body and therefore the outer surface of the analysis model is completely fixed. In this analysis, the center metal piece is to be moved in vertical downward direction until it hits the bumper. The finite element analysis diverged due to mesh distortion near the lower corners of the lower rubber legs as shown in Figure 5.14b.

A RKPM model generated using the existing nodes in finite element model, is used to analyze the same problem. The background mesh in RKPM analysis utilizes the existing mesh in finite element model. In other words, the major difference in FEM and RKPM models for this analysis is in the displacement interpolation functions. The RKPM analysis, as shown in Figure 5.14c, also diverged in a similar fashion as that of the finite element analysis. A refined RKPM model is generated simply by adding more particles near the highly deformed area. This refined model successfully complete the analysis as plotted in Figure 5.14d. This example demonstrates the convenience in model refinement using RKPM.

6. A STABILIZED PARTICLE INTEGRATION METHOD FOR RKPM

6.1 Introduction

The computational efficiency of RKPM can be improved by introducing a stabilized particle integration method for the construction of the stiffness matrix. Consider the integration of a function $f(\mathbf{x})$ over the domain Ω

$$I(f) = \int_{\Omega} f(\mathbf{x}) d\Omega \tag{6.1}$$

In general, the numerical integration of $f(\mathbf{x}) \equiv f(x_1, x_2, x_3)$ is performed by finding an approximation of $f(\mathbf{x})$, denoted by $\tilde{f}_n(\mathbf{x})$, where n is the order of approximation, and defining

$$I_n(f) \equiv \int_{\Omega} \tilde{f}_n(\mathbf{x}) d\Omega = I(\tilde{f}_n) \tag{6.2}$$

where $I_n(f)$ is the numerical integration of $f(\mathbf{x})$. For the method to be effective, the domain is divided into N subdomains such that:

$$I_n(f) \equiv \int_{\Omega} \tilde{f}_n(\mathbf{x}) d\Omega = \sum_{k=1}^{N} \int_{\Omega^k} \tilde{f}_n^k(\mathbf{x}) d\Omega \tag{6.3}$$

In the current approach, we employ a Taylor series approximation of $f(\mathbf{x})$

$$\begin{aligned} f(\mathbf{x}) &\approx \sum_{p=0}^{n} \frac{1}{p!} \left[(x_1 - x_1^k)\frac{\partial}{\partial x_1} + (x_2 - x_2^k)\frac{\partial}{\partial x_2} + (x_3 - x_3^k)\frac{\partial}{\partial x_3} \right]^p f(\mathbf{x})|_{x=x^k} \\ &= \sum_{p=0}^{n} \frac{1}{p!} \left\{ \sum_{r+s+t=p} \left[\frac{p!}{r!s!t!} (x_1 - x_1^k)^r (x_2 - x_2^k)^s (x_3 - x_3^k)^t \left(\frac{\partial^p f}{\partial x_1^r \partial x_2^s \partial_3^t} \right) \Big|_{x=x^k} \right] \right\} \\ &= \tilde{f}_n^k(x) \end{aligned} \tag{6.4}$$

The Taylor series approximation of $f(\mathbf{x})$ in Ω^k, denoted by $\tilde{f}_n^k(\mathbf{x})$, is expressed with respect to the centroid of Ω^k, $\mathbf{x}^k$, as shown in Figure 6.1. The integration of $f(\mathbf{x})$ over the

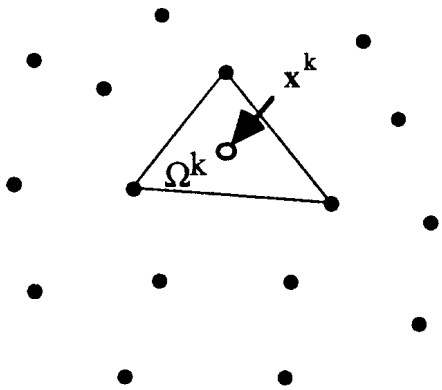


Figure 6.1. Two-dimensional subdomain Ω^k and its centroid $\mathbf{x}^k$

subdomain Ω^k is now approximated by the integration of $\tilde{f}_n^k(\mathbf{x})$ over Ω^k to yield:

$$I(\tilde{f}_n^k) = \sum_{p=0}^{n} \frac{1}{p!} \left\{ \sum_{r+s+t=p} \left[\frac{p!}{r!s!t!} M_{rst}^k \left(\frac{\partial^p f}{\partial^r x_1 \partial^s x_2 \partial^t x_3} \right) \Bigg|_{x=x^k} \right] \right\} \tag{6.5}$$

where

$$M_{rst}^k = \int_{\Omega^k} (x_1 - x_1^k)^r (x_2 - x_2^k)^s (x_3 - x_3^k)^t d\Omega \tag{6.6}$$

Consequently, the integration of $f_n(\mathbf{x})$ over the whole domain Ω is obtained by:

$$I_n(f) = \sum_{k=1}^{N} I(\tilde{f}_n^k) = \sum_{k=1}^{N} \left\{ \sum_{l=0}^{n} \frac{1}{l!} \left[\sum_{r+s+t=l} \frac{l!}{r!s!t!} M_{rst}^k \left(\frac{\partial^l f}{\partial x_1^r \partial x_2^s \partial^t x_3} \right) \Bigg|_{x=x^k} \right] \right\} \tag{6.7}$$

Note that in Eq. (6.7), only the integration of the moments M_{rst}^k is required, and they can be integrated numerically using the following particle integration technique:

$$M_{rst}^k = \sum_{j=1}^{NPk} (x_1^j - x_1^k)^r (x_2^j - x_2^k)^s (x_3^j - x_3^k)^t \Delta V_j \tag{6.8}$$

where x_i^j and ΔV_j are the i-coordinate and volume of particle j respectively, and NPk is the number of particles associated with the subdomain Ω^k. The selection of NPk and ΔV_j for various subdomain geometries are summarized in Table 6.1.
A^k and V^k are the area and volume of subdomain Ω^k in two-dimensional and three-dimensional problems, respectively.

	2D Triangle	2D Quadrilateral	3D Tetrahedron	3D Brick
NPk	3	4	4	8
ΔV_j	$\frac{1}{3}A^k$	$\frac{1}{4}A^k$	$\frac{1}{4}V^k$	$\frac{1}{8}V^k$

Table 6.1. Particle integration parameters for various subdomain geometries

6.2 Stabilized Particle Integration Method for RKPM

Liu *et al.* (1985a, 1985b, 1988b) and Liu and Hu (1992, 1993, 1994) proposed an incompressible plane strain element using a Taylor series approximation of the gradient matrix $\mathbf{B}$ to avoid volumetric and shear locking. Later, Liu *et al.* (1994) and Liu and Hu (1993, 1994) introduced a multiple quadrature underintegrated element to further improve the accuracy of the method. In the present work, we introduce the same technique for the construction of the stiffness matrix in conjunction with the integration procedures described in the previous section to yield:

$$\mathbf{K} = I_n(\mathbf{B}^T\mathbf{C}\mathbf{B}) = \sum_{k=1}^{N} I(\tilde{\mathbf{B}}_n^{k^T}\mathbf{C}\tilde{\mathbf{B}}_n^k) \tag{6.9}$$

where

$$\tilde{\mathbf{B}}_n^k = \sum_{p=0}^{n}\frac{1}{l!}\left\{\sum_{r+s+t=p}\left[\frac{p!}{r!s!t!}(x_1-x_1^k)^r(x_2-x_2^k)^s(x_3-x_3^k)^t\left(\frac{\partial^p}{\partial x_1^r\partial x_2^s\partial^t x_3}\mathbf{B}\right)\bigg|_{x=x^k}\right]\right\} \tag{6.10}$$

For simplicity, we demonstrate the method using a first order approximation of $\mathbf{B}$, i.e.,

$$\tilde{\mathbf{B}}_l^k = \mathbf{B}(\mathbf{x}^k) + (x_1-x_1^k)\left(\frac{\partial}{\partial x_1}\mathbf{B}\right)\bigg|_{x=x^k} + (x_2-x_2^k)\left(\frac{\partial}{\partial x_2}\mathbf{B}\right)\bigg|_{x=x^k} + (x_3-x_3^k)\left(\frac{\partial}{\partial x_3}\mathbf{B}\right)\bigg|_{x=x^k} \tag{6.11}$$

Substitute Eq. (6.11) into Eq. (6.9) to yield

$$\mathbf{K} = \mathbf{K}_0 + \mathbf{K}_I + \mathbf{K}_{II} \tag{6.12}$$

where

$$\mathbf{K}_0 = \sum_{k=1}^{N}\mathbf{M}_{000}^k(\mathbf{B}^T\mathbf{C}\mathbf{B})|_{x=x^k} \tag{6.13a}$$

$$\begin{aligned}\mathbf{K}_I = \sum_{k=1}^{N}[&\mathbf{M}_{100}^k(\mathbf{B}_{,x_1}^T\mathbf{C}\mathbf{B} + \mathbf{B}^T\mathbf{C}\mathbf{B}_{,x_1})|_{x=x^k} + \mathbf{M}_{010}^k(\mathbf{B}_{,x_2}^T\mathbf{C}\mathbf{B} + \mathbf{B}^T\mathbf{C}\mathbf{B}_{,x_2})|_{x=x^k}\\ &+ \mathbf{M}_{001}^k(\mathbf{B}_{,x_3}^T\mathbf{C}\mathbf{B} + \mathbf{B}^T\mathbf{C}\mathbf{B}_{,x_3})|_{x=x^k}]\end{aligned} \tag{6.13b}$$

$$\begin{aligned}\mathbf{K}_{II} = \sum_{k=1}^{N}[&\mathbf{M}_{200}^k(\mathbf{B}_{,x_1}^T\mathbf{C}\mathbf{B}_{,x_1})|_{x=x^k} + \mathbf{M}_{020}^k(\mathbf{B}_{,x_2}^T\mathbf{C}\mathbf{B}_{,x_2})|_{x=x^k} + \mathbf{M}_{002}^k(\mathbf{B}_{,x_3}^T\mathbf{C}\mathbf{B}_{,x_3})|_{x=x^k}\\ &+ \mathbf{M}_{110}^k(\mathbf{B}_{,x_1}^T\mathbf{C}\mathbf{B}_{,x_2} + \mathbf{B}_{,x_2}^T\mathbf{C}\mathbf{B}_{,x_1})|_{x=x^k} + \mathbf{M}_{101}^k(\mathbf{B}_{,x_1}^T\mathbf{C}\mathbf{B}_{,x_3} + \mathbf{B}_{,x_3}^T\mathbf{C}\mathbf{B}_{,x_1})|_{x=x^k}\\ &+ \mathbf{M}_{011}^k(\mathbf{B}_{,x_2}^T\mathbf{C}\mathbf{B}_{,x_3} + \mathbf{B}_{,x_3}^T\mathbf{C}\mathbf{B}_{,x_2})|_{x=x^k}]\end{aligned} \tag{6.13c}$$

$$\mathbf{B}_{,x_i} \equiv \frac{\partial}{\partial x_i}\mathbf{B} \tag{6.14}$$

The term $\mathbf{K}_0$ in Eq. (6.12) is equivalent to uniformly reduced integration of $\mathbf{K}$ and $\mathbf{K}_I$ and $\mathbf{K}_{II}$ are the stabilization matrices.

The computational algorithm for constructing a stiffness matrix using the stabilized particle integration method is listed as follows:

1. Loop on the sub-domain $k = 1, N$.
2. Compute the sub-domain centroid $\mathbf{x}^k$ and determine ΔV_j according to Table 6.1.
3. Integrate the moments $\mathbf{M}^k_{rst}$ according to Eq. (6.8).
4. compute derivatives of the displacement interpolation functions at $\mathbf{x}^k$.
5. Form $\mathbf{B}|_{x=x^k}$ and $\mathbf{B}_{,x_i}|_{x=x^k}$.
6. Form $\mathbf{K}_0$, $\mathbf{K}_I$, $\mathbf{K}_{II}$ and construct the stiffness matrix of sub-domain Ω^k according to Eqs. (6.12)-(6.13c) and assemble into the global matrix.

6.3 Extension to Nonlinear and Dynamic Problems

The stabilized particle integration for the stiffness matrix and internal force vector along with the definition of a mass matrix is given below:

$$\mathbf{K} = \mathbf{K}^G + \mathbf{K}^M \tag{6.15}$$

$$\mathbf{K}^G = \mathbf{I}_1(\mathbf{B}^T\mathbf{H}\mathbf{B}) \tag{6.16}$$

$$\mathbf{K}^M = \mathbf{I}_1(\mathbf{B}^T\mathbf{C}\mathbf{B}) \tag{6.17}$$

$$\Theta = \mathbf{I}_1(\rho\mathbf{N}\mathbf{N}^T) \tag{6.18}$$

$$\mathbf{f}^{\text{int}} = \mathbf{I}_1(\mathbf{B}^T\boldsymbol{\Xi}) \tag{6.19}$$

where $\mathbf{K}^M$, $\mathbf{K}^G$, Θ and $\mathbf{f}^{\text{int}}$ are the material parts of the stiffness matrix, geometric part of the stiffness matrix, mass matrix and internal force vector, respectively. Matrices $\mathbf{H}$, $\mathbf{C}$, $\mathbf{N}$ and $\boldsymbol{\Xi}$ are the initial stress matrix, material response matrix, RKPM interpolation function matrix, and stress vector, respectively. The particle integration of $\mathbf{K}^M$, $\mathbf{K}^G$, $\boldsymbol{\Theta}$ and $\mathbf{f}^{\text{int}}$ is derived as follows:

$$\mathbf{K}^M = \mathbf{I}_1(\mathbf{B}^T\mathbf{C}\mathbf{B}) = \sum_{k=1}^{N}\mathbf{I}(\tilde{\mathbf{B}}_1^{k^T}\mathbf{C}\tilde{\mathbf{B}}_1^k) = \mathbf{K}_0^M + \mathbf{K}_I^M + \mathbf{K}_{II}^M \tag{6.20}$$

$$\mathbf{K}^G = \mathbf{I}_1(\mathbf{B}^T\mathbf{H}\mathbf{B}) = \sum_{k=1}^{N}\mathbf{I}(\tilde{\mathbf{B}}_1^{k^T}\mathbf{H}\tilde{\mathbf{B}}_1^k) = \mathbf{K}_0^G + \mathbf{K}_I^G + \mathbf{K}_{II}^G \tag{6.21}$$

$$\Theta = \mathbf{I}_1(\rho\mathbf{N}\mathbf{N}^T) = \sum_{k=1}^{N}\mathbf{I}(\rho\tilde{\mathbf{N}}_1^k\tilde{\mathbf{N}}_1^{k^T}) = \Theta_0 + \Theta_I + \Theta_{II} \tag{6.22}$$

$$\mathbf{f}^{\text{int}} = \mathbf{I}_1(\mathbf{B}^T\boldsymbol{\Xi}) = \sum_{k=1}^{N}\mathbf{I}(\tilde{\mathbf{B}}_1^{k^T}\boldsymbol{\Xi}) = \mathbf{f}_0^{\text{int}} + \mathbf{f}_I^{\text{int}} \tag{6.23}$$

$$\tilde{\mathbf{N}}_1^k = \mathbf{N}(\mathbf{x}^k) + (x_1 - x_1^k)\left(\frac{\partial}{\partial x_1}\mathbf{N}\right)\bigg|_{x=x^k} + (x_2 - x_2^k)\left(\frac{\partial}{\partial x_2}\mathbf{N}\right)\bigg|_{x=x^k} + (x_3 - x_3^k)\left(\frac{\partial}{\partial x_3}\mathbf{N}\right)\bigg|_{x=x^k} \tag{6.24}$$

where $\mathbf{K}_0^M$, $\mathbf{K}_0^G$, Θ_0 and $\mathbf{f}_0^{\text{int}}$ are equivalent to the uniformly reduced integration of $\mathbf{K}^M$, $\mathbf{K}^G$, Θ and $\mathbf{f}^{\text{int}}$, respectively. Their explicit expressions are listed in Table 6.2. The matrices $\mathbf{K}_I^M$ and $\mathbf{K}_{II}^M$, $\mathbf{K}_I^G$ and $\mathbf{K}_I^G$ and $\mathbf{K}_{II}^G$, Θ_I and Θ_{II} and $\mathbf{f}_I^{\text{int}}$ are the stabilization matrices of $\mathbf{K}^M$, $\mathbf{K}^G$, Θ and $\mathbf{f}^{\text{int}}$, respectively. Their explicit expressions are listed in Table 6.3.

	Reduced Particle Integration	Integration of $\bar{\mathbf{C}}^k_{rst}$, $\bar{\mathbf{S}}^k_{rst}$, $\bar{\mathbf{R}}^k_{rst}$, $\bar{\mathbf{\Xi}}^k_{rst}$
$\mathbf{K}_0^M$	$\sum_{k=1}^{N}[\mathbf{B}^T\vert_{x=x_k}\bar{\mathbf{C}}^k_{000}\mathbf{B}\vert_{x=x_k}]$	$\bar{\mathbf{C}}^k_{rst} = \sum_{j=1}^{NPk}\mathbf{C}(\mathbf{x}^j)(x_1^j-x_1^k)^r(x_2^j-x_2^k)^s(x_3^j-x_3^k)^t\Delta V_j$
$\mathbf{K}_0^G$	$\sum_{k=1}^{N}[\mathbf{B}^T\vert_{x=x_k}\bar{\mathbf{H}}^k_{000}\mathbf{B}\vert_{x=x_k}]$	$\bar{\mathbf{H}}^k_{rst} = \sum_{j=1}^{NPk}\mathbf{H}(\mathbf{x}^j)(x_1^j-x_1^k)^r(x_2^j-x_2^k)^s(x_3^j-x_3^k)^t\Delta V_j$
Θ_0	$\sum_{k=1}^{N}[\bar{\mathbf{R}}^k_{000}(NN^T)\vert_{x=x_k}]$	$\bar{\mathbf{R}}^k_{rst} = \sum_{j=1}^{NPk}\boldsymbol{\rho}(\mathbf{x}^j)(x_1^j-x_1^k)^r(x_2^j-x_2^k)^s(x_3^j-x_3^k)^t\Delta V_j$
$\mathbf{f}_0^{\text{int}}$	$\sum_{k=1}^{N}[\mathbf{B}^T\vert_{x=x_k}\bar{\mathbf{\Xi}}^k_{000}]$	$\bar{\mathbf{\Xi}}^k_{rst} = \sum_{j=1}^{NPk}\mathbf{\Xi}(\mathbf{x}^j)(x_1^j-x_1^k)^r(x_2^j-x_2^k)^s(x_3^j-x_3^k)^t\Delta V_j$

Table 6.2. Explicit expressions of $\mathbf{K}_0^M$, $\mathbf{K}_0^G$, Θ_0 and $\mathbf{f}_0^{\text{int}}$

Remark:

If higher order RKPM displacement interpolation functions are used, higher order approximations of $\mathbf{B}$ and $\mathbf{N}$ are required. The same principle applies to linear problems.

7. KERNEL PARTICLE FINITE ELEMENTS

The concept of RKPM is also applicable to the development of higher order finite elements. The finite element shape functions can be constructed by using the kernel function at each node in conjunction with the correction function.

7.1 Element Shape Functions

The element shape function can be constructed by considering the element domain as an RKPM domain. We begin with the development of one-dimensional kernel particle element shape functions. The mapping of a one dimensional N-node element from global Cartesian coordinates to element natural coordinates is shown in Figure 7.1.

In each one-dimensional element, the element reproducing equation in natural coordinates can be written as:

$$u^R(\xi) = \int_{-1}^{1}\mathbf{\Phi}_a(\xi-s)C(\xi;\xi-s)u(s)ds \tag{7.1}$$

where $\mathbf{\Phi}_a(\xi-s)$ and $C(\xi;\xi-s)$ are the window function and correction function expressed in natural coordinates. The correction function $C(\xi;\xi-s)$ is constructed on the element level by following the procedures discussed in Section 2 to guarantee consistency.

The discretization of the element reproducing equation, Eq. (7.1), leads to:

$$u^R(\xi) = \sum_{I=1}^{N}\boldsymbol{\phi}_a(\xi-\xi_I)C(\xi;\xi-\xi_I)u_I\Delta\xi_I \equiv \sum_{I=1}^{N}\tilde{N}_I^{[N]}(\xi)u_I \tag{7.2}$$

where $\tilde{N}_I^{[N]}$ is the kernel particle shape function of an N-node element,

$$\tilde{N}_I^{[N]}(\xi) = \mathbf{\Phi}_a(\xi-\xi_I)C(\xi;\xi-\xi_I)\Delta\xi_I \tag{7.3}$$

	Stabilization matrixes of Particle Integration
$\mathbf{K}_I^M$	$\sum_{k=1}^{N}[\mathbf{B}_{,x_1}^T\vert_{x=x^k}\bar{\mathbf{C}}_{100}^k\mathbf{B}\vert_{x=x^k}+\mathbf{B}^T\vert_{x=x^k}\bar{\mathbf{C}}_{100}^k\mathbf{B}_{,x_1}\vert_{x=x^k}+\mathbf{B}_{,x_2}^T\vert_{x=x^k}\bar{\mathbf{C}}_{010}^k\mathbf{B}\vert_{x=x^k}+$ $+\mathbf{B}^T\vert_{x=x^k}\bar{\mathbf{C}}_{010}^k\mathbf{B}_{,x_2}\vert_{x=x^k}+\mathbf{B}_{,x_3}^T\vert_{x=x^k}\bar{\mathbf{C}}_{001}^k\mathbf{B}\vert_{x=x^k}+\mathbf{B}^T\vert_{x=x^k}\bar{\mathbf{C}}_{001}^k\mathbf{B}_{,x_3}\vert_{x=x^k}]$
$\mathbf{K}_{II}^M$	$\sum_{k=1}^{N}[\mathbf{B}_{,x_1}^T\vert_{x=x^k}\bar{\mathbf{C}}_{200}^k\mathbf{B}_{,x_1}\vert_{x=x^k}+\mathbf{B}_{,x_2}^T\vert_{x=x^k}\bar{\mathbf{C}}_{020}^k\mathbf{B}_{,x_2}\vert_{x=x^k}+\mathbf{B}_{,x_3}^T\vert_{x=x^k}\bar{\mathbf{C}}_{002}^k\mathbf{B}_{,x_3}\vert_{x=x^k}+$ $+\mathbf{B}_{,x_1}^T\vert_{x=x^k}\bar{\mathbf{C}}_{110}^k\mathbf{B}_{,x_2}\vert_{x=x^k}+\mathbf{B}_{,x_2}^T\vert_{x=x^k}\bar{\mathbf{C}}_{110}^k\mathbf{B}_{,x_1}\vert_{x=x^k}+\mathbf{B}_{,x_1}^T\vert_{x=x^k}\bar{\mathbf{C}}_{101}^k\mathbf{B}_{,x_3}\vert_{x=x^k}+$ $+\mathbf{B}_{,x_3}^T\vert_{x=x^k}\bar{\mathbf{C}}_{101}^k\mathbf{B}_{,x_1}\vert_{x=x^k}+\mathbf{B}_{,x_2}^T\vert_{x=x^k}\bar{\mathbf{C}}_{011}^k\mathbf{B}_{,x_3}\vert_{x=x^k}+\mathbf{B}_{,x_3}^T\vert_{x=x^k}\bar{\mathbf{C}}_{011}^k\mathbf{B}_{,x_2}\vert_{x=x^k}]$
$\mathbf{K}_I^G$	$\sum_{k=1}^{N}[\mathbf{B}_{,x_1}^T\vert_{x=x^k}\bar{\mathbf{H}}_{100}^k\mathbf{B}\vert_{x=x^k}+\mathbf{B}^T\vert_{x=x^k}\bar{\mathbf{H}}_{100}^k\mathbf{B}_{,x_1}\vert_{x=x^k}+\mathbf{B}_{,x_2}^T\vert_{x=x^k}\bar{\mathbf{H}}_{010}^k\mathbf{B}\vert_{x=x^k}+$ $=\mathbf{B}^T\vert_{x=x^k}\bar{\mathbf{H}}_{010}^k\mathbf{B}_{,x_2}\vert_{x=x^k}+\mathbf{B}_{,x_3}^T\vert_{x=x^k}\bar{\mathbf{H}}_{001}^k\mathbf{B}\vert_{x=x^k}+\mathbf{B}^T\vert_{x=x^k}\bar{\mathbf{H}}_{001}^k\mathbf{B}_{,x_3}\vert_{x=x^k}]$
$\mathbf{K}_{II}^G$	$\sum_{k=1}^{N}[\mathbf{B}_{,x_1}^T\vert_{x=x^k}\bar{\mathbf{H}}_{200}^k\mathbf{B}_{,x_1}\vert_{x=x^k}+\mathbf{B}_{,x_2}^T\vert_{x=x^k}\bar{\mathbf{H}}_{020}^k\mathbf{B}_{,x_2}\vert_{x=x^k}+\mathbf{B}_{,x_3}^T\vert_{x=x^k}\bar{\mathbf{H}}_{002}^k\mathbf{B}_{,x_3}\vert_{x=x^k}+$ $+\mathbf{B}_{,x_1}^T\vert_{x=x^k}\bar{\mathbf{H}}_{110}^k\mathbf{B}_{,x_2}\vert_{x=x^k}+\mathbf{B}_{,x_2}^T\vert_{x=x^k}\bar{\mathbf{H}}_{110}^k\mathbf{B}_{,x_1}\vert_{x=x^k}+\mathbf{B}_{,x_1}^T\vert_{x=x^k}\bar{\mathbf{H}}_{101}^k\mathbf{B}_{,x_3}\vert_{x=x^k}+$ $+\mathbf{B}_{,x_3}^T\vert_{x=x^k}\bar{\mathbf{H}}_{101}^k\mathbf{B}_{,x_1}\vert_{x=x^k}+\mathbf{B}_{,x_2}^T\vert_{x=x^k}\bar{\mathbf{H}}_{011}^k\mathbf{B}_{,x_3}\vert_{x=x^k}+\mathbf{B}_{,x_3}^T\vert_{x=x^k}\bar{\mathbf{H}}_{111}^k\mathbf{B}_{,x_2}\vert_{x=x^k}]$
Θ_I	$\sum_{k=1}^{N}[\bar{\mathbf{R}}_{100}^k(\mathbf{N}\mathbf{N}_{,x_1}^T\vert_{x=x^k}+\mathbf{N}_{,x_1}\mathbf{N}^T)+\bar{\mathbf{R}}_{010}^k(\mathbf{N}\mathbf{N}_{,x_2}^T\vert_{x=x^k}+\mathbf{N}_{,x_2}\mathbf{N}^T)+$ $+\bar{\mathbf{R}}_{001}^k(\mathbf{N}\mathbf{N}_{,x_3}^T\vert_{x=x^k}+\mathbf{N}_{,x_3}\mathbf{N}^T)]$
Θ_{II}	$\sum_{k=1}^{N}[\bar{\mathbf{R}}_{200}^k(\mathbf{N}_{,x_1}\mathbf{N}_{,x_1}^T)\vert_{x=x^k}+\bar{\mathbf{R}}_{020}^k(\mathbf{N}_{,x_2}\mathbf{N}_{,x_2}^T)\vert_{x=x^k}+\bar{\mathbf{R}}_{002}^k(\mathbf{N}_{,x_3}\mathbf{N}_{,x_3}^T)\vert_{x=x^k}+$ $+\bar{\mathbf{R}}_{110}^k(\mathbf{N}_{,x_1}\mathbf{N}_{,x_2}^T+\mathbf{N}_{,x_2}\mathbf{N}_{,x_1}^T)\vert_{x=x^k}+[\bar{\mathbf{R}}_{101}^k(\mathbf{N}_{,x_1}\mathbf{N}_{,x_3}^T+\mathbf{N}_{,x_3}\mathbf{N}_{,x_1}^T)\vert_{x=x^k}+$ $+\bar{\mathbf{R}}_{011}^k(\mathbf{N}_{,x_2}\mathbf{N}_{,x_3}^T+\mathbf{N}_{,x_3}\mathbf{N}_{,x_2}^T)\vert_{x=x^k}]+$
$\mathbf{f}_I^{\text{int}}$	$\sum_{k=1}^{N}[\mathbf{B}_{,x_1}^T\vert_{x=x^k}\bar{\mathbf{\Xi}}_{100}^k+\mathbf{B}_{,x_2}^T\vert_{x=x^k}\bar{\mathbf{\Xi}}_{010}^k+\mathbf{B}_{,x_3}^T\vert_{x=x^k}\bar{\mathbf{\Xi}}_{001}^k]$

Table 6.3. Explicit expressions of $\mathbf{K}_I^M$, $\mathbf{K}_{II}^M$, $\mathbf{K}_I^G$, $\mathbf{K}_{II}^G$, Θ_I, Θ_{II} and $\mathbf{f}_I^{\text{int}}$

The extension to multi-dimensional element shape functions can be done simply by taking product of one-dimensional shape functions. In a two-dimensional element, the mapping from global Cartesian coordinates to natural coordinates is shown in Figure 7.2.

For a two-dimensional $N1 \times N2$-node element, the shape function, $\tilde{N}^{[N1\times N2]}(\xi,\eta)$ is obtained by:

$$\tilde{N}_I^{[N1\times N2]}(\xi,\eta)=\tilde{N}_I^{[N1]}(\xi)\tilde{N}_I^{[N2]}(\eta) \tag{7.4}$$

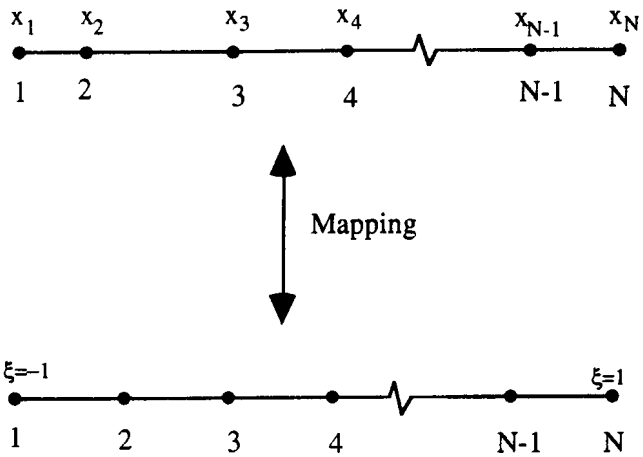


Figure 7.1. One-dimensional mapping from cartesian coordinates to natural coordinates

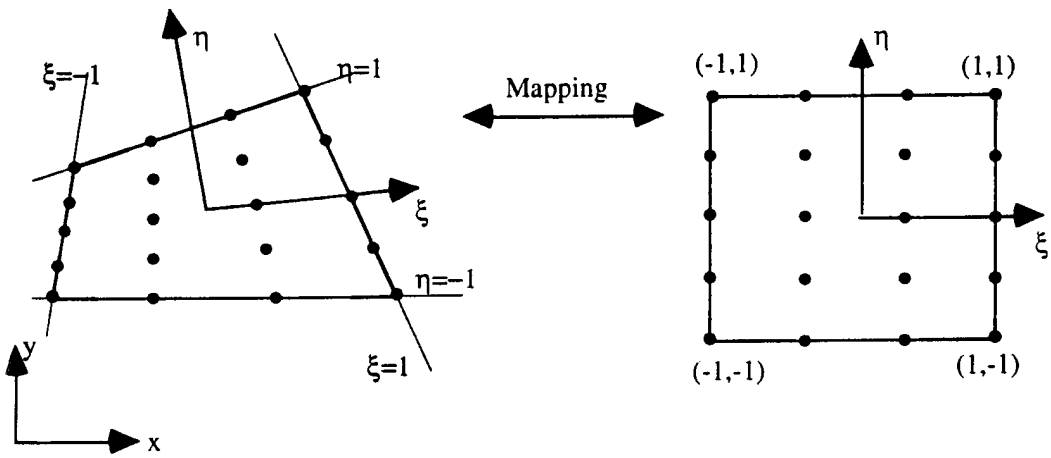


Figure 7.2. Two-dimensional mapping from cartesian coordinates to natural coordinates

where $\tilde{N}_I^{[N]}(\xi)$ is the shape function of the one-dimensional N-node element. Similarly, for a three-dimensional $N1 \times N2 \times N3$-node element, the shape function is:

$$\tilde{N}_I^{[N1\times N2\times N3]}(\xi,\eta,\zeta) = \tilde{N}_I^{[N1]}(\xi)\tilde{N}_I^{[N2]}(\eta)\tilde{N}_I^{[N3]}(\zeta) \tag{7.5}$$

The continuity of shape function across element boundaries can be examined by evaluating u along the element boundary, say at $\xi = 1$

$$u(1,\eta,\zeta) = \sum_{I=1}^{[N1\times N2\times N3]} \tilde{N}_I^{[N1\times N2\times N3]}(1,\eta,\zeta)u_I \tag{7.6}$$

Since $N^{[N1\times N2\times N3]}(\xi,\eta,\zeta)$ does not possess Kronecker delta properties, the value of u on the element boundary is dependent on the all u_I's and all the kernel functions defined inside the element domain. This leads to the situation where the distribution of u on element boundary is not continuous across element boundary. Therefore, it is desired to develop a kernel particle finite element shape function that satisfies continuity conditions across element boundaries.

7.2 Kernel Particle Finite Elements

Consider a function $u(\xi)$ that is interpolated by a set of interpolation functions $\{F_1(\xi), F_2(\xi), \ldots, F_N(\xi)\}$ in the following equation:

$$u(\xi) = \sum_{I=1}^{N} F_I(\xi) a_I \tag{7.7}$$

The function $u(\xi)$, evaluated at the sampling point ξ_J is obtained as:

$$u(\xi_J) = \sum_{I=1}^{N} F_I(\xi_J) a_I \tag{7.8}$$

or

$$u(\xi_J) = \sum_{I=1}^{N} T_{JI}(\xi_J) a_I \tag{7.9}$$

where

$$T_{JI} = F_I(\xi_J) \tag{7.10}$$

If the number of sampling points is selected to be the same as the number of interpolation functions, then $a_I = T_{JI}^{-1} u(\xi_J)$ and

$$u(\xi) = \sum_{I=1}^{N} F_I(\xi) a_I = \sum_{I=1}^{N} \sum_{J=1}^{N} F_I(\xi) T_{JI}^{-1} u(\xi_I) \equiv \sum_{I=1}^{N} G_I(\xi) u(\xi_I) \tag{7.11}$$

where

$$G_I(\xi) = \sum_{J=1}^{N} T_{IJ}^{-1} F_J(\xi) \tag{7.12}$$

Note that $G_I(\xi_J) = \delta_{IJ}$. We now construct a new shape function $N_I^{[N]}(\xi)$ defined by:

$$N_I^{[N]}(\xi) = \sum_{J=1}^{NEN} \tilde{T}_{IJ}^{-1} \tilde{N}_J^{[N]}(\xi) \tag{7.13}$$

where

$$\tilde{T}_{IJ} = \tilde{N}_J^{[N]}(\xi_I) \tag{7.14}$$

and $\tilde{N}_I^{[N]}(\xi_J) = \delta_{IJ}$. The multi-dimensional shape function of a $N1 \times N2 \times N3$-node element is simply constructed by using the product rule

$$N_I^{[N1 \times N2 \times N3]}(\xi, \eta, \zeta) = N_I^{[N1]}(\xi) N_I^{[N2]}(\eta) N_I^{[N3]}(\zeta) \tag{7.15}$$

The continuity condition is now verified by examining the distribution of u on element boundary at $\xi = 1$

$$u(1, \eta, \zeta) = \sum_{I=1}^{N1 \times N2 \times N3} N_I^{[N1 \times N2 \times N3]}(1, \eta, \zeta) u(1, \eta_I, \zeta_I) \tag{7.16}$$

Due to the Kronecker delta properties of the shape function, Eq. (7.16) can be reduced to:

$$u(1,\eta,\zeta) = \sum_{I=1}^{N2\times N3} N_1^{[N1\times N2\times N3]}(1,\eta,\zeta)u(1,\eta_I,\zeta_I) \tag{7.17}$$

Eq. (7.17) states that the distribution of u on an element boundary is dependent only on the shape functions and nodal values of u associated with the nodes along the same element boundary. Clearly, this kernel particle finite element shape function satisfies continuity.

This kernel finite element shape function is smooth inside the element domain if the transformation matrix $\tilde{\mathbf{T}}$ is invertible. Furthermore, the shape functions are complete if the correction functions are constructed according to the reproducing conditions. All of these properties ensure the convergence of the kernel particle finite element solutions [Hughes (1987)].

7.3 Application to Incompressible Problems

For constraint problems such as incompressible hyperelasticity, the pressure projection method proposed by Chen *et al.* (1995) and Chen and Pan (1995) can be incorporated into this kernel particle finite element formulation to avoid volumetric locking and pressure oscillations. The pressure calculated from displacement (interpolated from the displacement shape functions developed above) is projected onto the appropriate pressure field at the element level by minimizing the pressure norm as follows:

Consider a problem of approximating a square integrable function $P = \partial W/\partial J$ at the element level, in a least-squares sense, by a linear combination of a sequence of functions $\{Q_1(x), Q_2(x), \ldots, Q_n(x)\}$ in $L^2(\Omega^e)$. That is, choose $\mathbf{p} = [p_1, p_2, \ldots, p_n]^T$ to minimize

$$\Phi(\mathbf{p}) = \|\frac{\partial W}{\partial J} - \mathbf{Q}\mathbf{p}\|^2_{L_2(\Omega^e_X)} \tag{7.18}$$

where $\|\cdot\|^2_{L_2}(\Omega^e)$ is the L_2 norm in the element domain Ω^e, W is the strain energy density function, and

$$\begin{aligned} \mathbf{Q}(x) &= [Q_1(x), Q_2(x), \ldots, Q_n(x)] \\ J &= \det(\mathbf{F}) \end{aligned} \tag{7.19}$$

where $\mathbf{F}$ is the deformation gradient calculated from the RKPM displacements. Similarly, for nonlinear problems, we consider a least-squares approximation of the hydrostatic pressure increment $\Delta P = \partial^2 W/\partial J^2 \; JG^{-1}_{kl}\Delta E_{kl}$ by a linear combination of a sequence of functions $\{\tilde{Q}_1(x), \tilde{Q}_2(x), \ldots, \tilde{Q}_n(x)\}$ in $L^2(\Omega^e)$. Determine $\Delta\mathbf{p} = [\Delta p_1, \Delta p_2, \ldots, \Delta p_n]^T$ to minimize

$$\Psi(\Delta\mathbf{p}) = \|\frac{\partial^2 W}{\partial J^2} JG^{-1}_{kl}\Delta E_{kl} - \tilde{\mathbf{Q}}\Delta\mathbf{p}\|^2_{L_2(\Omega^e_X)} \tag{7.20}$$

where

$$\tilde{\mathbf{Q}}(x) = [Q_1(x), Q_2(x), \ldots, Q_n(x)] \tag{7.21}$$

Brezzi-Babuška conditions [Babuška (1973), Brezzi (1974)] need to be used to determine the appropriate pressure fields according to the order of the displacement shape functions.

8. REPRODUCING KERNEL DOMAIN DECOMPOSITION PARTICLE METHODS

As discussed in section 5.3, the treatment of boundary conditions in RKPM using the transformation method involves inversion of the transformation matrix $\mathbf{T}$ at a global level. The efficiency of this method can be improved by introducing a Reproducing Kernel Domain Decomposition Particle Method (RKDDPM).

Consider that the domain of interest is partitioned into a union of sub-domains: then, the displacement reproducing equation can be decomposed into the following equation

$$u^{Ra}(\mathbf{x}) = \sum_{e=1}^{ND} \int_{\Omega^e} u(\mathbf{y}) \bar{\mathbf{\Phi}}_a^e(\mathbf{x};\mathbf{x}-\mathbf{y}) d\mathbf{y} \tag{8.1}$$

where ND is the number of decomposed sub-domains. In each sub-domain, or cell, the support of the modified kernel function does not extend beyond the boundaries and

$$\bar{\mathbf{\Phi}}_a^e(\mathbf{x};\mathbf{x}-\mathbf{y}) = \mathbf{\Phi}_a(\mathbf{x}-\mathbf{y}) C^e(\mathbf{x};\mathbf{x}-\mathbf{y}) \tag{8.2}$$

where $\bar{\mathbf{\Phi}}_a^e(\mathbf{x};\mathbf{x}-\mathbf{y})$ and $C^e(\mathbf{x};\mathbf{x}-\mathbf{y})$ are the modified kernel function and correction function of cell e, respectively. The correction function is defined by:

$$C^e(\mathbf{x};\mathbf{x}-\mathbf{y}) = \sum_{i=0}^{N} b_i^e(\mathbf{x}) \|\mathbf{x}-\mathbf{y}\|^i \tag{8.3}$$

and the $b_i^e(\mathbf{x})$'s are obtained from the cell level reproducing conditions:

$$\mathbf{M}^e(x)\mathbf{b}^e(x) = \mathbf{g} \tag{8.4}$$

where

$$\mathbf{M}^e(x) = \begin{bmatrix} m_{(0)}^e(x) & m_{(1)}^e(x) & \dots & m_{(N)}^e(x) \\ m_{(1)}^e(x) & m_{(2)}^e(x) & \dots & m_{(N+1)}^e(x) \\ \vdots & \vdots & & \vdots \\ m_{(N)}^e(x) & m_{(N+1)}^e(x) & \dots & m_{(2N)}^e(x) \end{bmatrix} \tag{8.5}$$

$$\mathbf{b}^e(\mathbf{x}) = \begin{bmatrix} b_1^e(\mathbf{x}) \\ b_2^e(\mathbf{x}) \\ \vdots \\ b_N^e(\mathbf{x}) \end{bmatrix} \qquad \mathbf{g} = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \tag{8.6}$$

The cell moments are defined by:

$$\mathbf{m}_{(i)}^e(\mathbf{x}) = \int_{\Omega^e} \|\mathbf{x}-\mathbf{y}\|^i \Phi(\mathbf{x}-\mathbf{y}) d\mathbf{y} \tag{8.7}$$

The discretization of Eq. (8.1) gives:

$$u^h(\mathbf{x}) = \sum_{e=1}^{ND} \sum_{I=1}^{NPe} \bar{\mathbf{\Phi}}_a^e(\mathbf{x};\mathbf{x}-\mathbf{x}_I) \Delta V_I^e u_I^e \tag{8.8}$$

$$\equiv \sum_{e=1}^{ND} u^{h^e}(\mathbf{x}) \tag{8.9}$$

where NPe is the number of points located inside cell e and $u^{h^e}(\mathbf{x})$ is defined by:

$$u^{h^e}(\mathbf{x}) = \sum_{I=1}^{NPe} \bar{\mathbf{\Phi}}_a^e(\mathbf{x};\mathbf{x}-\mathbf{x}_1)\Delta V_I^e u_I^e = \sum_{I=1}^{NPe} \tilde{N}_I^e(\mathbf{x}) u_I^e \tag{8.10}$$

where

$$\tilde{N}_I^e(\mathbf{x}) = \bar{\mathbf{\Phi}}_a^e u(\mathbf{x};\mathbf{x}-\mathbf{x}_I)\Delta V_I^e \tag{8.11}$$

Note that $\tilde{N}_I^e(\mathbf{x})$ does not possess the Kronecker delta property, i.e., $\tilde{N}_I^e(\mathbf{x}_J) \neq \delta_{IJ}$. Using Eq.(8.10), one can establish the relation between the nodal reproduced displacement $u^h(\mathbf{x}_I)$ and u_I by:

$$u_I^{h^e} \equiv u^{h^e}(\mathbf{x}_I) = T_{IJ}^e u_J^e \tag{8.12}$$

or

$$u_I^e = T_{IJ}^{e^{-1}} u_J^{h^e} \tag{8.13}$$

where

$$T_{IJ}^e \tilde{N}_J^e(\mathbf{x}_I) \tag{8.14}$$

Due to the construction of the modified cell kernel function $\bar{\mathbf{\Phi}}_a^e(\mathbf{x};\mathbf{x}-\mathbf{y})$, the transformation matrix T_{IJ}^e can be inverted at the cell level. Based on the transformation in Eq. (8.12), a displacement interpolation function that possesses the Kronecker delta property can be developed:

$$u^{h^e}(\mathbf{x}) = \sum_{J=1}^{NPe} \left(\sum_{I=1}^{NPe} \tilde{N}_I^e(\mathbf{x}) T_{JI}^{e^{-1}} \right) u_J^{h^e} = \sum_{I=1}^{NPe} \tilde{N}_I^e(\mathbf{x}) u_I^{h^e} \tag{8.15}$$

where

$$N_I^e(\mathbf{x}) = \sum_{J=1}^{NPe} T_{IJ}^{e^{-1}} \tilde{N}_J^e(x) \tag{8.16}$$

$$N_I^e(\mathbf{x}_J) = \delta_{IJ} \tag{8.17}$$

In this method, the displacement continuity conditions are not always maintained across cell boundaries. Special arrangement of the node locations is required to preserve continuity.

Partitioned Particle Integration Method

We restrict ourselves to a two-dimensional problem where a set of particles is generated by a triangular decomposition of the domain as illustrated in Figure 8.1. Each triangle is first partitioned into three parts. In this approach, the partition lines are defined by the connection of the centroids of the triangles. The domain associated with particle k, designated as Ω^k is defined by:

$$\Omega^k = \Omega_1^k \cup \Omega_2^k \cup \ldots \cup \Omega_M^k \tag{8.18}$$

where M is the total number of triangles surrounding of particle k. The domain Ω is then expressed as the union of the particle domains:

$$\Omega = \Omega^1 \cup \Omega^2 \cup \ldots \cup \Omega^{NP} \tag{8.19}$$

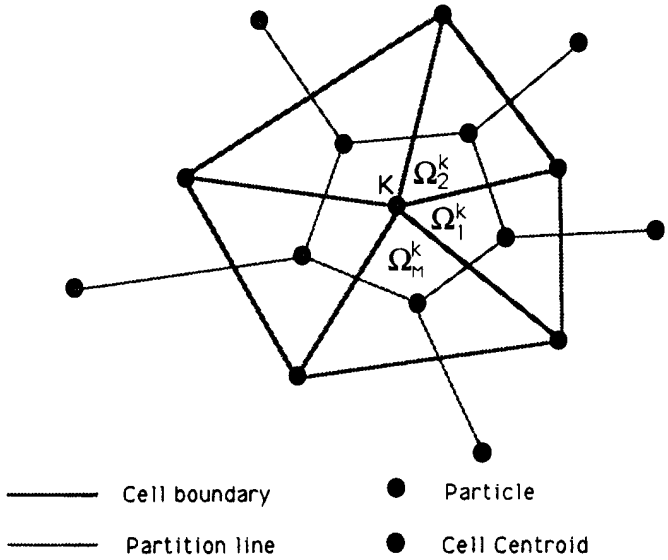


Figure 8.1. Two-dimensional geometric representation of particle domain

where NP is the total number of particles in the domain Ω. With this domain decomposition, we have

$$\mathbf{K} = \int_{\Omega} \mathbf{B}^T \mathbf{D} \mathbf{B} d\Omega = \sum_{k=1}^{NP} \int_{\Omega^k} \mathbf{B}^T \mathbf{D} \mathbf{B} d\Omega \tag{8.20}$$

A particle integration of Eq. (8.20) over the particle subdomain Ω_k can be implemented by introducing a Taylor series expansion of $\mathbf{B}$ around the particle location x^k to yield:

$$\mathbf{B} \approx \mathbf{B}\|_{x=x^k} + (x_1 - x_1^k)\mathbf{B}_{,x_1}\|_{x=x^k} + (x_2 - x_2^k)\mathbf{B}_{,x_2}\|_{x=x^k} \tag{8.21}$$

Consequently, the stiffness matrix is:

$$\begin{aligned}\mathbf{K} = \sum_{k=1}^{NP} \big[& \mathbf{M}_{00}^k (\mathbf{B}^T \mathbf{D} \mathbf{B})|_{x=x^k} + \mathbf{M}_{10}^K (\mathbf{B}_{,x_1}^T \mathbf{D} \mathbf{B} + \mathbf{B}^T \mathbf{D} \mathbf{B}_{,x_1}|_{x=x^k} + \\ & + \mathbf{M}_{01}^K (\mathbf{B}_{,x_2}^T \mathbf{D} \mathbf{B} + \mathbf{B}^T \mathbf{D} \mathbf{B}_{,x_2}|_{x=x^k} + \mathbf{M}_{20}^k (\mathbf{B}_{,x_1}^T \mathbf{D} \mathbf{B}_{,x_1})|_{x=x^k} + \\ & + \mathbf{M}_{02}^k (\mathbf{B}_{,x_2}^T \mathbf{D} \mathbf{B}_{,x_2})|_{x=x^k} + \mathbf{M}_{11}^k (\mathbf{B}_{,x_1}^T \mathbf{D} \mathbf{B}_{,x_2} + \mathbf{B}_{,x_2}^T \mathbf{D} \mathbf{B}_{,x_1}|_{x=x^k} \big]\end{aligned} \tag{8.22}$$

where

$$\mathbf{B}_{,x_i} \equiv \frac{\partial}{\partial x_i} \mathbf{B} \tag{8.23}$$

$$M_{rs}^k = \int_{\Omega^k} (x_1 - x_1^k)^r (x_2 - x_2^k)^s d\Omega \tag{8.24}$$

The particle integration method is used to calculate M_{rs}^k :

$$M_{rs}^k = \sum_{i=1}^{M} (x_1^i - x_1^k)^r (x_2^i - x_2^k) A_i^k \tag{8.25}$$

where A_i^k is the area of the sub-domain Ω_i^k.

9. MULTIRESOLUTION ANALYSIS AND hp–ADAPTIVITY

9.1 Multiresolution Analysis

This section is devoted to a review of multiresolution analysis [Beklkin *et al.*(1992), Chui (1992), Liu and Chen (1995), Mallat (1989), Strang (1989) and references therein]. The framework of a multi-resolution analysis consists of a sequence of nested closed subspaces, V_m, such that:

$$\{0\} = V_\infty \subset \ldots \subset V_1 \subset V_0 \subset V_{-1} \subset \ldots \subset V_{-\infty} = L^2(R) \tag{9.1}$$

The basis functions which generate these subspaces are scaling functions. Scaling functions are a family of functions which are generated by the translation and dilation of a single function $\phi(x)$

$$\phi_{mn}(x) = 2^{-m/2}\phi(2^{-m}x - n) \tag{9.2}$$

where m and n are the dilation and translation parameters, respectively.

As the basic characteristics of the multiresolution analysis dictates, an orthonormal wavelet basis exists for a collection of closed subspaces presented in Eq. (9.1):

$$V_{m-1} = V_m + W_m \tag{9.3}$$

where W_m is the wavelet basis which complements V_m, $m \in Z$ where

$$z = \{\ldots, -1, 0, 1, \ldots\} \tag{9.4}$$

A *Wavelet* generates the spaces W_m in the same manner as a scaling function $\phi(x)$ generates the spaces V_m. From Eq. (9.3), this family of subspaces of $L^2(R)$ yields a direct-sum decomposition of $L^2(R)$ such that:

$$L^2(R) = \ldots + (V_{-2} - V_{-1}) + (V_{-1} - V_0) + (V_0 - V_1) + \ldots = \ldots + W_{-1} + W_0 + W_1 + \ldots \tag{9.5}$$

A wavelet ψ in $L^2(R)$ is called an orthogonal wavelet if $\{\psi_{mn}\}$ satisfies:

$$\langle \psi_{mn}, \psi_{kl} \rangle = \delta_{mk}\delta_{nl} \tag{9.6}$$

where $\langle \bullet \rangle$ is the inner product operator and δ_{mk} is the Kronecker delta. If ψ is an orthogonal wavelet, then the direct sum in Eq. (9.5) becomes an orthogonal sum, so that the reconstruction formula is given by:

$$u = \sum_{m,n \in Z} \langle u, \psi_{mn} \rangle \psi_{mn}(x) \tag{9.7}$$

9.2 Multiple Scale RKPM

The reproducing kernel particle method [Liu and Oberste (1993), Liu *et al.* (1995b,1993), Liu and Chen (1994)] is given by:

$$u^h(x) = \int_\Omega C_a(x; x-y)\phi\left(\frac{x-y}{a}\right) u(y)dy \tag{9.8}$$

where $C_a(x; x-y)$ is the boundary correction term. Its dependence on the dilation parameter is implied by the subscript **a**. With the aid of the boundary correction term, the reproduction of a given function can be treated as a finite convolution of the function $u(y)$ and the kernel function $\phi(x/a)$. From the Fourier transform analysis, we note that this kernel function

serves as a low-pass filter in the reconstruction procedure. The multiple scale RKPM is defined by a family of kernel functions such that the wavelets corresponding to these kernel functions are defined by:

$$\psi_{m+1}(x) = \phi_m(x) - \phi_{m+1}(x) \tag{9.9}$$

Therefore, a multiple scale decomposition of a given response is given as:

$$\begin{aligned} u^h(x) &= u_0(x) && \text{finest scale} \\ &= w_1(x) + u_1(x) && \text{two-level decomposition} \\ &= w_1(x) + w_2(x) + u_2(x) && \text{three-level decomposition} \\ &= \sum_{i=1}^{n} w_i(x) + u_n(x) && n\text{-level decomposition} \end{aligned} \tag{9.10}$$

where

$$u_m(x) = \int_V C(2^m a_0, x, y,)\phi_m(x-y)u(y)dy \tag{9.11}$$

$$w_m(x) = \int_V \psi_m(x-y)u(y)dy \tag{9.12}$$

with

$$\psi_m(x-y) = C(2^{m-1}a_0, x, y)\phi_{m-1}(x-y) - C(2^m a_0, x, y,)\phi_m(x-y) \tag{9.13}$$

Similar to the multiresolution analysis, the multiple scale RKPM always starts from the finest scale. The subsequent levels of the response can be derived by the decomposition algorithm in Eq.(9.10). It is emphasized that all computations are performed in the *space-time domain*. However, because of the multiple scale characteristics of the *new time-frequency and space-wave number basis* for representing functions, the physical interpretation of the computed results is apparent.

9.3 Choosing Optimal Dilation Parameter

Aliasing control approach

A practical approach for choosing the dilation parameter **a** for the reproducing kernel window is to determine the interaction between the size of the given window and the sampling rate of the given nodal system in the Fourier transform domain. The Fourier transform of the reproducing kernel window function $\phi(x)$ is also a window $\hat{\phi}(x)$ in the transform domain. A large window (in function domain) filters out the fine scales (small wave numbers or high frequencies) which can be caused by sampling aliasing; consequently, oscillation in the solution may occur. A small window (in function domain), on the contrary, may introduce aliasing, but will cover wider scale/frequency bands.

hp-like adaptivity

The multiresolution concept can further be used to construct hp-adaptive algorithms. The multiple scale adaptive refinements are performed simply by inserting nodes into the highest wavelet scale solution region, and at the same time, narrowing the size of the window function to pick up the fine-scale structure of the response. This zooming in process, together with the addition of nodes, will result in an hp-like adaptive refinement algorithm. Hence, hp-like adaptive refinements can be performed without a mesh. Examples in Figure 9.1 show that RKPM almost reaches the resolution limit: $k = p/\Delta x$ and the ratio of the harmonic wave number, n, and the number of nodes (particles), p, approaches one.

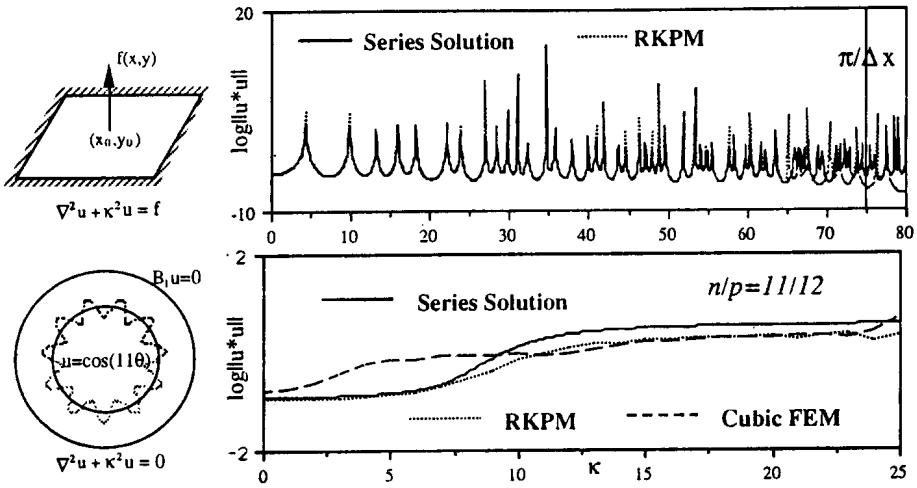


Figure 9.1. RKPM yields to the resolution limit, $k = p/\Delta x$

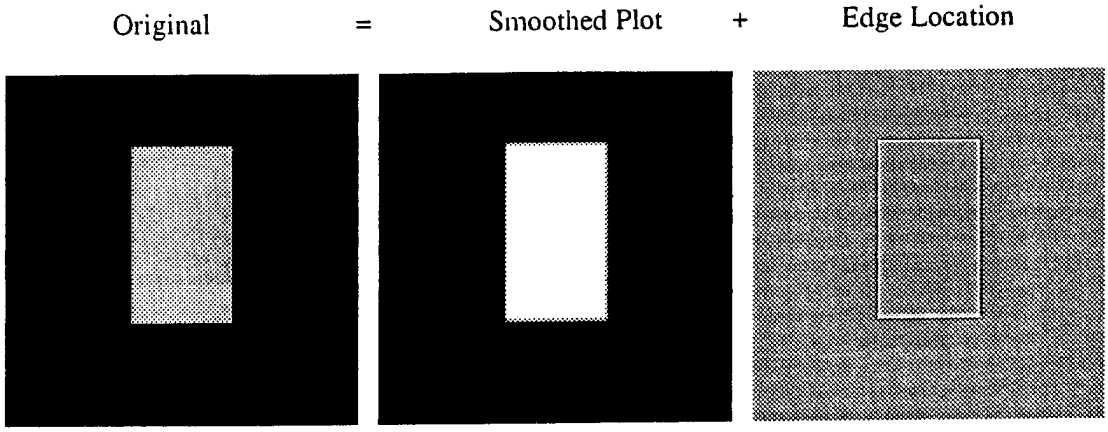


Figure 9.2. Application of RKPM in edge detection

Based on the multiple scale decomposition and the highest wavelet scale response, a convergence parameter or an error estimation indicator can be determined to locate the adaptivity regions. The adaptive refinement is accomplished by a *single p-order* (or C^∞)flexible space-scale window function that translates and dilates covering all the nodes in the computational domain.

An application of the multiresolution analysis for edge detection in the image processing is illustrated in Figure 9.2.

If we depict the gray level of the edge in a 1-D plot, it has a structure similar to a shock front (see Figure 9.3.)

Through the multiresolution analysis, the edge (shock front) can be separated into two parts: the low scale (scaling function solution) and the high scale (wavelet solution). The high scale part provides the information to locate the position of the edge (shock front) in the picture. The edge detection technique can also be used to construct hp-adaptive algorithms for computational mechanics. An example of the meshless hp-adaptivity is given in Figure 9.4. As can be seen in the figure, only a set of nodes is required for adaptivity, and there is no apprehension of compatibility problem. Through the use of the integral window transform, as given in Eqs.(2.7) and (2.10), the projected low scale solution $P_{2a}u$ is determined via aliasing control; and the peeled off high scale solution $Q_{2a}u$

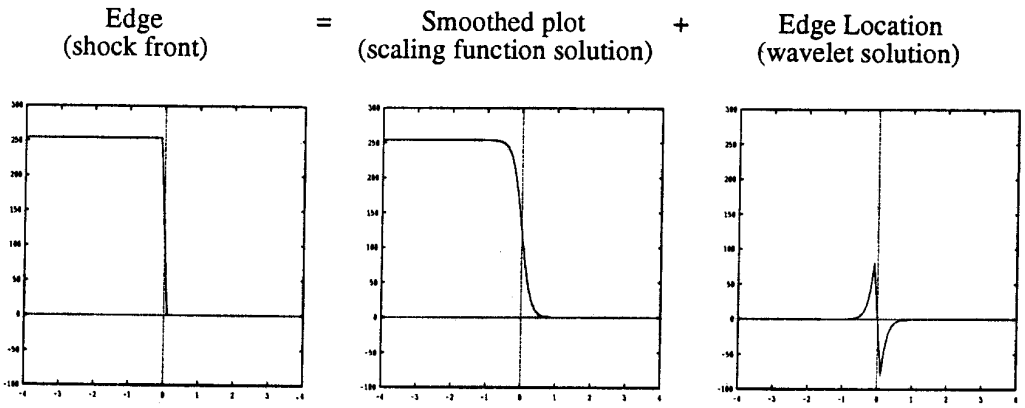


Figure 9.3. Multiresolution analysis of shock front structure

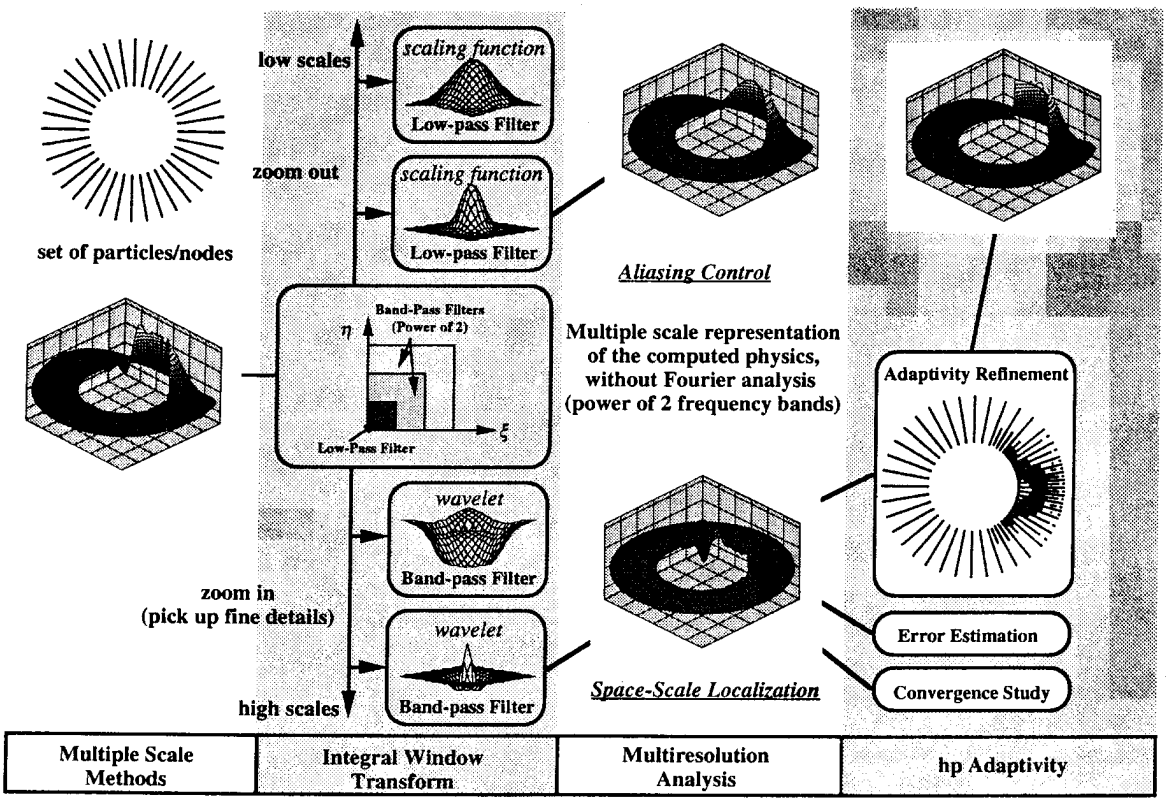


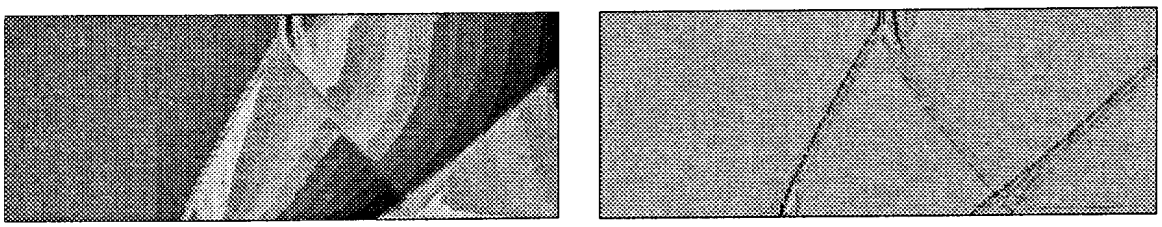
Figure 9.4. hp-like adaptivity in an acoustic problem

is obtained via a space-scale localization. In this multiresolution analysis, the quality of the solution is examined through error estimation and convergence study. In addition, physical interpretation of the computed solution can be synthesized by multiple scale representation without Fourier analysis. In this example, the multiple scales are represented by various regions covered by $\hat{\bar{\phi}}(\xi,\eta)$ and $\hat{\bar{\Psi}}(\xi,\eta)$ in the two dimensional Fourier domain (ξ,η). Based on the wavelet solution, the high gradient or non-convergent region is identified. A set of particles is then inserted into this wavelet scale region, and an hp-adaptive analysis is performed to give a finer scale solution.

9.4 Multiresolution Analysis

The compressible Euler equations are solved within three different flow regimes: subsonic, transonic, and supersonic. As an illustration, Figure 9.5 shows the results of a thin biconvex airfoil problem with Mach number $(Ma) = 1.4$. The superiority of the multiple scale RKPM decomposition is clearly illustrated since the aliasing wave numbers are removed via the multi-resolution analysis.

The multiresolution analysis via a multiple scale RKPM decomposition can be demonstrated. The total response of the high gradient shock problem is separated into two different wave number bands by the multiple scale RKPM. The low wave number plot shows the fundamental mode of the total response (Figure 9.5a). The high wave number plot (Figure 9.5b) is dominated by a shock structure. From the medium wave number plot, it can be shown that there is no significant contribution from this wave number region. Via this decomposition, the physical interpretation of the computed results can readily be synthesized.



(a) Low scale RKPM solution for flow (b) High scale RKPM solution for shock resolution

Figure 9.5. Multiresolution analysis of RKPM for 2-D compressible flow ($Ma = 1.4$)

9.5 Ability to Capture Shocks and Boundary Layers

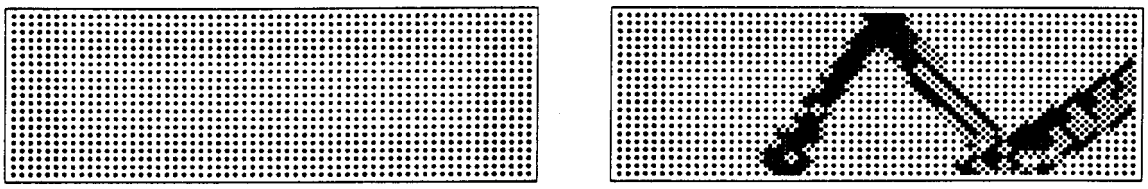
The n-level decomposition, specified by scaling functions, $\phi_n(x)$ and $\phi_m(y)$, and window functions, $\psi_n(x)$ and $\psi_m(y)$ is given as:

$$\phi_0(x,y) = \psi_1(x)\psi_1(y) + \ldots + \psi_2(x)\psi_2(y) + \ldots + \phi_n(x)\phi_n(y) \tag{9.14}$$

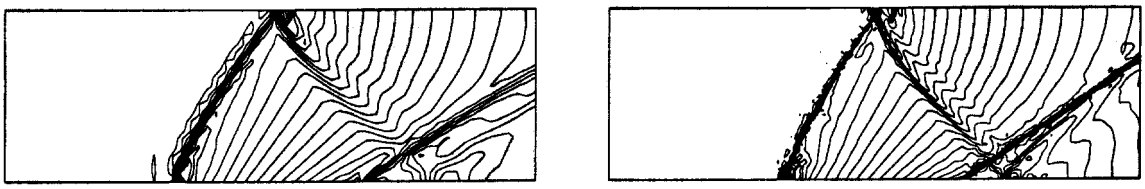
In the Fourier transform domain, an example of three-level decomposition is given as:

$$\begin{aligned}\hat{\phi}_0(\xi,\eta) = \hat{\phi}_0(\xi)\hat{\phi}_0(\eta) &= [\hat{\phi}_2(\xi) + \hat{\psi}_2(\xi) + \hat{\psi}_1(\xi)][\hat{\phi}_2(\eta) + \hat{\psi}_2(\eta) + \hat{\psi}_1(\eta)] = \\ &= \hat{\phi}_2(\xi)\hat{\phi}_2(\eta) + \hat{\phi}_2(\xi)\hat{\psi}_2(\eta) + \hat{\phi}_2(\xi)\hat{\psi}_1(\eta) + \\ &\quad + \hat{\psi}_2(\xi)\hat{\phi}_2(\eta) + \hat{\psi}_2(\xi)\hat{\psi}_2(\eta) + \hat{\psi}_2(\xi)\hat{\psi}_1(\eta) + \\ &\quad + \hat{\psi}_1(\xi)\hat{\phi}_2(\eta) + \hat{\psi}_1(\xi)\hat{\psi}_2(\eta) + \hat{\psi}_1(\xi)\hat{\psi}_1(\eta)\end{aligned} \tag{9.15}$$

The distribution of particles of a supersonic ($Ma = 1.4$) flow problem using 1281 particles and the successive multiresolution analysis based on the 2630 nonuniform particles is depicted in Figure 9.6(a). The 2×2 decomposition pressure contour plots of the 1281-particle solution and the corresponding results of the second multiresolution analysis (2630 particles) are depicted in Figure 9.6(b). The result from the original 1281 particles is decomposed in 2×2 scales as given in Figure 9.5 of the previous section, and additional points are added along the shock region based on the high scale wavelet solution, illustrated in Figure 9.5(b). As can be seen in Figure 9.6(b), the results of the wavelet multiresolution analysis, in which only particles are added to the high gradient region, enhance the resolution



(a) original distribution of particles (1281 nodes) and adaptive particles (2630 nodes)



(b) pressure contours for 1281 nodes and 2630 nodes

Figure 9.6. Multiresolution results for a compressible inviscid flow problem: Mach 1.4

of the shock layer. This multiresolution analysis is equivalent to that of a 5124-particle result. The low scale (scaling function) component $\phi_2(x)^*\phi_2(y)$ filters out the oscillation; therefore, it gives a very smooth solution around the shock front. The medium and high-scale (wavelets) components $\psi_2(x)^*\psi_2(y)$ and $\psi_1(x)^*\psi_1(y)$ show the mixture of high wave-number response and aliasing. However, the medium scale displays mostly the high wave-number approximation of the shock front, whereas, the high scale displays mostly the aliasing wave-number that cannot be resolved with a 1281-particle resolution. This is a typical example of the Gibbs phenomenon. Nevertheless, the high wavelet component $\psi_1(x)^*\psi_1(y)$ picks up the location of the high gradient region.

In defining the adaptive multiresolution analysis, the upper 20% of high gradient particles of the $\psi_1(x)^*\psi_1(y)$ wavelet components are chosen to locate the shock region. It is noted that not only does the width of the high gradient region become narrower, but its maximum value decreases when the adaptive multiresolution analysis is applied. This example shows that the integral window transform process can zoom in to pick up the high gradient of the response, and zoom out if no magnification of the response is necessary. The zoom in and zoom out capability of the multiple scale reproducing kernel method shows great promise in meshless unstructured multi-grid or hp-like multiresolution. Moreover, the physical interpretation of the computed results can further be understood.

10. FUTURE RESEARCH

10.1 Generalized RKPM

p-FE with RKPM enrichment

If we select ν^h as in Eq.(2.22), which is a set of neq global finite element shape functions $N_j(x)$ and we replace $N_p(x)$ by a higher order $\phi_p(x)$ at node p as illustrated in Figure 10.1a, then the consistency conditions Eqs.(2.24) and (2.25) are satisfied in the whole domain except under the support of $\phi_p(x)$, $\bar{\Omega}$. The p-FE with RKPM enrichment is achieved by enforcing the reproducing conditions only in the region $\bar{\Omega}$.

hp-FE with RKPM enrichment

The hp-FE with RKPM enrichment can be obtained by inserting an additional node in the domain of interest and a higher order $\phi_{neq+1}(x)$ window function in ν^h as shown in Figure 10.1b. Similar to the p-FE with RKPM enrichment, reproducing conditions are required in $\bar{\Omega}$. However, unlike the traditional hp finite elements, the additional window $\phi_{neq+1}(x)$ can cover some of the finite element nodes and no special adjustment of the higher order window function is needed along the boundary of elements. To accomplish better resolution from the given mesh and particles, a hp-refinement should be followed with a p-refinement in its neighborhood. Thus, we can control the dilation of these higher order window functions to achieve an optimal accuracy. A simple demonstration of this idea is given in Figure 10.1c.

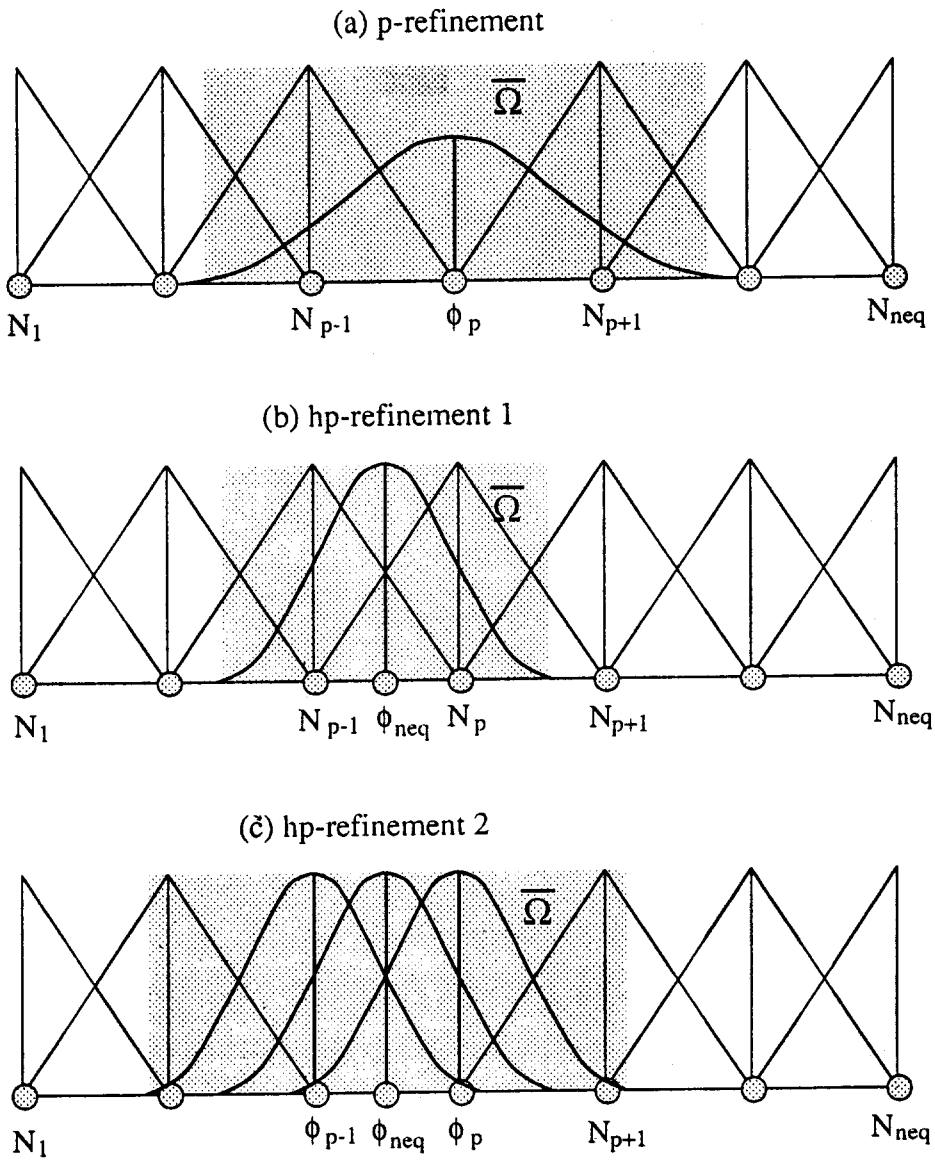


Figure 10.1. p-FE and hp-FE refinement with RKPM

Remark:

In a large deformation analysis, the finite element nodes and finite elements can be replaced by RKPM shape functions to avoid mesh distortion.

Numerical Example

Advection in a rotating flow field.:

$$a \cdot \nabla u - \kappa \cdot \nabla^2 u = 0 \tag{10.1}$$

where **a** is the advection velocity and κ denotes the diffusion coefficient.

The problem statement is shown in Figure 10.2. The flow is rigid rotation about the center of the bi-unit square domain, with velocity components given by:

$$a_1 = -y \qquad a_2 = x \tag{10.2}$$

The computational domain is specified as:

$$-0.5 \leq x \leq 0.5 \qquad \text{and} \qquad -0.5 \leq y \leq 0.5 \tag{10.3}$$

The problem is advection dominated, with diffusivity, κ, of 10^{-6}. Along the external boundary, u is set to zero, and on the internal "boundary" OA, u is prescribed to be a cosine hill.

$$u = \frac{1}{2}\left[\cos(4\pi y + \pi) + 1\right] \qquad (-0.5 \leq y \leq 0) \tag{10.4}$$

A 10-by-10 linear finite element method was used to perform the first computation step. After that, the multiresolution RKPM was employed to generate adaptivity, and the generalized RKPM was applied to derive the new shape functions. Four levels of adaptivity were computed, and the initial, the second and the fourth adaptivity refinements are depicted in Figure 10.3. The finite element solution shows severe oscillations, which is caused by the stability of the coarse mesh. After applying the hp-refinement, the solution is stabilized.

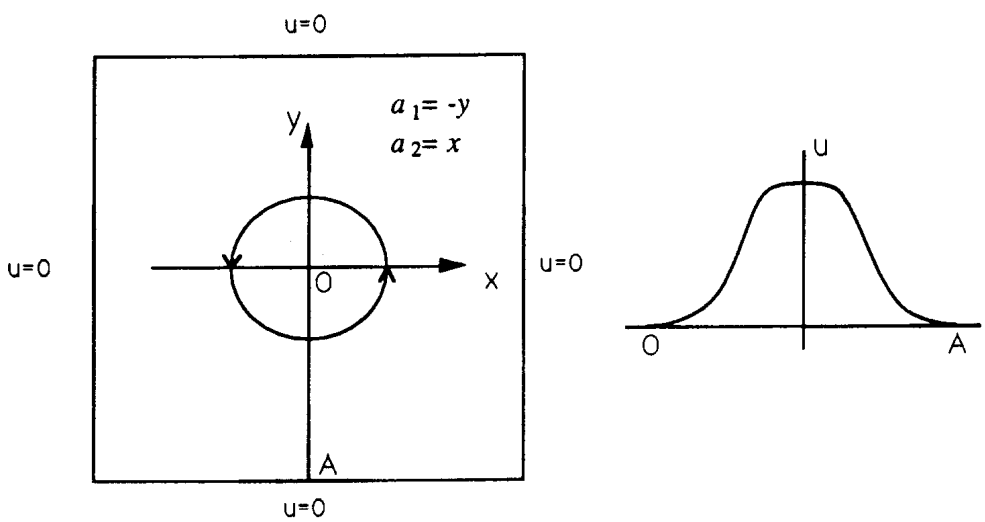


Figure 10.2. Rotating cone: problem statement

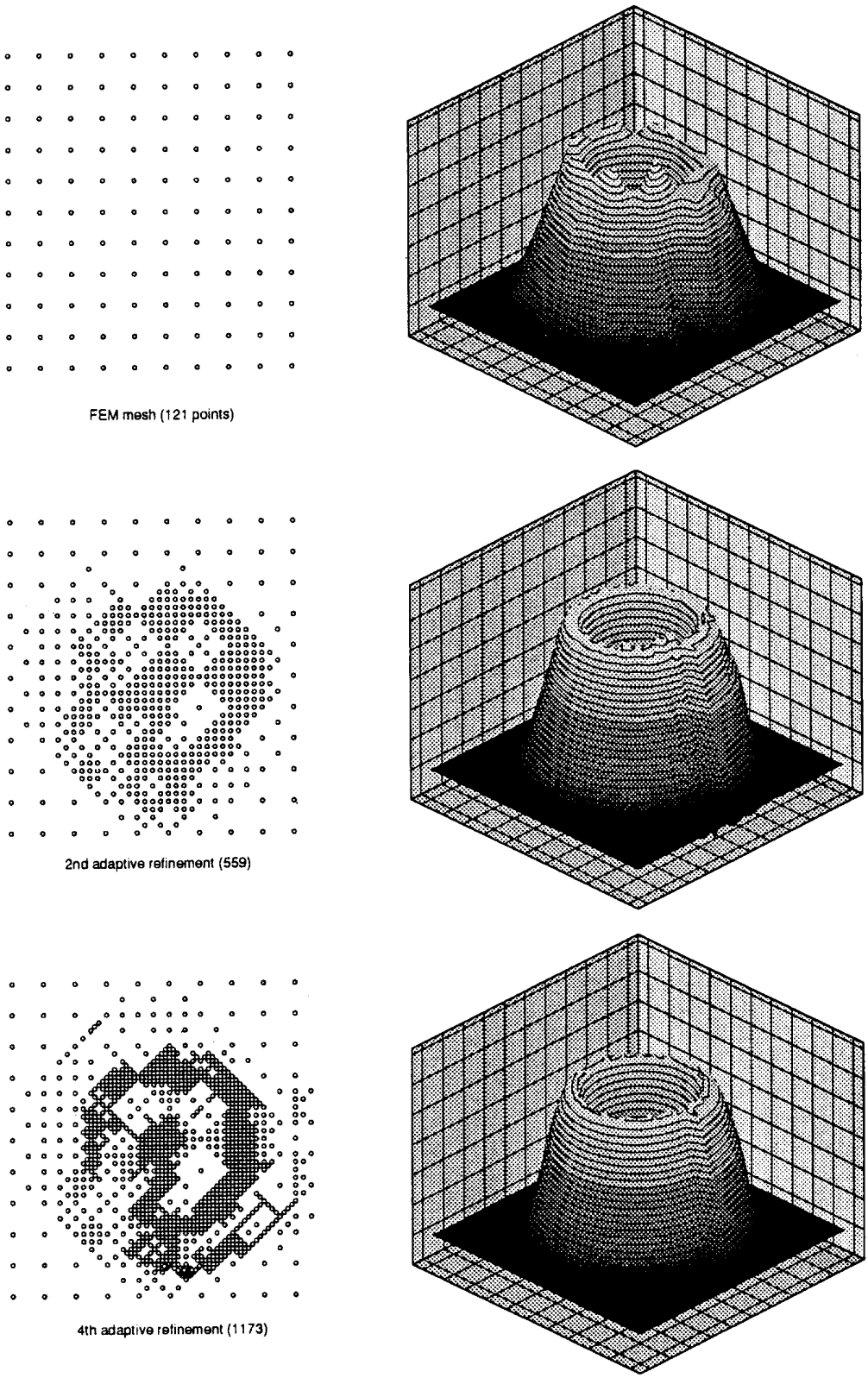


Figure 10.3. Adaptive solutions of the rotating cone by the generalized RKPM

10.2 Multiple Field RKPM

Liu and Chen (1995) and Liu *et al.* (1993) have developed a multiple scale method based on Wavelet Particle Method (WPM) and RKPM. This formula can be generalized to other multiple scale and multiple field methods. Consider the two scale decomposition of $u(x)$:

$$u(x) = v(x) + w(x) \tag{10.5}$$

where $v(x)$ and $w(x)$ are the different scales or different fields. Borrowing the concept of reproducing kernel projection, if v is the **2a** scale solution such that $P_{2a}v = v$ then

$$u(x) = P_{2a}u(x) - P_{2a}w(x) + w(x) \tag{10.6}$$

where $P_{2a}u$ is the projected solution, and $P_{2a}w$ is the interaction term. Eq.(10.6) is the general expression for a multiple scale analysis. In particular, if there is no overlapping of scales between $v(x)$ and $w(x)$, then by the property of the projection operator, the interaction term is zero. By repeat application of Eq. (10.6), multiple scale methods can be developed within the framework of a multilevel data structure.

Example 1: Wavelet series and reproducing kernel methods

If $w(x)$ is the orthogonal wavelet series, and $P_{2a}u$ is the reproducing kernel projection operator as defined in Eq.(2.7), then the reproducing kernel and wavelet methods developed in Liu and Oberste (1993) and Liu and Chen (1995) are obtained:

$$u(x) = \int_{\Omega} \bar{\phi}_a(x; x-y)u(y)dy - \sum_{m=0}^{+\infty}\sum_{n=n_1}^{n_2} \langle u, \psi_{mn}\rangle \left[\psi_{mn}(x) - \int_{\Omega_a} \bar{\phi}_a(x; x-y)\psi_{mn}(y)dy\right] \tag{10.7}$$

It is noted that if non-overlapping perfect low scale and high scale projections (perfect low and high pass filters) are applied, the interaction term becomes zero. The numerical simulation is given in Figure 10.4.

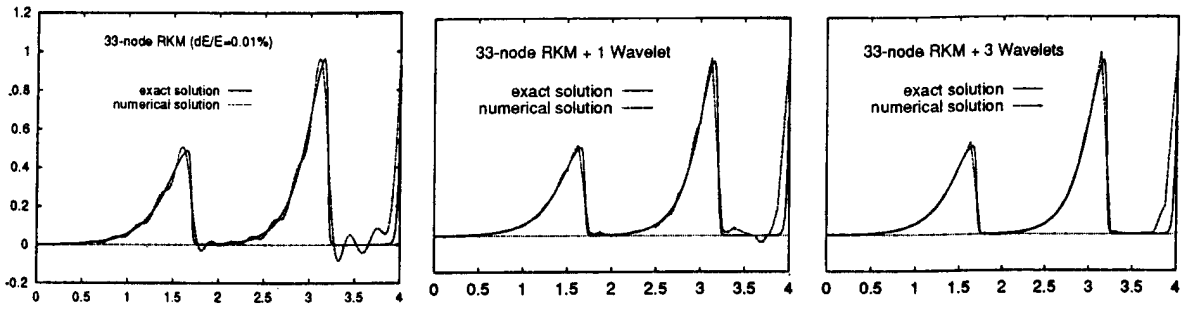


Figure 10.4. RKPM with 0, 1 and 3 wavelets enrichment

Example 2: Enriched projection method

If $P_{2a}u$ is the reproducing kernel projection operator, and $w(x) = cf(x)$, where c is a coefficient, and $f(x)$ is a given analytical function, Eq.(10.6) becomes:

$$u(x) = \int_{\Omega} \bar{\phi}_a(x; x-y)u(y)dy + c\left[f(x) - \int_{\Omega} \bar{\phi}_a(x; x-y)f(y)dy\right] \tag{10.8}$$

Boundary Layer Problem

The advection-diffusion equation is used to demonstrate the concept of the multiple field RKPM

$$u_{,xx}(x) - au_{,x}(x) = b(x) \tag{10.9}$$

with boundary conditions

$$u(0) = 0 \qquad \text{and} \qquad u(4) = 1 \tag{10.10}$$

and body force

$$b(x) = \left[\frac{1}{2}\delta(x) - \delta(x-2)\right] - a\left[\frac{1}{2}\langle x\rangle^0 - \langle x-2\rangle^0\right] \tag{10.11}$$

where $< \bullet >$ is the Macaulay bracket and it is defined as

$$\langle x-y\rangle^n = \begin{cases} 0, & x \le y \\ (x-y)^n, & x > y \end{cases} \tag{10.12}$$

and $\delta(x)$ is the Dirac delta function. The advection parameter, a, is chosen as 75 to ensure a boundary layer formation at the boundary.

The exact solution is given as:

$$u(x) = \frac{1}{1-e^{4a}}\left(1-e^{ax}\right) + \frac{1}{2}\langle x\rangle^1 - \langle x-2\rangle^1 \tag{10.13}$$

The analytical form to be coupled with the RKPM is the boundary layer term:

$$w(x) = cf(x) = c\left(1-e^{ax}\right) \tag{10.14}$$

where c is the extra unknown to be solved for.

As it can be seen in Figure 10.5 and Table 10.1, the absolute error in L^2 Norm for the multiple field RKPM is approximately half of that for the regular RKPM.

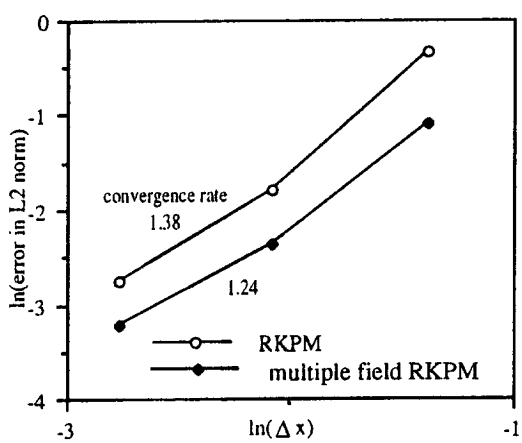


Figure 10.5. Error plot of the boundary layer problem

	RKPM	Multiple field RKPM
17 nodes	0.72062	0.33436
33 nodes	0.16797	9.41136E-2
65 nodes	6.45532E-2	3.97927E-2

Table 10.1. Errors of the boundary layer problem

Example 3: Wavelet RKM

If $P_{2a}u$ is the reproducing kernel projection operator defined in Ω_{2a} , and $w = Q_{2a}u$ in Ω_a, $\Omega_{2a} \supseteq \Omega_a$, then Eq.(10.6) becomes:

$$\begin{aligned} u(x) = &\int_{\Omega_{2a}} \bar{\phi}_{2a}(x; x-y)u(y)dy \\ &+ \int_{\Omega_a} \left[\bar{\psi}_{2a}(x; x-y) - \int_{\Omega_{2a}} \bar{\phi}_{2a}(x; x-z)\bar{\psi}_{2a}(z; z-y)dz \right] u(y)dy \end{aligned} \tag{10.15}$$

This is the same equation described in Liu and Chen (1995), if we choose $\Omega_a = \Omega_{2a}$. Examples are shown in Figure 9.5 in Section 9.

Example 4: hp-finite element with adaptive wavelets

If we choose $u^{FEM}(x) = P_{2a}u(x)$ in Ω_{2a}, and $w(x) = Q_{2a}w(x)$ is defined in Ω_a, then the multiple scale solution becomes:

$$u(x) = \sum_{j=1}^{NP} N_j(x)u(x_j) + \int_{\Omega_a} \left[\bar{\psi}_{2a}(x; x-y) - \sum_{j=1}^{NP} N_j(x)\bar{\psi}_{2a}(x_j; x_j - y) \right] w(y)dy \tag{10.16}$$

where $N_j(x)$ is the FEM shape function, and NP is the number of FEM nodes. To take full advantage of the adaptive hp-finite element method with wavelet enrichment, $w(x)$ should not intersect with any essential boundary conditions.

11. CONCLUSION

We have presented an overview of various meshless and wavelet methods. Because of the close relationship between this class of methods and finite element methods, we have presented our view of coupling of these methods using the concept of multiple scales together with a multilevel data structure. With the appropriate combination of multiple scale and multiple field methods, we believe the reliability of the computed response can be greatly increased. Although these methods seem very promising, significant advances need to be made before robust treatments for a large class of computational mechanics problems will be available.

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